CAPSTONE S25

Five-anomaly model evaluation

Complete technical runthrough

Data collection, annotation audits, standardized testing,
supervised CLIP and SmolVLM adaptation, temporal modeling,
component improvements and results

Targets	Smoking; littering; loitering; public rough sleeping; unattended items
Image models	MobileCLIP2-S2; SigLIP2-B/16-224; SmolVLM-500M-Instruct; MiniCPM-V 4.0
Video baseline	Separately supervised offline VadCLIP smoking adaptation
Experiment coverage	Historical work and completed studies, 11–19 October 2026
Report prepared	19 October 2026, Asia/Jakarta

Status at the reporting cutoff. Several recognition and object-detection components improved. Reliable generalization across all five target events has not been established. Previously inspected tests, provisional labels, source proxies, numerical corrections and unsuccessful strategies are identified throughout this document. This reporting task performs no new model training, quality inference or latency benchmark.

Reading guide and executive findings

This report reconstructs the project from the files actually present in the shared workspace. It explains what data was found, what additional data was acquired, how labels and splits were checked, how the models were evaluated, which learning strategies were tried, what changed in the results, and what remains unresolved. The account includes the early benchmark and its limitations, the corrected GPU comparison, the expanded image study, six actual visual fine-tuning runs, the 330-configuration strategy study, the 50-head video pilot, the 17 October ensemble, state-memory and tiled-detection experiments, and the 19 October SmolVLM LoRA training and controlled evaluation.

The principal outcome is a better research workflow and several stronger components, rather than a validated five-event deployment model. Waste-dumping supervision improves clip ranking substantially. Smoking photographs become easy to distinguish within their data domain, while surveillance-camera recall remains poor at conservative operating points. Geometry helps the historical loitering diagnostic, but thermal supplied-track results and a single RGB camera do not establish automatic tracking across cameras. Public-sleep evidence is small and provisionally reviewed. Explicit unattended-item state is more useful than a global image score on two clean AVSS events, but normal exposure is far too short to validate a low false-alert rate.

The 17 October component result is concrete: fixed overlapping image tiles increase recall against 136 CAVIAR item reference boxes from 66/136 (48.5%) to 116/136 (85.3%) at intersection-over-union 0.5. A SigLIP ensemble raises AP on 205 retrospectively evaluated HMDB clips from approximately 0.778 to 0.9645. Neither result demonstrates a comparable gain on unseen surveillance-camera events. Additional item-state memory produces no improvement on the two-event AVSS primary subset and increases repeat alarms on the separately censored diagnostic.

The newly completed SmolVLM study trains three language-layer LoRA adapters with 4,341,760 parameters each, using 3,973 training examples, 1,812 development examples and ten held-out source/task cohorts totaling 3,792 unique cases. Primary routed AP improves on IWDD from 0.5922 to 0.7660 and historical unattended endpoints from 0.4762 to 0.6933, but loitering falls from 0.4250 to 0.1822. At the development FPR-budget cutoff, unattended recall is 94.6% with 70.8% held-out FPR. CCTV smoking recall is only 1.8%. The optional rule restoring the original loitering model was introduced retrospectively and is identified as exploratory. These results reinforce the distinction between ranking gains, source transfer and usable event decisions.

How evidence is described

Published/source annotation. Labels or boxes released with a dataset. Their meaning is retained: a primary action, a posture, a box or an abandonment action is not automatically a complete label for the target event.

Provisional assistant review. Visual interpretation performed during preparation. This includes historical unattended intervals and apparent public-sleep photographs. It is not independent human verification.

Held out within a collection. Inputs excluded from fitting under the recorded split. Their collection or domain may still contribute training data.

New-source transfer. A source not used in the study's learning/selection when first scored. Once its results have been inspected, later experiments on it are retrospective.

Component diagnostic. A test of a box detector, crop coverage, posture, supplied trajectory or action classifier. Its units and target differ from end-to-end event discovery.

Recommended reading order

Read Chapters 1–3 for scope, data and evaluation rules; Chapters 4–10 for the initial and CLIP/-component experiments; Chapter 11 for SmolVLM adaptation; Chapter 12 for the current five-target assessment; and the final main chapter for reproducibility. Appendices provide all 330 strategy-development entries, all 50 final video-head selections, all 38 latest smoking-policy summaries, seed-specific fine-tuning operating points, the event table, pinned identities and the source registry. These appendices preserve alternatives instead of selecting whichever test column happens to look best.

Count discipline. Images, cropped views, examples, overlapping windows, predictions, clips, events and camera groups are different units. For example, 55,156 predictions are four models answering multiple input protocols; they are not 55,156 anomaly events. Three seeds add optimization replications on the same data, not three new sources. The 936 prepared video-pilot source files are dominated by movie/web action clips, not 936 surveillance cameras.

The document is dated 19 October 2026 and supersedes the preserved 18 October edition. The documentation update runs no new model experiment; it incorporates the completed 19 October training artifacts. Run-directory names often carry their preparation/start date rather than finalization date. All experiment figures and conclusions retain their original stage-specific protocols. The accompanying source manifest records the local files and SHA-256 identities used to build the tables and figures.

Contents

Reading guide and executive findings **i**

1 Scope, research questions and project history **1**

1.1 Original objective and changing emphasis . . . 1

1.2 What each target requires . . . 1

1.3 Model families and comparison boundaries . . . 2

1.4 The early small benchmark and why it was superseded . . . 3

1.4.1 Correctness problems identified before the new quality claims . . . 3

1.5 Experiment chronology . . . 4

1.6 What was preserved . . . 4

2 Data discovery, collection and annotation preparation **6**

2.1 Audit the local corpus before increasing its size . . . 6

2.2 Data identity and standardization . . . 7

2.3 Expanded historical unattended intervals . . . 7

2.4 Supervised training sources acquired on 15 October . . . 8

2.4.1 Stanford40 and CCTV smoking . . . 8

2.4.2 MIVIA illegal waste dumping . . . 8

2.4.3 Street Scene supervised loitering re-split . . . 8

2.4.4 Apparent public sleep and matched controls . . . 9

2.5 Further source acquisition on 16 October . . . 9

2.5.1 Additional smoking photos and HMDB action clips . . . 9

2.5.2 Thermal human trajectories . . . 9

2.5.3 MEVA abandonment actions and further public-sleep photos . . . 10

2.6 AVSS author annotations: a new source at first evaluation . . . 10

2.7 17 October author release: reject source inflation . . . 11

3 Common evaluation and generalization methodology **12**

3.1 Define the unit of evaluation . . . 12

3.2 Grouping and leakage controls . . . 12

3.3 Shared image protocols and response parsing . . . 12
3.4 Classification metrics . . . 13
3.5 Development-only operating points . . . 13
3.6 Model selection and multiple comparisons . . . 14
3.7 Event metrics and unknown time . . . 14
3.8 Uncertainty and numerical reproducibility . . . 14

4 Corrected workflow and the first complete GPU comparison 15
4.1 Make quality testing use the available hardware . . . 15
4.2 Workflow changes implemented before the full comparison . . . 15
4.3 The completed 13 October accuracy cohort . . . 16
4.4 Prompt and sampling diagnostics . . . 17
4.5 Separate supervised offline VadCLIP smoking study . . . 17
4.6 Historical CPU latency pilot: recorded, not rerun . . . 17
4.7 Validation at this stage . . . 18

5 Expanded four-model accuracy evaluation 19
5.1 Why this expansion was necessary . . . 19
5.2 GPU execution policy . . . 19
5.3 Expanded unattended endpoint results . . . 19
5.4 Frozen development cutoffs and interval alarm screen . . . 20
5.5 What the three weak target cohorts measured . . . 20
5.6 Oracle lying-versus-standing posture diagnostic . . . 21
5.7 What this study changed in the next plan . . . 21

6 Actual supervised CLIP visual fine-tuning 23
6.1 Learning objective and representations . . . 23
6.2 Probe, visual parameters and regularization . . . 23
6.3 Selection, replications and execution details . . . 24
6.4 Numerical audit and final evaluation . . . 24
6.5 Held-out ranking gains and regressions . . . 24
6.6 Why AP did not make the models safe to threshold . . . 25
6.7 Checkpoint and interface validation . . . 26

7 Expanded sources and the 330-configuration strategy study 27
7.1 Motivation and controlled strategy families . . . 27
7.2 Visual weight interpolation and classifier fitting . . . 27
7.3 Selection timing and source protection . . . 28

7.4 Historical source results . . . 28
7.5 New photo holdout and source-transfer results . . . 29
7.6 Separate thermal trajectory component . . . 30
7.7 Integrity checks and outcome . . . 30

8 Causal video pilot: failure audit, regions and learned heads 32
8.1 Start from observed failure modes . . . 32
8.2 Video preparation and identity repair . . . 32
8.3 Automatic detection and causal tracking . . . 33
8.4 Frozen visual encoders and automatic regions . . . 33
8.5 Fifty final candidate heads . . . 34
8.6 Smoking results and the fair within-pilot controls . . . 34
8.7 Completion and recovery . . . 35

9 Continuous unattended-event evaluation and calibration 36
9.1 Freeze global alarm policies before opening the new source . . . 36
9.2 Historical windows and actual event discovery . . . 36
9.3 Clean AVSS primary events . . . 36
9.4 Known annotation criteria: same item, stationary and owner away . . . 37
9.5 Censored HARD and why it stays separate . . . 37
9.6 Disjoint normal-recording calibration . . . 38
9.7 Quantify the short-exposure limitation . . . 38

10 17 October improvements: ensembles, item memory and tiling 40
10.1 A focused round rather than another broad fine-tune . . . 40
10.2 Nineteen smoking policies per encoder . . . 40
10.3 Causal stationarity smoothing . . . 41
10.4 Per-item alarm memory . . . 41
10.5 New-source compatibility review . . . 42
10.6 Fixed overlapping tiles: a measurable component gain . . . 42
10.7 Aggregation correction and independent replay . . . 43

11 19 October: adapting SmolVLM to the five definitions 44
11.1 Purpose and the controlled comparison . . . 44
11.2 Manifest, units and source separation . . . 45
11.2.1 Main study and additional transfer . . . 45
11.2.2 Leakage controls and the limits of these controls . . . 45
11.2.3 Label semantics and masked unknown objectives . . . 46

11.3 Image processing, view order and frozen visual features . . . 46
11.4 What is learned and how the binary score is defined . . . 47
11.4.1 Low-rank updates in language projections . . . 47
11.4.2 Answer-only training with a full-vocabulary loss . . . 47
11.5 Optimization, balancing and selection timing . . . 48
11.5.1 Coverage and source-balanced resampling . . . 48
11.5.2 Development metric and primary routes . . . 48
11.5.3 Early overfitting is visible in all candidates . . . 49
11.6 Held-out discrimination: gains and preserved regressions . . . 50
11.7 Fixed decisions and threshold-transfer failure . . . 51
11.7.1 The development FPR budget does not transfer automatically . . . 51
11.7.2 Raw margin-zero decisions answer a different question . . . 51
11.8 Seed stability and the optional retrospective safeguard . . . 52
11.9 Context against the existing CLIP strategies . . . 53
11.10 Verification, exported adapters and inference limits . . . 54

12 What the results establish and what remains unresolved 56
12.1 Smoking . . . 56
12.2 Littering . . . 56
12.3 Loitering . . . 57
12.4 Rough sleeping / apparent public sleep . . . 57
12.5 Unattended items . . . 57
12.6 Generalization lessons across all stages . . . 58
12.7 Proposed next validation gates, not work already completed . . . 58

13 Software, artifacts, reproducibility and reporting 59
13.1 Repository organization and portable execution . . . 59
13.2 Reproduce a new study without overwriting completed output . . . 59
13.3 What is saved for every major stage . . . 60
13.4 SmolVLM reproduction and controls . . . 60
13.5 Verification progression . . . 61
13.6 Presentations and this LaTeX report . . . 61

A Historical audit and superseded adapter experiments 63

B Fine-tuning replications and operating points 65

C All 330 strategy-development configurations 67

D All 50 final video-pilot heads **74**

E All 38 latest smoking policies **76**

F Detailed event results and source accounting **78**

G Complete SmolVLM histories and source results **82**

H Pinned identities, artifact registry and acquisition references **85**

H.1 Preserved run-directory inventory 85

H.2 Source registry 86

H.3 Recorded public source locations 91

H.4 Final scope statement 91

List of Tables

1.1 Definitions 2
1.2 Models 2
1.3 Chronology 4

2.1 Expanded cohorts 7
2.2 Training counts 9
2.3 Avss sources 10

4.1 Baseline smoking 16
4.2 Baseline unattended 16
4.3 Vadclip 17
4.4 Historical latency 18

5.1 Expanded unattended 20
5.2 Weak diagnostics 20
5.3 Oracle pose 21

6.1 Finetuning main 25
6.2 Finetuning operating 25

7.1 Strategy routes 28
7.2 Strategy historical 28
7.3 Strategy transfer 29
7.4 Thermal 30

8.1 Crop audit 32
8.2 Pilot smoking 34

9.1 Pilot avss primary 37

10.1 Latest smoking 40
10.2 Latest events 41

10.3 Latest components 42

11.1 Smol counts 45
11.2 Smol settings 47
11.3 Smol checkpoints 49
11.4 Smol routes 49
11.5 Smol ap 50
11.6 Smol threshold transfer 51
11.7 Smol raw 52
11.8 Smol seeds 52
11.9 Smol repeated gains 53
11.10Smol clip context 53

13.1 Verification 61

A.1 Historical audit 63
A.2 Historical vadclip ablation 63

B.1 Finetuning all operating 65
B.2 Finetuning seeds 66

C.1 Strategy all candidates 67

D.1 Pilot all heads 74

E.1 Latest all policies 76

F.1 Pilot all events 78
F.2 Latest all event groups 80

G.1 Smol history 82
G.2 Smol all metrics 82

H.1 Model revisions 85
H.2 Run inventory 85
H.3 Source registry 86

List of Figures

6.1 Same-study pretrained and development-routed AP for the primary fine-tuning seed. The cohorts differ across targets; bars do not form a cross-task difficulty ranking. . 25

7.1 Strategy-study transfer AP, comparing the previous bundle with the development-selected strategy. Photos share a training collection; HMDB is movie/web action evidence; MEVA is an abandonment-action proxy; public-sleep labels are provisional. 30

9.1 Conditional one-sided 95% Poisson upper false-alert rate after zero alerts, as a function of reviewed normal exposure. The AVSS exposure is far below the approximately three-hour crossing for one alert/hour. This is an exposure calculation, not a measured deployment-rate curve. 39

10.1 Full-image versus full-plus-tiles reference-box recall at IoU .5. CAVIAR is a training-source component diagnostic; highway debris is an out-of-domain stress test. Neither bar measures object precision or event detection. 42

11.1 All three LoRA training histories. The original development baseline and selected first epochs distinguish optimization progress from source-transfer progress. These curves reuse the same development cohorts. 49

11.2 Original, joint and per-task routed SmolVLM AP on ten source/task cohorts. All use the same inputs and binary score within this study. Joint and routed results differ for public sleep. 50

Chapter 1

Scope, research questions and project history

1.1 Original objective and changing emphasis

The project began as a comparison of compact contrastive and generative vision–language models for public-space anomaly recognition. The requested targets are smoking, littering, loitering, rough sleeping and unattended personal items. The requested model families are MobileCLIP2-S2, SigLIP2-B/16-224, SmolVLM-500M-Instruct, MiniCPM-V 4.0 and VadCLIP. The early repository emphasized a constrained CPU deployment setting. The subsequent user direction explicitly prioritized model quality, allowed available GPU compute without pacing, and eventually removed the need for further latency testing.

That change affects interpretation of the work. Accuracy experiments should use available compute to test the models properly; a slow CPU replay does not make the quality test more valid. Conversely, CPU latency requires a separately controlled protocol. The later experiments therefore use GPU training and extraction where applicable, with no artificial camera-rate sleeps. The historical CPU pilot remains part of the record, but no new latency study is added after the change in priorities.

The research questions become progressively more specific. First, can the frozen models produce useful categorical decisions using common prompts and images? Second, does a larger and more carefully reviewed corpus change those conclusions? Third, can supervised binary heads and actual visual adaptation improve task discrimination? Fourth, do crops, explicit temporal order, geometry and controlled weight blending transfer better to other data sources? Fifth, can causal item history and event accounting address the gap between high image AP and excessive alerts? The 17 October experiments then isolate ensembling, source-wise operating points, stationarity smoothing, alarm memory and image tiling. The 19 October study asks whether language-layer LoRA can improve a task-conditioned SmolVLM decision function across all five definitions, while preserving a matched original-model control and independently repeating the winning learning rate.

1.2 What each target requires

Table 1.1: Recorded visible-evidence target definitions.

Label	Definition
normal	A public area with ordinary activity and none of the defined anomalies visibly present.
littering	A person visibly discarding rubbish on the ground instead of using a bin.
loitering	A person remaining in the same public area for an extended time without an apparent activity. Duration requires evidence across time.
rough sleeping	A person apparently sleeping or lying down with bedding in a public area. Describe visible evidence only; do not infer housing status.
smoking	A person visibly smoking a cigarette, cigar, or pipe, with a smoking object near the mouth or visible smoke.
unattended items	A bag, box, or other personal item apparently left behind without its owner nearby. Ownership and duration require evidence across time.

The definitions are the repository definitions accepted during the work. There is no newly imposed universal loitering duration. Source-provided event criteria remain explicit. In particular, AVSS's 30-second stationary-object criterion is a source-specific diagnostic, not a retrospective relabelling of every historical unattended interval.

Several targets require evidence absent from a single global image. A person near a cigarette-sized object is spatially subtle; a person remaining in an area is a temporal identity problem; a bag becoming unattended is a relationship and duration problem. Apparent public sleep needs visible posture/context and, for a persistent event, time evidence. The model interface should therefore permit unknown when duration, ownership or visibility is insufficient. A scene classifier alone cannot establish object localization or ownership.

1.3 Model families and comparison boundaries

Table 1.2: Models and comparison scope; pinned revisions appear in the appendix.

Model	Family	Checkpoint / feature source	Role
MobileCLIP	openclip	`timm/MobileCLIP2-S2-OpenCLIP`	Frozen image comparison; CLIP later adapted
SigLIP	siglip	`google/siglip2-base-patch16-224`	Frozen image comparison; CLIP later adapted
SmolVLM	vlm	`HuggingFaceTB/SmolVLM-500M-Instruct`	Original prompted comparison; 19 Oct language-layer LoRA with binary scoring
MiniCPM	minicpm	`openbmb/MiniCPM-V-4`	Prompted image comparison only
VadCLIP	Supervised video adapter	Frozen CLIP-B/16 features	Separate offline smoking protocol

The CLIP-family adapters produce continuous text-image scores. For the original shared image protocol, each model compares the same selected inputs and definitions. Generative VLMs answer a deterministic classification prompt and may emit a valid label, explicit unknown or invalid response. Hard labels do not provide a continuous ranking score from which a meaningful AP or AUROC can be invented.

VadCLIP is a separate supervised offline smoking adaptation over frozen CLIP-B/16 features. It receives sequence supervision, uses a different split and can use context within an offline sequence. It is therefore informative about a temporal supervised baseline, but it is not an equal-supervision

five-task competitor in the image-model table. The 15–17 October learning studies adapt the two CLIP backbones and fit subsequent heads. On 19 October, a separate study actually adapts SmolVLM language projections with LoRA under matched binary prompting. MiniCPM and the historical VadCLIP adapter receive no additional training in that round.

1.4 The early small benchmark and why it was superseded

The historical 11 October narrative describes an 80-image benchmark with 16 examples each of littering, loitering, smoking, rough sleeping and normal, and no unattended-item examples. It reported a six-label multiclass comparison and various CPU and VadCLIP ablations. Later audits established that some apparent target labels were only person, action or synthetic posture proxies. The historical binary anomaly prevalence was 64/80; an always-anomaly classifier already obtains binary F1 of approximately 0.889. Such a base rate can make a high F1 look more convincing than the event evidence supports.

The report is retained as historical context, not used as the main five-event conclusion. The subsequent validation summary explicitly narrows its scope to a real MobileCLIP CPU pipeline check, rather than a full-model/full-event ranking. This is an important correction in the project: unsupported event labels and incomplete model execution must not be carried forward merely because an earlier report presented a leaderboard.

The 12 October run originally open in the IDE is another distinct artifact. It is marked execution status `error`; its prepared smoking cohort has only 236 frames, and the MiniCPM smoking result is missing. Some saved response counts include three repetitions of the same selected input. A repeated inference is not a new labelled sample. This run is useful for diagnosing caps, CPU execution and response behavior, but the completed uncapped GPU study replaces it as the principal baseline comparison.

1.4.1 Correctness problems identified before the new quality claims

The archived 11 October audit records 15 findings, not merely a request to add more samples. It reproduces fractional-positive labels being truncated to zero after temporal compression: a mixed segment label .5 becomes integer zero at evaluation. Changing visual length therefore changes both targets and prevalence, invalidating a clean length-ablation interpretation. It also reproduces a precision–recall threshold-index mismatch, so the reported threshold can disagree with the reported F1; subsequent evaluation recomputes decisions at actual development-selected cutoffs.

A CPU “best checkpoint” created using a detached CPU tensor without a clone shares storage with ongoing training weights. Later optimizer steps can change the saved in-memory best state. The new training paths retain cloned snapshots and checkpoint identity checks. Other audit concerns include validation reused for epoch/branch/threshold selection without a final test, padded frames influencing real predictions, cache reuse without complete image/annotation/weight identity, and accepting stale notebook summaries after incomplete configuration checks.

The image-label audit also shows that normal meant not-smoking in one corpus, not verified absence of every event, while proxy class membership could be trivially confounded with the source. A post-hoc metadata-only dimension rule is reported as reaching 80% accuracy on the old 80-image bundle; this is a source-confounding demonstration, not a legitimate held-out detector baseline. The historical macro-F1 class denominator could change with predicted classes, so the old claim of a fixed five-sixths upper bound was not a reliable scoring description. Abstentions were excluded from some conditional scores, and row-wise rather than source-group bootstrap exaggerated independence.

These findings explain the later emphasis on source-qualified binary tasks, fixed supported label sets, all-input coverage, cloned checkpoints, raw-frame mappings, development-frozen cutoffs, immutable experiment directories and independent metric replay. The appendix preserves the original finding inventory and the historical VadCLIP ablation records as superseded evidence. Their displayed APs are not promoted into the corrected final leaderboard.

Evidence: `reports/anomaly_benchmark_report.md`

Evidence: `reports/archive/2026-10-11/evaluation_validation_summary.json`

Evidence: `reports/archive/2026-10-11/evaluation_audit.md`

Evidence: `results/evaluation_20261012T165345_130231Z/report.md`

1.5 Experiment chronology

Table 1.3: Distinct stages of the recorded project.

Dates	Stage	Completed evidence / important qualification
11 Oct	Early benchmark and audit	80 images; proxy labels and no unattended class. Historical, superseded.
12 Oct	Corrected CPU-era run	236 smoking frames; run status error; MiniCPM smoking missing.
13 Oct	Full GPU comparison	13,400 predictions across four models; separate supervised VadCLIP and CPU pilot.
14 Oct	Expanded accuracy	55,156 predictions; 1,770 unattended endpoints and qualified weak diagnostics.
15–16 Oct	Visual fine-tuning	Two backbones x three seeds; six actual changed-visual-weight runs.
16 Oct	Strategy screen	330 configurations, new-source transfer and provided thermal trajectories.
16 Oct	Causal video pilot	50 heads; real tracks/regions; two clean new AVSS events, one censored.
17 Oct	Focused improvements	38 policies; state/memory ablations; 830 tiled-detector GPU views.
18 Oct	First full technical report	83-page account through 17 October; no new training for documentation.
19 Oct	SmolVLM LoRA and updated documentation	Three trained adapters; ten held-out cohorts; repeat-seed audit; no latency testing.

The chronology should be read as a sequence of distinct studies. The earlier test images become historically inspected after their first evaluation. Later development-only selection is still preferable to choosing on test results, but repeated experimentation against familiar source families cannot restore an untouched test claim. The report consequently uses “retrospective” whenever later changes reuse those results.

1.6 What was preserved

Every completed study has its own result directory, selection records and raw predictions. Original AMP outputs remain beside corrected float32 evaluations. Earlier source-prefix video fits remain in a diagnostic subdirectory after the known-film guard. The initial incorrect tiled-metric export remains explicitly labelled invalid after an aggregation bug was corrected from the identical cached detections. Earlier slide decks remain historical; the 17 October deck consolidates 32 slides without

replacing their source results. The 19 October edition extends the technical paper and presentation to include the completed SmolVLM study while retaining the historical versions. This preservation makes the progression auditable and prevents a later narrative from erasing regressions.

Chapter 2

Data discovery, collection and annotation preparation

2.1 Audit the local corpus before increasing its size

The initial problem was not simply a low sample cap. The local `data/` tree mixed COCO/YOLO exports, renamed images, augmented copies, raw video, MAT video volumes, foreground masks, action labels, person boxes and posture labels. A valid event test required separating those meanings. File totals could be large while usable task-specific negative labels, recording identities and temporal intervals were absent.

The 13 October local audit counted 4,214 smoking images across two COCO exports. The smoking-video export contains 3,278 frames with Smoking/Not Smoking boxes and 16 camera/day sessions. The other CCTV export contains 936 images that are all smoking-positive; a `no_vest` category does not establish smoking-negative scenes. The 3,278-frame export supports an exploratory development comparison, whereas the 936-image collection cannot supply a matched binary benchmark without additional genealogy and normal examples.

Littering material contained 6,721 image files and 7,134 text label files. Its class names included dropping, dumping, putting-down, throwing and tossing. These are related actions rather than a uniform annotation of rubbish being discarded. The literal filename audit found 46 images lacking same-name labels and 45 labels lacking same-name images across train/test. Rename receipts can recover aliases, so these are review candidates rather than proof that every unmatched file is corrupt. Seven empty test labels are not sufficient grounds for declaring all those scenes event-negative. The exported validation labels lacked corresponding validation images, and train/test class distributions differed.

The local loitering material contained 1,675 images, 95 videos and 37 MAT files. The image exports annotate people. Inspected Avenue MAT files contain grayscale video volumes under `vol`, with spatial shape 120 by 160 and a frame axis; they are not loitering interval annotations. VIRAT material requires its event/track mapping. A visible person box cannot label excessive dwell time, and a convenient abnormal-video label cannot automatically be mapped to loitering.

The rough-sleeping exports contained 224 image files but only 112 unique file/RGB identities. Their labels concerned laying and standing, with duplicated source exports. They do not verify apparent sleep in public, bedding context or duration. The unattended material included 16 raw videos plus acquired CAVIAR/ABODA sources; reviewed CAVIAR leaving/pickup/walking intervals were usable for an initial small event/control study, whereas ABODA needed interval preparation. D2 foreground masks were auxiliary, not complete target event truth.

Evidence: `reports/data_readiness_20261013.json`

Evidence: `reports/workflow_review_20261013.md`

2.2 Data identity and standardization

Preparation used a common example schema with a target, label, source, recording/group identity, ordered input views and evidence qualifications. File hashes, decoded EXIF-corrected RGB pixel hashes and checksum-qualified source moments detect aliases at different levels. Byte-identical files are not the only duplicates: different filenames or encodings may decode to the same pixels. Conversely, two files sharing an old original name but different contents must not be merged silently.

The corrected baseline preparation removed 104 aliases from 3,454 collected input rows. Its retained 3,350 inputs include repeated endpoint identities across frame/window protocols, so the total is a protocol-input count. The lookup path for original download names was also indexed. Previously, each lookup reparsed a roughly 3 MB rename receipt and scanned entries; the indexed implementation performs one receipt parse per file version and invalidates it when the receipt changes. A live check recorded more than 30,000 lookups with a single parse. This changes preparation efficiency, not model quality.

Source moments, labels, hashes and split membership are checked together. Conflicting labels on duplicate pixels and cross-split aliases fail validation. Missing labels are unknown. Source grouping is conservative where camera identities are incomplete; perceptual similarity can identify related files but cannot prove all physical camera, person or film identities independent.

2.3 Expanded historical unattended intervals

The first reviewed unattended corpus had 36 endpoints: 20 development and 16 test. Expansion reviewed intervals across 18 recordings and retained 1,770 endpoints: 808 development and 962 test. The test labels comprise 242 positives and 720 normals, and five positive test recordings. Entire ABODA classroom and walkway families remain test; development draws from entrance, lobby and CAVIAR families. Thus the apparent 49.2-fold endpoint increase still yields only two held-out camera families.

Uncertain cases were excluded rather than forced into binary truth: an occluded CAVIAR bag, an owner who remains present, ambiguous or occluded street placements, and a pre-existing item without visible deposition/owner history. Source copies were deduplicated. Positive CAVIAR intervals were checked against source XML, while broader timeline labels are provisional assistant visual review. They were prepared before the model predictions used in that study, but independent human verification is still absent.

Each endpoint also generated a past-only, four-frame contact sheet over 20 seconds. This extends the earlier six-second sheet. It is a different input protocol: temporal coverage grows while each subimage occupies less spatial area. It must not be described as adding independent events, as native video processing, or as an isolated data-volume intervention.

Table 2.1: Expanded study cohort sizes; frame/window unattended inputs share endpoints.

Cohort	Dev	Test	Evidence
unattended expanded frame	808	962	Provisional intervals
unattended expanded window	808	962	Provisional intervals
littering weak	6667	0	Weak / component diagnostic
loitering weak	1675	0	Weak / component diagnostic
person presence	1675	0	Weak / component diagnostic

Continued on next page

Table 2.1 continued

Cohort	Dev	Test	Evidence
rough sleeping weak	111	0	Weak / component diagnostic
lying pose oracle	73	48	Oracle posture crops

2.4 Supervised training sources acquired on 15 October

2.4.1 Stanford40 and CCTV smoking

Stanford40 contributes smoking photographs and confusable actions: drinking, blowing bubbles, brushing teeth, phoning and texting. The original author dataset is the source; a recorded public mirror was used for the image archive because the original image server was slow. Official test membership remains intact. Official training examples are split into grouped train/development pools after perceptual screening. Two previously used debug-test images are excluded from fresh holdout claims.

The final combined smoking data has 2,097 training examples, 1,135 development examples, 646 CCTV test frames, and 806 separate official-test photographs. Camera c4 trains, c2 develops and c1 tests across dates. Historical CCTV results were already visible, so the camera holdout is retrospective. Stanford40 is an image holdout within a domain that contributes training photographs, not an entirely unseen dataset. Its primary-action labels are not exhaustive smoking annotations for every person.

2.4.2 MIVIA illegal waste dumping

The author-labelled MIVIA-IWDD release supplies action videos and hard negatives. Of 500 listed videos, 498 were acquired; two missing files, `vid0325.mp4` and `vid0346.mp4`, belong to training. All 100 official test videos were obtained. Conservative perceptual-source clustering excluded another 63 acquired training videos connected to official test material. The final manifest retains 435 clips: 274 train, 61 development and 100 official test.

Eight frames are sampled uniformly across a full clip without using its event onset to choose test inputs. The source onset annotations are retained, but this experiment is offline video-level action classification. A sampled frame before the action is not independently labelled a dumping event. The release includes bulky waste dumping as well as small rubbish, so its relationship to pedestrian littering must remain explicit.

2.4.3 Street Scene supervised loitering re-split

Track/type annotations are mapped using the authors' event-index tables. The source paper numbers events from one whereas annotation track identifiers are zero-based; preparation accounts for that offset. Twenty seconds of past context is represented by four chronological views on a 15-second endpoint grid plus target-track endpoints. Other anomaly types are excluded as normal controls, avoiding training loitering to distinguish merely any abnormal scene.

Whole recordings remain together. The result is a supervised research re-split of recordings originally belonging to the source Test collection, not the official unsupervised Street Scene benchmark. A single camera limits the generalization claim even when recording identities are kept separate. The final held-out subset has 75 windows and only 11 positives.

2.4.4 Apparent public sleep and matched controls

The first new photo study uses openly licensed Commons images. Positive provisional labels require a person visibly apparently asleep in a public area, or lying with bedding there. Controls include awake people on benches, awake people lying on grass, indoor sleepers and clear empty context. Ambiguous rest versus sleep, uncertain public/private context, paintings and animals are omitted. Author and perceptual-neighbor groups stay together. The retained 113 photos provide 74 train, 21 development and 18 test cases, with six positive test photos.

These labels describe apparent visible sleep/posture/context; they neither label nor infer housing status. They were assistant-reviewed, not independently human-reviewed. They support a small image diagnostic and supervised learning experiment, not a strong persistent rough-sleeping event benchmark.

Table 2.2: First supervised study: normal/positive counts, with a separate smoking photo holdout.

Target	Train N/+	Dev N/+	Test N/+	External N/+
Smoking	1381/716	722/413	253/393	666/140
Dumping / littering	130/144	29/32	50/50	–
Loitering	282/41	85/13	64/11	–
Public sleep	49/25	14/7	12/6	–
Unattended	416/74	137/181	720/242	–

Evidence: `data/clip_training_20261015/dataset.json`

Evidence: `data/clip_training_20261015/iwdd_source_audit.json`

2.5 Further source acquisition on 16 October

2.5.1 Additional smoking photos and HMDB action clips

A public smoking/control photo archive supplied 1,120 candidates. Two related images were rejected by source screening, retaining 1,118: 715 training (357 positive/358 normal), 179 development (89/90) and 224 official photo holdout (112/112). New training examples were screened against higher-priority and existing holdouts before acceptance. This grows the strategy training pool from 3,258 to 3,973 examples and development from 1,633 to 1,812.

The author-hosted HMDB51 archive was acquired, and 906 clips were extracted for smoking and six confusable actions: drink, eat, chew, talk, sit and stand. The subset includes 109 smoking and 797 other-action clips. It is not the official 51-class HMDB benchmark. Primary action labels can omit incidental smoking. A first indexed-seek preparation appeared to fail on clips; sequential decoding successfully decoded all 906. The failed preparation is retained diagnostically and is not used for selecting models.

In the strategy study the complete 906-clip subset is an unseen-source transfer test with eight views per whole clip. The later video pilot assigns title/recording groups to adaptation folds and uses 16 views. Ten known film-part aliases are then purged from training; the final pilot has 530 training, 161 development and 205 test clips. Its test was already inspected in the transfer study and is consequently retrospective.

2.5.2 Thermal human trajectories

The published thermal loitering annotations supply 874,936 human observations and 19,737 track identifiers across 31 dates. Files are checked against source Git blob identities at a recorded revision.

Tracks with conflicting labels or fewer than eight observations are excluded, leaving 17,878 eligible trajectories. Dates split chronologically into 19 train, six development and six test dates.

This supplies duration and movement evidence missing from person-only images. It still represents one thermal camera and complete provided tracks. It is a separate trajectory-component benchmark, not a test of RGB CLIP, automatic detection, identity recovery or online loitering onset.

2.5.3 MEVA abandonment actions and further public-sleep photos

MEVA annotations and 13 relevant source videos were obtained; one additional annotated video was unavailable in the public prefix. Fifteen abandonment-action windows and twelve matched pre-event controls are grouped across ten synchronized sessions and seven cameras. The windows are anchored using annotation-end knowledge. That is appropriate for the stated offline activity-transfer diagnostic, but it does not measure finding events in a continuous stream or certify a persistent unattended item.

A focused public-sleep search downloaded 40 further human-context photographs. Provisional review retained 17 positives and 12 controls, omitting 11 ambiguous candidates. Excluding previously used photographers or related images leaves 28 new-author transfer photos: 17 positive, 11 normal and 25 author groups. These do not enter training in the strategy round. Explicitly linked public author copies and their license metadata are recorded.

Additional littering collections were investigated. Some required an access form or did not provide an immediately accessible matching event release. No communication to dataset owners was sent as part of the recorded work. Garbage-object photographs were not substituted for disposal-action annotations simply to increase the count.

2.6 AVSS author annotations: a new source at first evaluation

The author AOD survey bundle supplies three AVSS2007 videos and temporal unattended-object annotations. The archive and extracted data have recorded hashes; downloaded source code was not executed. Actual encoded FPS determines time conversion rather than replacing it with nominal annotation FPS. The MEDIUM XML's unused schema descriptor is malformed; preparation reads the unchanged data block rather than silently rewriting event spans.

EASY and MEDIUM event ends fall inside their decoded streams and form the two-event primary subset before scoring. HARD's annotated tail extends beyond its 5,311 decoded frames, and its actual FPS is 25 rather than the nominal 30. Its onset is retained without rescaling; only the end is clipped to the observed prefix, and the sequence is reported as right-censored separately. This is a data-quality exclusion specified before model scores, not a favorable subset chosen afterward.

Physical camera identities are not verified; all three clips are conservatively one source group. A post-selection contact-sheet review indicates the same apparent railway-platform viewpoint. AVSS was new when first scored in the 16 October pilot. By the 17 October policy ablations it had already been inspected and no longer supports an untouched-test claim.

Table 2.3: AVSS source timing and censoring before pilot scoring.

Source	Encoded FPS	Decoded frames	Original event frames	Censored?
AVSSS07_EASY	29.97003	5474	[[2870, 4727]]	no
AVSSS07_HARD	25.00000	5311	[[2958, 5670]]	yes
AVSSS07_MEDIUM	29.97003	4834	[[2310, 4593]]	no

2.7 17 October author release: reject source inflation

The additional author archive is pinned to a Git revision and SHA-256 and contains three bundled videos plus three entries explicitly pointing to ABODA footage already represented in the project. Those aliases do not count as independent new videos. The bundled sequences are short highway-debris scenes, approximately 7.6–14.6 seconds long. Their VoTT annotations provide small object boxes and timestamps, without owner histories, full normal intervals or 30-second stationary evidence.

Ten annotated frames per bundled video are decoded using actual FPS, checked against annotated image dimensions, and paired with unchanged reference boxes. Beyond-stream annotation timestamps are exposed and omitted rather than rescaled. The resulting 30-frame sample is a small-object stress test, not three newly verified unattended-personal-item events. A 136-frame CAVIAR training-source item sample provides a compatible second component reference. Together they produce 166 frames for fixed full-image/tiling comparison.

Finding. The data became more extensive in domains and raw volume. Independent labelled events, compatible negative controls and physical camera diversity remain much less extensive than the number of files or adjacent windows suggests.

Chapter 3

Common evaluation and generalization methodology

3.1 Define the unit of evaluation

A single image, a four-image contact sheet, four native chronological views, eight whole-clip views and a continuous 2 Hz stream answer different questions. Within a stated protocol, competing models receive the same planned examples, order, definitions and split identities. Across protocols, a change in AP or recall cannot automatically be attributed to the model: temporal coverage, image detail, training exposure or label meaning may also change.

The report keeps clip classification separate from event discovery. A whole-clip dumping classifier may use frames after the action begins; it does not establish online onset detection. A supplied complete trajectory may use its entire path; it cannot be advertised as a causal loitering alert. The later GRU windows end at the actual decision timestamp, and their event alarms use only prior evidence. The CCTV smoking test remains static where no original continuous video is available.

3.2 Grouping and leakage controls

Source recordings, cameras, synchronized views, author groups and known near-duplicates remain in one fold. Hash validation checks exact input identity; conservative perceptual clustering checks related exports. Official partitions are retained when available, and related training sources are excluded instead of moving official test videos into training. The split search uses annotation counts to ensure both classes exist and is not driven by model predictions.

These safeguards have limits. Supplied title prefixes do not verify all film or actor aliases. Perceptual hashes do not prove camera independence. A camera that contributes only a different day is not a new physical camera. Reusing a previously inspected holdout makes subsequent experiments retrospective even if its labels are not directly loaded into optimization. The correct next confirmation is a new frozen source block rather than another random frame split.

3.3 Shared image protocols and response parsing

Original four-model evaluation letterboxes the image to a common square with maximum side 768 pixels before model-native preprocessing. CLIP models use their pinned processors; image embeddings and text scores are model-specific. Generative VLM generation is deterministic with a 64-token budget. MiniCPM retains a one-slice baseline setting. This is a reproducible com-

mon baseline rather than a complete search of each model's best native high-resolution or video capability.

The classification prompt asks for one label from the defined set, permits unknown when evidence such as ownership or duration is insufficient, and requests a JSON object. Parsing accepts case-insensitive exact labels or the documented structured form. It does not search arbitrary explanatory prose for a convenient class word. An explicit unknown and an invalid response remain distinct reasons for abstention. Categorical quality denominators retain failures/abstentions; otherwise a model could appear stronger merely by declining difficult cases.

3.4 Classification metrics

For a fixed positive target, let TP, FP, FN, TN denote confusion counts. The reported operating-point metrics are

$$\text{precision} = \frac{TP}{TP + FP}, \qquad \text{recall} = \frac{TP}{TP + FN}, \tag{3.1}$$

$$F_1 = \frac{2TP}{2TP + FP + FN}, \qquad \text{FPR} = \frac{FP}{FP + TN}. \tag{3.2}$$

Implementation handles empty denominators explicitly rather than producing undefined favorable scores. Coverage is reported separately. An always-normal model can have high ordinary accuracy in a rare-event corpus and zero event recall. An always-positive model can have apparently high F1 when positives dominate. Both constant baselines are therefore preserved.

Average precision summarizes the precision–recall ranking from genuine continuous scores, with grouped score ties handled consistently. AUROC assesses positive-versus-negative rank discrimination. AP depends on prevalence and cohort composition, so a .96 AP on photographs cannot be compared directly with .72 AP on surveillance frames as if only the model changed. Hard VLM labels are not assigned fabricated probability rankings. The later SmolVLM adaptation instead records genuine first-answer-token logit margins under a matched binary protocol; its AP is valid for that protocol and is not inferred from old categorical outputs. Positive-only weak cohorts report target-prediction rates, not AP, precision, FPR or verified recall.

3.5 Development-only operating points

Three classification operating points recur: a fixed raw decision (zero for binary logits), a maximum-development-F1 cutoff, and a cutoff constrained to no more than 5% empirical development FPR. The classification selector enumerates observed score cutoffs, includes a strict above-maximum option and breaks ties consistently. Test labels never refit the threshold. In the first small unattended development set, 16 normal endpoints mean one false positive changes FPR by 6.25 percentage points; a 5% budget therefore permits zero observed false positives but does not establish a population guarantee.

The 17 October smoking selector is stronger at the development-source level: one global cutoff must satisfy the empirical budget in every source. If n_s normal examples exist in source s and $k_s = \lfloor 0.05 n_s \rfloor$, the cutoff must strictly exceed that source's first disallowed descending normal score, with ties treated conservatively. The final cutoff is the maximum across sources. This constrains development data and can still fail after a camera/domain shift.

3.6 Model selection and multiple comparisons

The first fine-tuning study averages AP within each task's sources, then across tasks; per-target routing among zero-shot, probe and fine-tuned scores is also development-selected. Seed 42 is primary in advance; 43/44 measure optimization variation. The strategy screen selects mean development-source AP within each task and preserves a secondary worst-source analysis with its timing qualification. The video pilot selects worst-source AP, then mean AP, retaining all three seeds. Latest smoking selection prioritizes worst-source recall under its source-wise FPR cutoff, then mean recall and AP ties.

These rules prevent selecting the best test score, but they do not remove all selection optimism. Trying 330 configurations against a small familiar development set can overfit that set. Seed variation measures training noise under fixed data, not variation across deployment cameras. A strong final claim requires a previously untouched source test with enough event/normal exposure.

3.7 Event metrics and unknown time

An event detector emits alarms at real decision times after persistence, rather than backdating them to the first high score. A labelled positive interval can match one alarm whose start falls inside it. Additional alarms inside it are duplicates. Alarms inside reviewed normal intervals are normal false alerts; alarms in unreviewed transition gaps are unknown-time alerts. An alarm emitted before onset and continuing into the positive interval is not automatically converted into a true positive later.

Let N_F be normal false alerts and T_N reviewed normal seconds. The measured rate is

$$\widehat{\lambda}_F = 3600\frac{N_F}{T_N} \quad \text{alerts per normal camera-hour.} \tag{3.3}$$

Unknown time contributes neither normal exposure nor invented negative labels. Delay is measured from the annotated event onset to actual alarm emission. Historical onset labels are provisional; an unknown-time alarm may be a useful early detection rather than a confirmed failure. The raw counts remain visible so that this uncertainty is not hidden in an aggregate metric.

3.8 Uncertainty and numerical reproducibility

The original comparison uses recording-group bootstrap and paired resampling, with 2,000 draws where recorded. Later HMDB AP intervals use supplied title/recording groups and 500 resamples. Adjacent frames do not become independent bootstrap subjects. With too few independently identified cameras, the reports explicitly withhold or qualify intervals rather than attach a precise but misleading frame-wise confidence estimate.

Numerical policy also matters. Fine-tuning uses float32 weights and bfloat16 autocast, but a consistency audit found AMP inference scores sensitive to batch shape. The final evaluation uses float32 and disables TF32 across methods/seeds. Learned weights, hyperparameters and selected routes stay fixed; only thresholds are recomputed from float32 development scores. Original AMP artifacts remain preserved. This correction addresses consistency and is not a new test-driven model search.

Exact identity metadata includes model revisions, source/pixel hashes, input definitions, preprocessing, pooling, code snapshots and raw predictions. Cached frozen features are keyed to those identities. Fine-tuned features cannot silently reuse pretrained-feature caches. Live checkpoint and feature replays test whether the saved artifacts reproduce the reported score, rather than only checking that a JSON file says a run completed.

Chapter 4

Corrected workflow and the first complete GPU comparison

4.1 Make quality testing use the available hardware

The original environment contained CPU-only Torch 2.14.1, while the host's NVIDIA kernel module and user-space NVML library disagreed: kernel 580.159.03 versus NVML 580.178. Consequently `nvidia-smi` failed, even though an RTX 5000 Ada GPU was physically present. The solution was an isolated CUDA environment rather than altering the system driver. Official matching NVIDIA user-space libraries were downloaded and checksum-verified under the project data tree. Torch 2.8.0+cu128 and torchvision 0.23.0 were installed in `.venv-gpu`; real CUDA tensor multiplication and CLIP inference verified the path.

The GPU launcher checks the loaded kernel module version before selecting matching local libraries. The original CPU environment and system installation remain unchanged. Initially only 4,275 MiB of the approximately 32,760 MiB device was free. After the user freed memory, 30,115 MiB became available and the complete unquantized MiniCPM GPU run succeeded. Its observed peak reserved tensor memory was 8,116 MiB. This explains the earlier interruption and why a failed small CPU-era run is not treated as the final quality result.

4.2 Workflow changes implemented before the full comparison

Device configuration was propagated through the image adapters and all VadCLIP stages. Explicit CUDA requests now fail clearly if unavailable; automatic placement records a CPU fallback rather than hiding it. Dtype selection keeps CPU inference float32 and uses supported BF16/FP16 for generative GPU models while preserving CLIP float32 defaults. Native CLIP batching and optional isolated model workers retain input order and distinguish batch-amortized timing from one-input latency.

The full accuracy runner removed its default host RAM cap and artificial sleeps, increased the timeout to four hours, and uses available CPU affinity unless explicitly constrained. A hidden per-group preparation cap was removed: all confirmed inputs are retained unless a cap is requested. Reusing an older preparation can still retain its old cap, which is why counts and preparation identity are recorded. Collector failures are isolated; reviewed protocol names and whitespace are validated; weight/cache paths are configurable; and the official VadCLIP feature encoder checksum is verified.

Reports were expanded to include confusion counts, precision, recall, F1, FPR, AP, AUROC, response coverage, calibrated operating points and qualified resource measurements. Assembly

of separate jobs checks ordered sample identities, pixel hashes, definitions, package code hashes, metric version and inference settings before combining them. It rejects duplicate model names and recomputes metrics rather than merely concatenating report text.

4.3 The completed 13 October accuracy cohort

All four image models process 3,350 confirmed protocol inputs each, totaling 13,400 predictions: 3,278 smoking development frames and 72 unattended frame/window inputs. The smoking set has 1,422 positives and 1,856 normals across 16 already inspected camera/day groups. The unattended frame and six-second-window protocols each have 20 development endpoints (4 positive/16 normal) and 16 test endpoints (4/12), with one positive source sequence per split.

The CLIP batch size is 32; VLMs use one input per call. Generation is deterministic with a 64-token budget; CLIP uses float32 and the unquantized VLMs use bfloat16. There is one quality prediction per selected input and no pacing. All runtime attempts succeed. SmolVLM returns 14 invalid smoking responses, which remain in the quality denominator.

Table 4.1: Initial complete GPU smoking development results: 3,278 frames, previously inspected sources.

Model	PPV	Recall	F1	FPR	AP	AUROC	TP/FP
MobileCLIP	66.7%	0.6%	0.011	0.2%	0.424	0.478	8/4
SigLIP	0.0%	0.0%	0.000	0.0%	0.443	0.512	0/0
SmolVLM	40.3%	5.4%	0.095	6.1%	–	–	77/114
MiniCPM	0.0%	0.0%	0.000	0.0%	–	–	0/0

Finding. At the recorded raw decision, MobileCLIP finds eight smoking positives, SmolVLM 77, and SigLIP/MiniCPM none. These are development results on previously inspected sessions, not new-camera performance.

The positive prevalence is approximately 0.434; a constant-score ranking has AP at that reference level. CLIP AP values close to it and paired confidence intervals crossing zero difference do not establish a reliable winner. An always-normal answer reaches approximately 56.6% accuracy while missing every event. The development diagnosis therefore focuses on recall and false positives rather than ordinary accuracy.

Table 4.2: Initial unattended raw test results: each protocol has four positives and twelve normals.

Model	Input	PPV	Recall	F1	FPR	AP
MobileCLIP	frame	0.0%	0.0%	0.000	0.0%	0.233
MobileCLIP	window	25.0%	25.0%	0.250	25.0%	0.281
SigLIP	frame	0.0%	0.0%	0.000	0.0%	0.189
SigLIP	window	0.0%	0.0%	0.000	25.0%	0.196
SmolVLM	frame	25.0%	100.0%	0.400	100.0%	–
SmolVLM	window	25.0%	100.0%	0.400	100.0%	–
MiniCPM	frame	0.0%	0.0%	0.000	0.0%	–
MiniCPM	window	0.0%	0.0%	0.000	0.0%	–

SmolVLM predicts unattended items for every small-cohort input. Its test F1 0.400 equals the always-positive baseline with FPR 100%. MiniCPM is always normal. A MobileCLIP contact sheet finds one of four positives but falsely flags three normals. Development threshold fitting can increase recall at severe false-positive cost or drive recall to zero under the empirical FPR budget. It does not solve the underlying source/decision problem.

4.4 Prompt and sampling diagnostics

A separate short-description CLIP check uses every smoking development frame on CPU float32 with the same images. It changes the text to ordinary activity versus a person smoking a cigarette. MobileCLIP recall rises to 0.533 and F1 to 0.490, but FPR rises to 0.494 and AP is only 0.460. SigLIP remains always normal. This exposes strong prompt-dependent decision bias; it is not a validated new detector and does not justify choosing prompts from test outcomes.

The recording-balanced CPU comparison has 308 inputs, and the full CLIP CPU run has 3,350. They are diagnostic artifacts for sampling/numerical differences. The uncapped completed GPU study is the principal frozen-model quality comparison. CPU float32 and GPU BF16 VLM decisions need not be identical, so model quality is not inferred from mixing those runs.

4.5 Separate supervised offline VadCLIP smoking study

VadCLIP uses nine training sessions (1,910 frames), four development sessions (312 frames) and three test sessions (1,056 frames, 560 positive/496 normal). It trains a temporal adapter over frozen CLIP-B/16 features. The selected dual-mean branch uses a development threshold of approximately 0.738177, giving 290 TP, 177 FP, 270 FN and 319 TN. The alternative align, visual, dual-max and feature baselines are retained.

Table 4.3: Separate supervised VadCLIP test: 1,056 frames; dual_mean is the development-selected branch.

Branch	PPV	Recall	F1	FPR	AP	AUROC
visual	53.0%	100.0%	0.693	100.0%	0.530	0.500
align	62.1%	51.8%	0.565	35.7%	0.633	0.587
dual_mean	62.1%	51.8%	0.565	35.7%	0.633	0.587
dual_max	53.0%	100.0%	0.693	100.0%	0.530	0.500

The visual and dual-max branches collapse to all-positive decisions, yielding an apparently higher F1 around .693 while AUROC is .5. Dual-mean has some ranking discrimination but 35.7% FPR at 51.8% recall is not a usable alert point. A development-only learning-rate sweep over $10^{-5}, 10^{-4}, 10^{-3}$ selects 10^{-3} and epoch one, with test AP approximately .632, close to the main GPU result. Only the chosen sweep candidate is evaluated on its test split. The small sweep does not exhaust architecture or training choices.

This is supervised, offline and split differently from the zero-shot image comparison. It is not the original weakly supervised benchmark protocol and does not establish five-target performance. Context within a sparse whole session may include future information relative to a frame; it is not a causal streaming detector.

4.6 Historical CPU latency pilot: recorded, not rerun

For completeness, a separate historical CPU replay used the Threadripper PRO 7975WX, four fixed logical CPUs, four Torch threads, float32, one worker and batch size one. All models saw the same eight development images, with three warmups and three measured repeats, giving 24 observations per model. Decode/preprocess/parse are included; model load/warmup are excluded from per-input latency. Peak RSS includes loading and warmup.

Table 4.4: Historical CPU pilot only: eight images, three measured repeats, four logical CPUs; not rerun.

Model	p50 (s)	p95 (s)	Peak RSS (GiB)	Load (s)
MobileCLIP	0.055	0.056	0.97	2.89
SigLIP	0.109	0.138	1.22	2.43
SmolVLM	7.505	7.563	3.27	2.41
MiniCPM	8.091	8.176	18.41	4.15

This is a small replay on the recorded host, not emulation of another CPU or a modern deployment estimate. It does not include camera queueing, longer answers or larger native inputs. After the user removed latency testing from the objective, later work did not repeat it. Event onset delay reported later is a detection-quality metric, not a replacement latency benchmark.

Evidence: `results/gpu_all_models_20261013/report.md`

Evidence: `reports/hardware_20261013.json`

4.7 Validation at this stage

The completed workflow review records 144 tests passing in both CPU and isolated GPU environments, including device placement, token dtypes, native batching, AP ties, uncapped preparation, rename-receipt invalidation, source/pixel leakage and incompatible-report rejection. All 13,400 planned predictions are checked and metrics recomputed. This supports execution and artifact integrity; it cannot compensate for the small number of event sources or missing task labels.

Chapter 5

Expanded four-model accuracy evaluation

5.1 Why this expansion was necessary

The initial unattended test had only four positive endpoints and a single positive sequence per split. More source footage was already available, but required interval review, grouping and duplicate handling. The expansion therefore retained eligible inputs after review rather than imposing an arbitrary fixed sample cap. It also attempted the other three requested targets with explicitly weaker diagnostics, avoiding an unsupported claim that all source labels were already standardized anomaly events.

Finalized on 14 October, the study evaluates 13,789 input/protocol instances per model and 55,156 predictions total across seven cohorts. The frame/window unattended protocols share the same 1,770 endpoints and have 20-second past context for sheets. Littering weak candidates contribute 6,667 development-only images, person/loitering 1,675, rough-sleeping weak candidates 111, and the synthetic oracle posture study 121 person crops across source-held-out development/test.

5.2 GPU execution policy

The four models run in isolated GPU workers, with eight CPU threads per worker. CLIP uses batch 64 and float32; VLMs use one input per call and unquantized automatic bfloat16. Shared 768-pixel letterbox preprocessing, deterministic generation, the 64-token budget and MiniCPM's one-slice baseline are retained. No default accuracy RAM cap, camera-rate sleep or latency repetitions are introduced. Model revisions, source hashes and raw responses remain available.

All 40 model/cohort/split summaries complete successfully. There are zero runtime errors. Across all new input/protocol calls, SmolVLM has 1,457 abstentions (approximately 10.6%) and MiniCPM 45 (approximately .33%). A valid unknown/invalid answer is not silently dropped when computing categorical quality.

5.3 Expanded unattended endpoint results

Table 5.1: Expanded unattended raw test decisions: 962 endpoints per input protocol, provisional labels.

Model	Input	PPV	Recall	F1	FPR	AP	AUROC
MobileCLIP	frame	25.3%	75.2%	0.379	74.6%	0.209	0.417
SigLIP	frame	0.0%	0.0%	0.000	2.6%	0.309	0.675
SmolVLM	frame	18.6%	63.6%	0.288	93.8%	–	–
MiniCPM	frame	77.1%	34.7%	0.479	3.5%	–	–
MobileCLIP	window	27.2%	76.0%	0.400	68.5%	0.362	0.580
SigLIP	window	12.5%	1.7%	0.029	3.9%	0.325	0.663
SmolVLM	window	24.9%	96.3%	0.396	97.4%	–	–
MiniCPM	window	96.3%	10.7%	0.193	0.1%	–	–

MiniCPM has the strongest raw frame F1, .479, with 77.1% precision, 34.7% recall and 3.47% FPR: 84 TP, 25 FP, 158 FN and 695 TN. Its earlier constant-normal behavior does not persist on the broader cohort. It still misses nearly two thirds of positive endpoints. MobileCLIP and SmolVLM flag many positive frames but also 74.6% and 93.8% of normal frames. SigLIP's raw frame decision finds no positives, although its continuous ranking has AP .309 and AUROC .675 against positive prevalence .252.

Contact sheets change the tradeoff. MiniCPM window recall falls to 10.74%, FPR to .14% and F1 to .193. SmolVLM window recall is 96.28% with 97.36% FPR and 14 unanswered inputs. This does not show that the architectures cannot model video; it shows that this particular reduced contact-sheet protocol is insufficient and entangles temporal context with spatial resolution.

5.4 Frozen development cutoffs and interval alarm screen

The development 5% FPR-budget cutoff gives MobileCLIP 20.3% frame-test FPR and 35.1% sheet-test FPR. SigLIP sheets obtain 7.0% recall at 5.14% FPR under the budget point, or 55.4% recall at 30.0% FPR with the max-dev-F1 point. None of these cutoffs is fitted to test labels. The transfer failure is part of the result rather than a reason to recalibrate on the test set.

An auxiliary interval screen counts whether any raw categorical alarm occurs inside each reviewed positive or normal interval. There are five positive and seven normal test intervals. It has no persistence rule and is sensitive to interval length; it is not event onset detection or alerts/hour. In the frame screen, MobileCLIP and SmolVLM hit 5/5 positive intervals but alarm in 7/7 normal intervals. MiniCPM hits 3/5 and alarms in 3/7 normals. The later causal pilot replaces this loose screen with explicit alarm-time accounting.

5.5 What the three weak target cohorts measured

Table 5.2: Weak/coarse labels: target-prediction rates, not verified event recall or precision.

Diagnostic	Model	Inputs	Target rate	Abstain	Errors
littering weak	MobileCLIP	6667	0.8%	0	0
littering weak	SigLIP	6667	26.7%	0	0
littering weak	SmolVLM	6667	74.9%	924	0
littering weak	MiniCPM	6667	55.8%	0	0
loitering weak	MobileCLIP	1675	10.6%	0	0
loitering weak	SigLIP	1675	84.5%	0	0
loitering weak	SmolVLM	1675	0.1%	319	0

Continued on next page

Table 5.2 continued

Diagnostic	Model	Inputs	Target rate	Abstain	Errors
loitering weak	MiniCPM	1675	4.6%	1	0
person presence	MobileCLIP	1675	28.5%	0	0
person presence	SigLIP	1675	17.4%	0	0
person presence	SmolVLM	1675	92.7%	120	0
person presence	MiniCPM	1675	96.1%	8	0
rough sleeping weak	MobileCLIP	111	9.9%	0	0
rough sleeping weak	SigLIP	111	95.5%	0	0
rough sleeping weak	SmolVLM	111	81.1%	13	0
rough sleeping weak	MiniCPM	111	66.7%	24	0

These rates describe model behavior on candidates, not verified event recall. The littering actions are not uniformly rubbish disposal; the person images lack duration; synthetic lying lacks established public-sleep context. No precision, FPR, AP or F1 is inferred from an all-positive proxy. The entire weak collection remains development-only because source identities or task labels are insufficient for a strong held-out claim.

The person-presence check is useful for localizing the problem. On the same 1,675 person-labelled images, SmolVLM/MiniCPM identify people at 92.7%/96.1% but predict loitering at only .12%/4.60%. That is not proof of loitering failure: a visible person is compatible with normal passage and missing duration. Conversely, high SigLIP loitering or rough-sleeping candidate rates cannot establish event accuracy.

5.6 Oracle lying-versus-standing posture diagnostic

Source boxes crop 111 lying/fallen people and ten standing people, keeping source images together. The test contains 44 lying and four standing crops. These boxes are supplied free to the diagnostic; they are not automatic detections and cannot be assumed available during deployment.

Table 5.3: Oracle posture test: 44 lying and four standing source-box crops; not rough-sleeping events.

Model	PPV	Recall	F1	FPR	AP	AUROC
MobileCLIP	63.6%	15.9%	0.255	100.0%	0.792	0.028
SigLIP	100.0%	100.0%	1.000	0.0%	1.000	1.000
SmolVLM	100.0%	59.1%	0.743	0.0%	–	–
MiniCPM	100.0%	90.9%	0.952	0.0%	–	–

SigLIP correctly separates all 48 test crops. MiniCPM finds 40/44 lying crops and all four standing controls, F1 .952. SmolVLM finds 26/44 and abstains on the remaining 18, with all four standing controls correct. MobileCLIP's FPR is 100% in this tiny control sample. These results identify a potentially useful posture component and crop sensitivity, not persistent rough-sleeping event accuracy. Four negative crops cannot provide a stable false-positive estimate.

5.7 What this study changed in the next plan

The larger cohort demonstrated that model behavior and ranking were not captured by the first small sequence test. It also showed that more endpoints alone did not stabilize thresholds or provide duration/ownership reasoning. The next step became source-qualified supervised learning on actual action/track labels, with a clean distinction between image posture, full-clip action, source-camera transfer and event detection.

Evidence: `results/expanded_accuracy_20261013/quality_summary.json`

Evidence: `results/expanded_accuracy_20261013/verification.json`

Chapter 6

Actual supervised CLIP visual fine-tuning

6.1 Learning objective and representations

The 15–16 October study is the first stage in this progression that actually updates the two requested CLIP visual backbones. Its frozen manifest has 7,498 target-specific examples and 12,478 unique view images: 3,258 train, 1,633 development, 1,801 test and 806 additional official-test photographs. Every example has one known binary target; the other four objectives are unknown, not presumed negative.

Let f_t be the normalized embedding for chronological view t. The probe and fine-tuned representation is

$$z = \operatorname{norm}\left[f_T;\ \frac{1}{T}\sum_t f_t;\ \max_t f_t;\ f_T - f_1\right]. \tag{6.1}$$

For a static image the change term is zero. Five binary heads allow simultaneous objective scores rather than forcing mutually exclusive event labels. The masked binary loss gathers only the head corresponding to the example's annotated target. Thus a smoking-negative example supplies no gradient asserting absence of loitering, sleep or unattended items.

The pretrained baseline is a last-frame text score. Improvements over it can combine supervised classification and temporal context. The frozen linear probe versus visual fine-tune comparison is more controlled because both use the same ordered-view representation. Unknown-label masking and temporal ordering have dedicated tests.

6.2 Probe, visual parameters and regularization

The frozen probe uses L2-regularized logistic classification with $C \in \{1, 10, 100\}$. The development-selected probe initializes the supervised heads. Fine-tuning then unfreezes the last MobileCLIP visual stage/final convolution/projection or the last SigLIP transformer block/pooling head, while freezing the text tower, early vision layers and batch-normalization statistics.

MobileCLIP updates 18,106,992 visual parameters; SigLIP updates 14,176,512, plus their binary heads. These are real changed visual tensors, verified by parameter-delta checks; this stage is not merely fitting a linear classifier to fixed embeddings. Horizontal flips at probability .5 and mild brightness/contrast .15 and saturation .1 preserve complete scene content. Random crops that might remove a cigarette or item are avoided.

Sampling balances task, class, dataset source and recording/author group. With feature teacher

z_0 and pretrained visual weights θ_0, the implemented loss is

$$\mathcal{L} = \mathcal{L}_{\text{BCE,known target}} + 0.1\,\text{MSE}(z, z_0) + 0.1 \sum_{k \in \mathcal{U}} \text{mean}(\theta_k - \theta_{0,k})^2, \tag{6.2}$$

where $\mathcal{U}$ is the unfrozen visual parameter set. The classifier sees dropout .1. Distillation and weight anchoring constrain drift from pretrained features; they are attempted generalization safeguards, not proof of robustness.

AdamW uses visual learning rates 10^{-5} and $3 \cdot 10^{-5}$ for primary seed 42, head learning rate $3 \cdot 10^{-4}$ and weight decay .01. The budget is ten epochs maximum with three stale development epochs, and gradient norm clipping at one. GPU training is unpaced with float32 weights and BF16 autocast, batch 32 target examples, six decoding workers and eight CPU threads per job.

6.3 Selection, replications and execution details

Checkpoints and visual learning rate maximize development macro task AP, averaging source AP within each task. Per-target routing chooses among pretrained, probe and fine-tuned scores using development AP, with simpler methods winning ties. Seed 42 is predesignated primary. Seeds 43/44 reuse its development-selected visual learning rate and report variability; their best test column cannot replace the primary.

Both primary models select visual learning rate $3 \cdot 10^{-5}$. MobileCLIP selects epoch three and SigLIP epoch two. MobileCLIP primary routes use its fine-tuned score for all five targets. SigLIP routes smoking/unattended to fine-tuning, dumping/public sleep to probes, and loitering to its pretrained score. This is why the selected five-target result is not identical to whichever fine-tuned AP is largest in a test table.

All six model/seed jobs complete. The scheduler eventually allows up to three concurrent GPU jobs. Initial jobs use convolution autotuning; later jobs disable it to avoid repeated searches under variable frame-batch shapes. Executed snapshots and commands preserve those differences without changing labels, splits or the selection plan.

6.4 Numerical audit and final evaluation

An AMP inference consistency audit finds batch-shape-sensitive scores. The final quality pass therefore scores all methods and seeds in float32 with TF32 disabled. Trained weights, hyperparameters and development-selected routes stay fixed. Only operating thresholds are recalibrated on float32 development predictions. The original AMP outputs remain beside each `float32_evaluation/` directory.

This distinction matters for exact threshold boundaries. A threshold fitted from one numerical representation should not be applied to a slightly different serialized model without checking it. The later strategy screen likewise refits cutoffs from serialized float32 classifier development scores while retaining fitted weights and route selection.

6.5 Held-out ranking gains and regressions

Table 6.1: Primary seed 42 AP and development routing; pretrained baseline is last-frame text scoring.

Model	Target	N	Zero	Probe	Tuned	Dev route	Route AP
MobileCLIP	Smoking	646	0.642	0.594	0.564	finetuned	0.564
MobileCLIP	Dumping / littering	100	0.424	0.861	0.858	finetuned	0.858
MobileCLIP	Loitering	75	0.320	0.113	0.093	finetuned	0.093
MobileCLIP	Public sleep	18	0.502	0.976	0.976	finetuned	0.976
MobileCLIP	Unattended	962	0.209	0.361	0.652	finetuned	0.652
SigLIP	Smoking	646	0.669	0.670	0.650	finetuned	0.650
SigLIP	Dumping / littering	100	0.499	0.885	0.864	linear probe	0.885
SigLIP	Loitering	75	0.097	0.107	0.110	zero shot	0.097
SigLIP	Public sleep	18	0.733	0.915	0.976	linear probe	0.915
SigLIP	Unattended	962	0.309	0.855	0.896	finetuned	0.896

Waste dumping improves strongly for both encoders relative to the same-study pretrained baseline. Official-test smoking photographs also improve, MobileCLIP .220 to .901 and SigLIP .499 to .954. However, CCTV smoking ranking regresses under both primary development-selected routes. Loitering remains unresolved: selected AP .093/.097 is below the .147 positive prevalence of the 75-window cohort. The six-positive sleep-photo test yields high point estimates but cannot establish broad coverage. Unattended AP improves substantially under provisional labels and two held-out camera families.

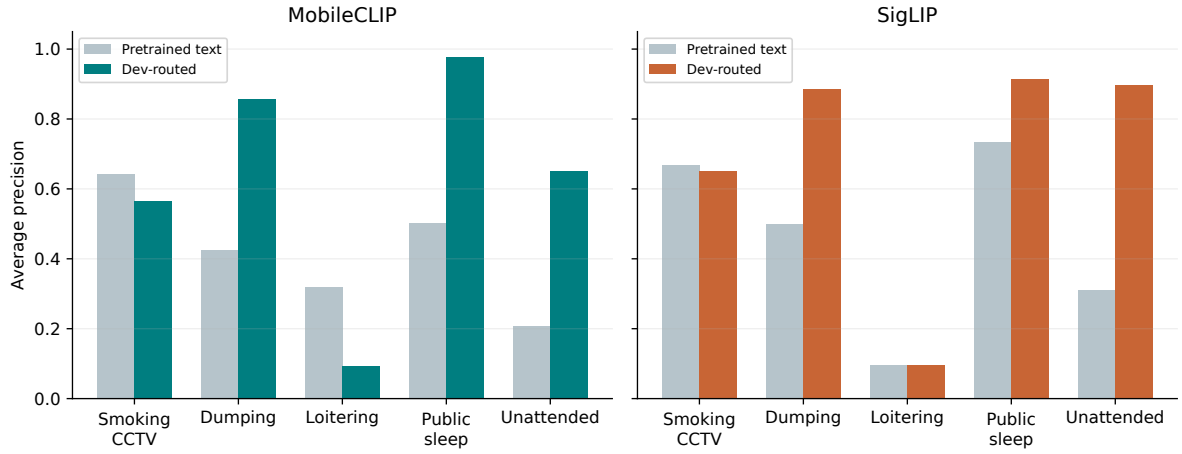


Figure 6.1: Same-study pretrained and development-routed AP for the primary fine-tuning seed. The cohorts differ across targets; bars do not form a cross-task difficulty ranking.

6.6 Why AP did not make the models safe to threshold

Table 6.2: Primary smoking/unattended threshold transfer: fixed weights, development-only cutoffs.

Model	Target	Cutoff rule	Cutoff	PPV	Recall	F1	FPR
MobileCLIP	Smoking	raw	0.000	0.0%	0.0%	0.000	1.2%
MobileCLIP	Smoking	dev max f1	-4.416	60.7%	93.1%	0.735	93.7%
MobileCLIP	Smoking	dev fpr budget	-0.760	50.0%	3.1%	0.058	4.7%
MobileCLIP	Unattended	raw	0.000	100.0%	25.6%	0.408	0.0%
MobileCLIP	Unattended	dev max f1	-5.249	25.6%	100.0%	0.408	97.6%

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Table 6.2 continued

Model	Target	Cutoff rule	Cutoff	PPV	Recall	F1	FPR
MobileCLIP	Unattended	dev fpr budget	-3.311	47.0%	96.3%	0.631	36.5%
SigLIP	Smoking	raw	0.000	68.3%	10.9%	0.189	7.9%
SigLIP	Smoking	dev max f1	-1.445	60.3%	91.3%	0.727	93.3%
SigLIP	Smoking	dev fpr budget	0.778	100.0%	0.3%	0.005	0.0%
SigLIP	Unattended	raw	0.000	96.7%	61.2%	0.749	0.7%
SigLIP	Unattended	dev max f1	-5.255	25.2%	100.0%	0.403	99.6%
SigLIP	Unattended	dev fpr budget	-4.791	25.3%	100.0%	0.403	99.4%

SigLIP unattended zero-logit decisions give F1 .749, recall 61.2%, precision 96.7% and FPR .7% on this holdout. Yet its maximum-development-F1 cutoff gives 99.6% test FPR, and the development 5% budget point gives 99.4%. MobileCLIP's corresponding unattended thresholds also transfer poorly. Both primary smoking max-dev-F1 cutoffs exceed 93% CCTV FPR. These are retained as failures of operating-point transfer rather than recalibrated from test negatives.

The all-seed appendix reports selected route AP and operating points for every task. Optimization variation can be material, but three seeds still evaluate the same cameras and provisional labels. Selecting a fortunate seed from its test score would overstate the evidence.

6.7 Checkpoint and interface validation

The final audit verifies finite changed visual tensors in all six checkpoints, recomputes 46,926 held-out and 29,394 development predictions, checks source/pixel identities and selections, and runs real saved-checkpoint inference on both encoders. Live-versus-saved differences are below one millionth in the recorded primary checks. At completion, 153 tests pass.

The exported interface classifies images or ordered clips; it is not a box detector. It requires chronological timestamps for duration objectives and returns unknown when sufficient history is absent. Scores are logits, not calibrated event probabilities. The default max-dev-F1 option is a faithful recorded protocol, not a validated deployment recommendation in light of the observed threshold failures.

Evidence: `reports/clip_training_protocol_20261015.md`

Evidence: `results/clip_finetuning_20261015/verification.json`

Chapter 7

Expanded sources and the 330-configuration strategy study

7.1 Motivation and controlled strategy families

The first supervised study established that useful learning was possible, but global-scene features and development thresholds did not transfer consistently. The next round therefore expanded source data and compared representations, frozen classifiers and controlled blends of the earlier fine-tuned visual weights. It did not launch 330 new CNN/transformer fine-tunes.

For each target and each backbone there are 33 configurations: one endpoint zero-shot reference, ten representation recipes each with linear $C \in \{.1, 1, 10\}$, and two MLP representations. Across five targets and two models this gives 330 configurations, including 320 newly fitted classifiers and ten zero-shot references. Primary seed 42 makes it an exploratory screen rather than a repeat of the three-seed visual study.

Full-scene pooled features use endpoint, mean, max and endpoint-minus-first embeddings. Ordered features concatenate four evenly indexed view embeddings plus mean and max. Automatic contextual region features are added to full-scene context. The fixed COCO Faster R-CNN ResNet50 FPN v2 detector uses confidence .5 and contextual padding; its selected person/item crop falls back to the scene when detections are absent. Ground-truth boxes never choose inference crops.

The sparse geometry representation records person/item counts and presence, minimum person-item distance, and nearest-view person-overlap persistence. It is not an identity tracker and does not know ownership. It supplies a controlled geometry hypothesis that may help timed tasks, not a full MOT implementation.

7.2 Visual weight interpolation and classifier fitting

For the visual parameters updated in the earlier study, interpolation forms

$$\theta_\alpha = (1-\alpha)\theta_{\text{pretrained}} + \alpha\theta_{\text{adapted}}, \qquad \alpha \in \{0, .25, .5, 1\}. \tag{7.1}$$

Each resulting full-scene representation receives a newly fitted classifier using the current training split. The older adapted weights used only their recorded training data. An interpolation coefficient is a development candidate, not an assumed robustness improvement.

The MLP uses one 64-unit ReLU hidden layer, dropout .3 and a binary output, AdamW learning rate .001/weight decay .01, batch 128, up to 80 epochs and patience eight. Task/source/class weights are retained. Full-scene pooled and full-plus-crop pooled representations supply the two

MLP recipes. Linear classifiers use L-BFGS with maximum 400 iterations; serialized coefficients are float32. All final visual extraction uses float32 with TF32 disabled.

7.3 Selection timing and source protection

Primary per-target routes maximize mean development-source AP. They freeze before the 224-photo holdout is scored. A secondary worst-development-source analysis is specified after those photo results are available but before HMDB/MEVA transfer predictions. Its photo result is therefore exploratory and does not replace the primary route. Transfer-family hyperparameters are fixed from saved development results.

Related new training images are screened against old holdouts. Inaccessible files, rejected related examples and source labels are recorded. No unlabeled objective is turned into normal and no unrelated anomaly type is used merely to fill a target class. The full candidate-development table is reproduced in the appendix, so the report shows how a route was selected rather than only its final test AP.

Table 7.1: Primary strategy routes and development scores; alpha suffixes denote visual interpolation.

Model	Target	Representation	Head	C	Mean AP	Worst AP
MobileCLIP	Smoking	full crop pool a0	linear	10	0.800	0.510
MobileCLIP	Dumping / littering	full pool a0.5	linear	0.1	0.923	0.923
MobileCLIP	Loitering	full geometry a0	linear	0.1	0.486	0.486
MobileCLIP	Public sleep	full pool a0.25	linear	0.1	1.000	1.000
MobileCLIP	Unattended	full ordered a1	linear	10	0.947	0.947
SigLIP	Smoking	full crop pool a0	mlp	–	0.821	0.505
SigLIP	Dumping / littering	full geometry a0	linear	10	0.926	0.926
SigLIP	Loitering	full geometry a0	linear	0.1	0.522	0.522
SigLIP	Public sleep	full pool a0	linear	0.1	1.000	1.000
SigLIP	Unattended	full ordered a1	linear	10	0.968	0.968

7.4 Historical source results

Table 7.2: Previously inspected source holdouts: previous bundle versus primary selected strategy; max-dev-F1 decisions.

Model	Source/cohort	N	+	Old AP	New AP	F1	Recall	FPR
Mobile CLIP	Smoking: historical CCTV c1	646	393	0.564	0.690	0.727	85.0%	75.9%
Mobile CLIP	Smoking: historical Stanford photos	806	140	0.901	0.880	0.709	95.7%	15.6%
Mobile CLIP	Littering/waste dumping: historical IWDD	100	50	0.858	0.806	0.830	88.0%	24.0%
Mobile CLIP	Loitering: historical Street Scene	75	11	0.093	0.489	0.000	0.0%	0.0%
Mobile CLIP	Public sleep: historical photos	18	6	0.976	0.931	0.727	66.7%	8.3%
Mobile CLIP	Unattended: historical camera holdout	962	242	0.652	0.980	0.403	100.0%	99.7%
SigLIP	Smoking: historical CCTV c1	646	393	0.650	0.722	0.757	94.1%	84.6%

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Table 7.2 continued

Model	Source/cohort	N	+	Old AP	New AP	F1	Recall	FPR
SigLIP	Smoking: historical Stanford photos	806	140	0.954	0.937	0.776	94.3%	10.2%
SigLIP	Littering/waste dumping: historical IWDD	100	50	0.885	0.893	0.828	82.0%	16.0%
SigLIP	Loitering: historical Street Scene	75	11	0.097	0.550	0.375	27.3%	3.1%
SigLIP	Public sleep: historical photos	18	6	0.915	0.883	0.667	50.0%	0.0%
SigLIP	Unattended: historical camera holdout	962	242	0.896	0.925	0.402	100.0%	100.0%

Automatic region/context features are selected for smoking, sparse geometry for loitering and ordered views on the adapted encoder for unattended items. Historical AP improves strongly in some targets: loitering rises from .093/.097 to .489/.550 and unattended to .980/.925. MobileCLIP dumping falls from .858 to .806, and both public-sleep photo routes regress. These are already inspected holdouts and remain retrospective diagnostics.

High AP again does not yield a stable global threshold. Historical unattended max-dev-F1 cutoffs produce 99.7% and 100% FPR. Their raw-zero F1 is .087 and .131, worse than the prior SigLIP raw-zero .749. The source ranking has improved while score scale and source offsets make the frozen operating point unsuitable. Historical CCTV smoking max-dev-F1 FPR remains 75.9%/84.6%.

7.5 New photo holdout and source-transfer results

Table 7.3: New-source/image-holdout results at first strategy evaluation; not later HMDB-adapted cohorts.

Model	Source/cohort	N	+	Old AP	New AP	F1	Recall	FPR
Mobile CLIP	Smoking: new photo holdout	224	112	0.953	0.997	0.945	100.0%	11.6%
Mobile CLIP	Smoking: new HMDB action source	906	109	0.605	0.576	0.496	58.7%	10.7%
Mobile CLIP	Public sleep: new author photos	28	17	0.925	0.952	0.640	47.1%	0.0%
Mobile CLIP	Unattended: new MEVA action proxy	27	15	0.569	0.545	0.683	93.3%	100.0%
SigLIP	Smoking: new photo holdout	224	112	0.995	1.000	0.966	100.0%	7.1%
SigLIP	Smoking: new HMDB action source	906	109	0.685	0.664	0.371	84.4%	37.0%
SigLIP	Public sleep: new author photos	28	17	0.943	0.933	0.522	35.3%	0.0%
SigLIP	Unattended: new MEVA action proxy	27	15	0.637	0.621	0.629	73.3%	75.0%

The new 224-photo holdout is nearly perfectly ranked by both encoders, but shares a collection with the added training photos. Both primary strategies regress slightly on the entirely new HMDB movie/web source before any HMDB training: MobileCLIP .605 to .576 and SigLIP .685 to .664. MEVA action-proxy AP also falls, .569 to .545 and .637 to .621. Public-sleep new-author point estimates are mixed: MobileCLIP .925 to .952, SigLIP .943 to .933.

This is evidence against a broad claim that crops or adapted weights alone solved source generalization. The strong photo score and weak video transfer are both retained. The later HMDB-adapted

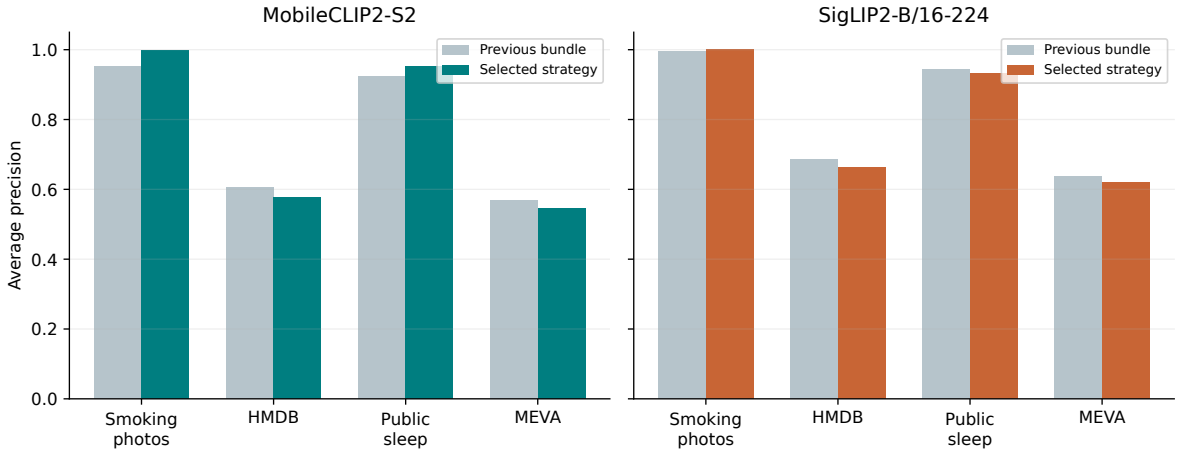


Figure 7.1: Strategy-study transfer AP, comparing the previous bundle with the development-selected strategy. Photos share a training collection; HMDB is movie/web action evidence; MEVA is an abandonment-action proxy; public-sleep labels are provisional.

pilot changes training data and evaluation cohort and must not be treated as a direct continuation of the same 906-clip AP curve.

7.6 Separate thermal trajectory component

Four candidates use the complete provided thermal trajectories: duration-only logistic regression, regularized linear descriptors, a random forest and histogram gradient boosting. The 14 descriptors include log duration/observation count, temporal coverage, path length, net displacement, straightness, spatial extent, spread, speed quantiles, stationary fraction and turning/reversal summaries. Frame-span units are retained rather than inventing an unverified source FPS.

Chronological splitting uses 19/6/6 dates. The development-selected 400-tree depth-eight forest reaches mean development-source AP .863 and test AP .577, compared with duration-only test AP .283. Its max-dev-F1 operating point gives F1 .664 and FPR .187. The date/season transfer drop is substantial despite a single physical camera and provided identities.

Table 7.4: Provided thermal tracks: mean development-date AP and held-out-date decisions.

Candidate	Dev AP	Test AP	Recall	F1	FPR
duration only	0.327	0.283	100.0%	0.355	100.0%
trajectory linear	0.804	0.554	74.6%	0.615	18.7%
trajectory forest	0.863	0.577	83.5%	0.664	18.7%
trajectory boosting	0.848	0.571	85.8%	0.647	21.9%

The serialized estimator and replayed held-out scores are exported separately. These results indicate that movement history contains useful information beyond duration alone; they do not validate an automatic RGB loitering pipeline, online onset detection or unseen-camera transfer.

7.7 Integrity checks and outcome

All 117,612 recorded strategy predictions are replayed, all 330 development selections and 20,980 image identities checked, and real saved-head inference exercised for all five targets on both backbones. The maximum observed live-versus-saved difference is below .000005. At completion, 159 tests pass. The outcome is a better set of representation candidates and a clearer source-transfer

failure diagnosis, not a five-task deployment guarantee.

Evidence: `reports/strategy_protocol_20261016.md`

Evidence: `results/strategy_study_20261016/verification.json`

Chapter 8

Causal video pilot: failure audit, regions and learned heads

8.1 Start from observed failure modes

The strategy results motivated a controlled temporal experiment instead of another large undirected visual fine-tune. An error audit exports 117 deliberately selected severe/source-balanced cases across all five targets. Provisional notes distinguish low spatial detail, misleading crops, confusable actions, unsupported duration and item/owner evidence. This is a review queue for investigation, not an independent or representative statistical estimate of error causes. No reviewed test mistake is silently corrected into a favorable new label.

For CCTV smoking the median largest annotated smoker is only 11.2 pixels tall after 224-pixel letterboxing. In 71.8% of held-out positive frames, all smokers are below 12 pixels. The previous automatic crop loses at least half of every target in 40.2% of positive frames. Multiple person/full-upper-body crops reduce that coverage failure to one of 393 frames (.254%). Detector resolution and crop policy change together, so this is a geometric coverage result rather than an isolated causal accuracy gain.

Table 8.1: Smoking geometric coverage, independent of recognition accuracy; positive frames with target boxes.

Split	+ frames	Median height at 224	All targets <12 px	Prior <half coverage	New <half coverage
train	636	15.1	8.6%	42.3%	0.9%
dev	393	11.2	67.4%	23.4%	0.3%
test	393	11.2	71.8%	40.2%	0.3%

The item audit exposes a separate bottleneck: the COCO detector/association pipeline matches none of 136 sampled CAVIAR leaving-item reference boxes at IoU .5. A classifier cannot recover reliable item state if its automatic upstream item proposals are absent. This training-camera finding motivates a pre-score Grounding DINO geometry amendment with unchanged CLIP visual representations.

8.2 Video preparation and identity repair

The preparation covers 936 actual video files: 906 HMDB clips, 17 historical continuous recordings and 13 MEVA sources. Static smoking/photo examples also receive recording identities for bookkeeping, but those identifiers are not additional decoded surveillance videos. Continuous

sources are decoded to real indices at 2 Hz; HMDB uses 16 sequentially decoded uniform views. Twenty-second windows end at the actual decision timestamp, with no decisions before a 20-second warmup. Unknown intervals use target −1 and never become normal examples.

The initial prepared modeling pool has 8,941 examples. Verification completes/repairs inherited pixel-hash fields in separate canonical manifests without changing labels, views, timestamps or splits. The known-source guard then finds a title-part alias: The Departed Part 1 and Part 2 cross train/development under raw prefixes. Ten training clips are purged, leaving 8,931 modeling examples; no test clip is removed. Earlier fits remain under `prefix_split_diagnostic/`, while all final selections use refitted guarded data.

The combined historical/new input verification covers 25,452 distinct view-image paths and no cross-fold decoded-pixel conflicts. The source guard covers explicit part/pt/clip/scene/episode suffixes; it does not independently verify every movie or actor identity. The new title grouping is therefore an improved safeguard with a stated limit.

8.3 Automatic detection and causal tracking

The common COCO detector is Faster R-CNN ResNet50 FPN v2, with minimum image size 640, maximum 1024, low confidence .2 and new-track threshold .5. Person and backpack/handbag/suitcase labels feed a documented Hungarian IoU/velocity association baseline. It retains tracks across gaps up to two seconds. Association can use IoU at least .05 or small predicted-center distance; velocity is smoothed from past displacements.

Stationarity originally resets when an observed center moves more than .015 normalized frame distance from its anchor. Ownership is a proximity hypothesis: an item must have a near person candidate observed for two seconds before association. Distance uses the item center relative to a person's rectangle normalized by person height. An owner becomes absent when missing or sufficiently far; the established relation is retained after departure. There is no appearance ReID or independent ownership truth. The algorithm is not ByteTrack.

The 18 geometry channels include clipped person/item counts, maximum age, stationary duration, owner-absence duration, owned-item fraction, presence, speed, minimum distance, area, detection confidence and visible-owner fraction. Source boxes do not choose tracks or inference crops. Missing detections affect the actual final output rather than being repaired with oracle annotations.

Grounding DINO tiny is pinned to revision `a2bb814dd30d776dcf7e30523b00659f4f141c71`, with a recorded safetensors hash. The fixed query is:

```
person . bag . backpack . suitcase . cardboard box . package .
```

Box threshold .25, text threshold .20, class-wise NMS .5 and maximum item area .2 are fixed. A private label 1001 denotes the generic bag/package class. Its new-track threshold is .25. Custom kernels are disabled and inference is float32. Geometry changes while the existing CLIP image features remain fixed, isolating this ablation.

Against partial CAVIAR item references, DINO tracks recover 66/136 (48.5%) at IoU .5; person reference recall rises from 36.0% to 67.2%. The references omit other carried bags, so unmatched item predictions cannot be called proven false positives or used as item precision.

8.4 Frozen visual encoders and automatic regions

This pilot uses frozen pretrained CLIP encoders, distinct from the six earlier visual fine-tunes. Each image supplies full-scene features, automatic region mean/max, and optional geometry. Up

to four persons each contribute contextual whole-body and upper-55%-body crops, plus up to two item/observed-owner contexts. This upper-body policy is a geometric heuristic, not a validated face/hand detector. Scene fallback is preserved when detections are absent.

Each backbone encodes 137,674 scene/crop views over the 25,452 input images in float32 with TF32 disabled. Frozen arrays preserve paths, full/crop features, geometry, region boxes, region-image indices and person flags. The extraction does not use anomaly labels to choose inference regions. Cached arrays are loaded once into memory for fitting/replay; repeatedly decompressing the same NPZ for every row was avoided.

8.5 Fifty final candidate heads

Each encoder fits 14 smoking and 11 unattended candidates, for 50 saved checkpoints total. Scene and multiple-region linear baselines use fixed $C = 1$. Causal GRUs project inputs through layer normalization and 128-unit GELU/dropout .2, then use a 64-unit GRU and linear output. Region GRUs are compared with and without the 18 geometry features, across seeds 42/43/44. Relative elapsed time and inter-view gaps are explicit channels, with packed variable-length sequences preserving causal padding semantics.

AdamW uses learning rate .0005 and weight decay .01; heads train up to 60 epochs with patience eight, batch 128 and gradient clipping one. Task/source/class-balanced sampling is retained. Checkpoints and routes maximize worst development-source AP, with mean AP as tie-breaker. Static examples and variable-length action clips remain separate supported input types, not fabricated CCTV sequences.

Smoking additionally compares a per-person multiple-instance linear head, max-pooled over automatic person regions/views. Training-camera COCO annotations supervise 1,460 matched automatic regions, including 277 positives, at IoU at least .3; ambiguous matches are omitted. The loss combines weak scene/clip BCE with .5 times the matched-region BCE. No test box chooses an inference region.

The hard-negative version starts mining at epoch four using training normals within each source, upweighting difficult cases without changing original class/source mass. Both person variants use three seeds. A checkpoint selected before epoch four has not benefited from mining merely because its run name includes hard negatives. SigLIP has such a recorded early-selected case. This metadata is retained in the appendix.

8.6 Smoking results and the fair within-pilot controls

Table 8.2: Pilot smoking: same-pilot scene AP versus selected route, with dev-FPR-budget recall/FPR.

Model	Source	N	Scene AP	Pick AP	Recall	FPR
MobileCLIP	smoking cctv	646	0.626	0.725	16.5%	7.9%
MobileCLIP	hmdb51 action transfer	205	0.842	0.908	86.7%	2.9%
MobileCLIP	kaggle sujaykapadnis smoking	224	0.996	0.995	92.9%	1.8%
MobileCLIP	stanford40	806	0.874	0.874	70.7%	2.9%
SigLIP	smoking cctv	646	0.717	0.688	2.3%	1.6%
SigLIP	hmdb51 action transfer	205	0.888	0.778	66.7%	1.7%
SigLIP	kaggle sujaykapadnis smoking	224	1.000	1.000	100.0%	3.6%
SigLIP	stanford40	806	0.930	0.938	86.4%	3.0%

MobileCLIP's selected person-supervised hard-negative route raises retrospective CCTV AP from

the previous bundle's .690 to .725, while SigLIP's selected region GRU drops .722 to .688. Both do well on adapted movie/web clips, but the SigLIP selected route's .778 AP trails its own .888 scene-only pilot baseline. Keeping this regression is necessary: the development selection is not replaced with whichever test ablation is best.

The previous bundle on the same 205 HMDB cases uses eight views and no HMDB training; the pilot uses 16 views and 530 grouped training clips. MobileCLIP .697 to .908 and SigLIP .762 to .778 therefore combine data/method changes. The scene baseline shares the new pilot's training data/views and is the better controlled reference. Neither establishes temporal CCTV generalization because the CCTV test remains static.

8.7 Completion and recovery

One concurrent SigLIP fit exhausts VRAM while the detector batches images. It is rerun after the detector exits; no feature extraction or labels change. Final verification replays 99,528 predictions, checks all 50 candidate selections and saved heads, and exercises four live encoder/ROI cases. Maximum saved-head difference is about $1.9 \cdot 10^{-6}$ and live-feature difference about $4.0 \cdot 10^{-7}$. At completion, 171 tests pass. A real checkpoint-backed replay CLI is available, but it reuses cached features and is not advertised as an arbitrary-video streaming application.

Evidence: `reports/video_pilot_protocol_20261016.md`

Evidence: `results/video_pilot_20261016/verification.json`

Chapter 9

Continuous unattended-event evaluation and calibration

9.1 Freeze global alarm policies before opening the new source

For the video pilot, selected global classification routes are MobileCLIP `region_linear` and SigLIP `grounded_track_temporal_seed_42`. Development alarm thresholds maximize event recall subject to at most one normal false alert per reviewed camera-hour, then prefer fewer duplicates/false alerts and a higher tied threshold. The alarm machinery requires two seconds above threshold, releases after one second below and uses a ten-second cooldown. Gaps over 1.1 seconds reset pending score evidence. Emission timestamps are actual decision times rather than backdated high-score onsets.

This formalizes a useful desired operating point, but development has only about 28 seconds of reviewed normal exposure. A single normal false alert would imply a large hourly rate. The selected zero-false-alert cutoff therefore has weak statistical support. The record distinguishes a development constraint from a proven deployment budget.

9.2 Historical windows and actual event discovery

On 698 known historical unattended windows (242 positive/456 normal), the selected MobileCLIP route has AP about .916 and SigLIP .865. Their fixed frame-budget behavior differs sharply: MobileCLIP recall .789/FPR .055 versus SigLIP recall 1.0/FPR .899. Those numbers alone do not describe alarm starts across time.

Under strict global event matching, MobileCLIP detects 1/5 historical events with zero normal false alerts and four unknown-time alerts; SigLIP detects 0/5 with six normal false alerts and three unknown-time alerts over 230 normal seconds. Some alarms occur in unreviewed transitions before positive windows. They cannot be counted as normal false alarms or silently matched later, and provisional onset labels prevent treating every such case as a certain practical miss. The report preserves the distinctions rather than showing only window AP.

9.3 Clean AVSS primary events

EASY and MEDIUM form a two-event primary subset, contributing 605 overlapping decision windows and 165.7389 reviewed normal seconds. They are one conservative source block, not two verified independent cameras. At first use they are new to the pilot selection; in later reuse they are retrospective.

Table 9.1: Clean AVSS primary source subset; global and known-source criteria remain distinct.

Model	Protocol/cohort	Head	Hit / N	FP / dup / unk	Normal s	Delay s
Mobile CLIP	fresh avss clean primary	region linear	2/2	1/3/0	165.7	25.8
Mobile CLIP	fresh avss clean known 30 second rule	region linear	2/2	0/0/0	165.7	36.3
Mobile CLIP	fresh avss clean state rule only	region linear	2/2	0/0/0	165.7	34.8
SigLIP	fresh avss clean primary	grounded track temporal seed 42	2/2	6/3/0	165.7	12.1
SigLIP	fresh avss clean known 30 second rule	grounded track temporal seed 42	2/2	0/1/0	165.7	34.8
SigLIP	fresh avss clean state rule only	grounded track temporal seed 42	2/2	0/0/0	165.7	34.8

The global score route detects 2/2 events for both encoders but generates one MobileCLIP and six SigLIP normal false alerts, plus three repeated event alerts each. Their measured normal-exposure rates are 21.7/hour and 130.3/hour. Median delays are 25.8 and 12.1 seconds after annotated abandoned-state onset. These are not compatible with a reliable low-false-alert deployment claim despite perfect two-event recall.

9.4 Known annotation criteria: same item, stationary and owner away

A separately frozen AVSS diagnostic requires the same detected item to be stationary for at least 30 seconds, its previously associated owner absent for at least .5 seconds, and the fixed two-second persistence for an alarm. Stationarity from one item cannot be combined with owner absence from another. Missing owner history produces no candidate alert and can cause a real miss; it does not create an inferred normal annotation.

Both classifier-plus-state gates detect 2/2 with zero normal false alerts. MobileCLIP has no repeats and median delay 36.3 seconds; SigLIP has one repeat and delay 34.8 seconds. The state rule alone detects 2/2, zero normal false or repeated alarms, and median delay 34.818 seconds. This rule is independent of CLIP, so its identical entries across backbones do not represent two separate confirmations.

The rule's useful result comes with a long delay after labelled onset, likely influenced by detector/track persistence and stationarity resets. The implementation's proximity owner relation can confuse passers-by, and pixel-anchor jitter can restart the stationary timer. A two-event result cannot establish robustness across crowded scenes, pickups, occlusion, camera motion or different item types.

9.5 Censored HARD and why it stays separate

HARD's annotation tail is outside the supplied stream and nominal/actual FPS disagree. The observed prefix is retained, with the original onset unscaled and a clipped right edge. The all-three global diagnostic detects 3/3 observed events but has four/nine normal false alerts and four/five duplicates for MobileCLIP/SigLIP, over approximately 264.06 normal seconds. It cannot be presented as three clean independent confirmations or silently pooled into the headline two-event result.

The timing issue is upstream data evidence, not a poor model result being excluded. The clean/-censored distinction was specified before pilot scoring and is carried through the tables. Later method changes evaluated on HARD remain retrospective diagnostic comparisons with the same qualification.

9.6 Disjoint normal-recording calibration

Calibration reserves entire ABODA classroom video7 and walkway video2 recordings, using only their existing reviewed normal portions. The same complete recordings are excluded from both the global and calibrated evaluation arms. This prevents fitting a camera offset from the negatives later used to judge that offset.

An offset is the difference between calibration and development normal-score 95th percentiles. After exclusions, evaluation has four events and 137 normal seconds. MobileCLIP remains 0/4 with no normal false alarms in both arms. SigLIP changes from 0/4 to 1/4 events while reducing normal false alerts from four to one, still 26.3/hour. Calibration data itself is tiny, about 84.5 seconds for the classroom and eight seconds for the walkway. AVSS has no disjoint normal recording with verified camera identity, so camera calibration is unavailable there.

This is a useful historical diagnostic rather than proof that normalization solves transfer. Camera-specific calibration, unchanged global transfer and a source-criteria rule are different protocols and remain separate.

9.7 Quantify the short-exposure limitation

For a homogeneous Poisson false-alert process, a one-sided exact 95% upper rate after k observed normal false alarms in T seconds is

$$\lambda_{U,0.95} = \frac{3600}{2T}\chi^2_{0.95,2(k+1)}. \tag{9.1}$$

For zero alerts this simplifies to $-3600\log(.05)/T$. At $T = 165.7389$ seconds, the upper bound is about 65.1 false alerts/hour. Approximately three zero-alert normal hours would be needed just for this conditional bound to drop below one/hour. Such a calculation assumes a homogeneous process; it does not protect against camera/domain shift or clustered alert behavior.

Finding. The source-rule pilot is promising on its two events. Its zero observed false alarms and high window AP provide far less assurance than an independently reviewed multi-camera event study with long normal streams.

Evidence: `results/video_pilot_20261016/events.csv`

Evidence: `results/video_pilot_20261016/mobileclip2-s2/event_policies.json`

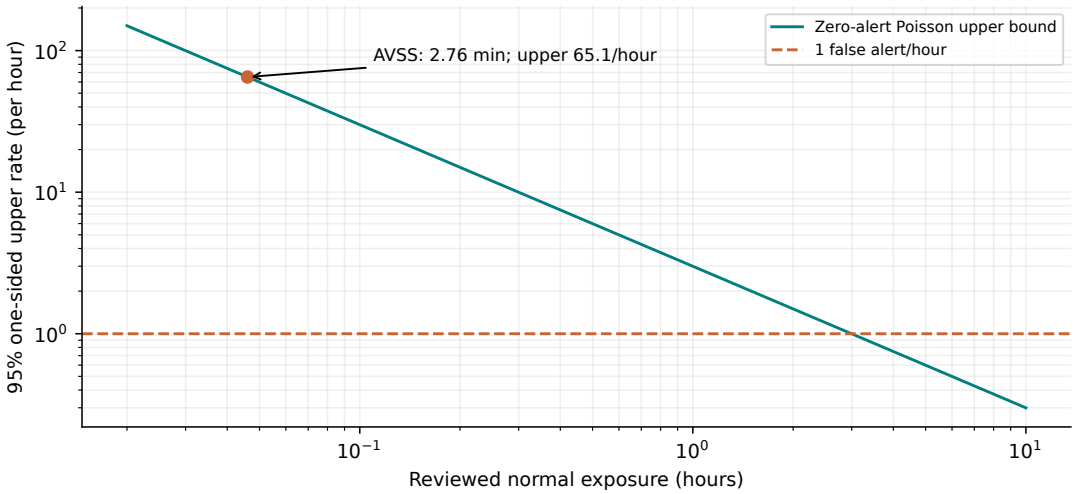


Figure 9.1: Conditional one-sided 95% Poisson upper false-alert rate after zero alerts, as a function of reviewed normal exposure. The AVSS exposure is far below the approximately three-hour crossing for one alert/hour. This is an exposure calculation, not a measured deployment-rate curve.

Chapter 10

17 October improvements: ensembles, item memory and tiling

10.1 A focused round rather than another broad fine-tune

This stage isolates three ideas suggested by the preceding failures: select smoking routes at a conservative source-aware operating point; preserve causal item/alarm identity through detector jitter and short track fragmentation; and use fixed tiling to preserve small-object detail. It also tests the compatibility of another author dataset. No visual backbone is updated, no VLM is retrained and no latency benchmark is run. Every historical smoking/CAVIAR/AVSS policy comparison is explicitly retrospective.

10.2 Nineteen smoking policies per encoder

For each backbone, 14 saved smoking heads form individual candidates. Four additional candidates average sigmoid-transformed scores within the three-seed region, region-plus-track, person-supervised and hard-negative families. A final candidate averages all 14 heads. This yields 19 policies per model, 38 total. Equal weights are fixed; there are no fitted stacking coefficients, and a sigmoid-transformed value is not claimed to be a calibrated probability.

One cutoff must meet no more than 5% empirical FPR separately in every development source. Policy selection maximizes worst-source recall at that cutoff, then mean recall, then worst/mean AP. Both encoders' complete selections are written before loading held-out predictions. An AP-selected alternative and the previous primary with its original pooled cutoff remain visible for comparison.

Table 10.1: Latest source-budget/ensemble policies versus previous primary policies, on the same retrospective cases.

Model	Policy	CCTV AP	CCTV recall/FPR	HMDB AP	HMDB recall/FPR
MobileCLIP	Previous	0.7249	16.5%/7.9%	0.9078	86.7%/2.9%
MobileCLIP	17 Oct pick	0.7249	10.9%/5.5%	0.9078	86.7%/2.3%
SigLIP	Previous	0.6876	2.3%/1.6%	0.7779	66.7%/1.7%
SigLIP	17 Oct pick	0.7195	1.8%/1.6%	0.9645	76.7%/1.7%

SigLIP selects the all-head ensemble. On the same retrospective 205 HMDB clips, AP rises from .7779 to .964485, and recall from 66.7% to 76.7% at the same observed 1.71% FPR. CCTV AP rises from .6876 to .7195, but recall falls from 2.29% to 1.78% while FPR stays 1.58%. The good

movie/web result therefore does not resolve surveillance smoking.

MobileCLIP retains the same person-supervised hard-negative head with a stricter cutoff. CCTV recall falls 16.54% to 10.94%, while FPR decreases 7.91% to 5.53%. HMDB ranking stays .9078 and recall 86.7%, with FPR reduced 2.86% to 2.29%. Source-wise development constraints are a more transparent operating-point design, but their test failure remains part of the evidence. Repeated reuse of a small development set also makes the policy screen exploratory.

10.3 Causal stationarity smoothing

The original tracker remains intact as a control. The alternative computes a trailing median of three observed centers. A displacement beyond the original .015 normalized-frame threshold must persist for one second before resetting the stationary evidence. Pending motion blocks an alert immediately. Gaps longer than two seconds discard that evidence. This rejects isolated box jitter while attempting to preserve a clear motion reset; it does not establish a physical stationary-object ground truth.

The new state policy still requires the same item stationary for 30 seconds and a previously associated owner away for .5 seconds, plus two seconds alarm persistence. Owner association remains the existing proximity heuristic, without appearance ReID or new ownership annotations. Thus the ablation addresses jitter, not the full owner-association problem.

10.4 Per-item alarm memory

The memory alternative assigns observed items one-to-one to spatial memories by box overlap or small center distance. This can preserve an emitted alarm across a short track-ID change while never inventing stationary duration or owner history for the new track. Simultaneous items have separate memories rather than collapsing to one global alarm.

An emitted item alarm remains latched until sustained owner return for five seconds, sustained movement for three seconds or memory expiry after 30 seconds without observation. Gaps reset pending persistence. These are fixed engineering choices rather than test-fitted hyperparameters. Synthetic regression tests check independent simultaneous items, ID fragmentation, missing observations, rearming, monotonic time, jitter and sustained motion.

Table 10.2: Latest AVSS primary state ablations; all results retrospective; upper bound assumes a Poisson process.

Policy	Hit / N	FP / dup / unk	Normal s	Delay s	95% upper/hr
original tracks item memory	2/2	0/0/0	165.74	34.82	65.1
robust tracks item memory	2/2	0/0/0	165.74	35.07	65.1
original tracks global debounce	2/2	0/0/0	165.74	34.82	65.1
robust tracks global debounce	2/2	0/0/0	165.74	35.07	65.1

Neither smoothing nor memory improves the two-event AVSS primary result. All four rule variants detect 2/2 with zero normal false or repeated alarms; median delay increases from 34.818 to 35.068 seconds with smoothing. The historical five-event state diagnostic stays 4/5 with no normal false and one unknown-time alert; it uses provisional boundaries and a different source criterion from the previous classifier metrics. On censored HARD, memory increases repeated alarms from one to two. The added complexity is therefore retained as a research alternative rather than promoted as an accuracy improvement.

10.5 New-source compatibility review

The pinned additional author archive is 28,582,066 bytes and is acquired without executing bundled code. Three entries explicitly alias ABODA; three bundled clips show highway debris. Their boxes are tiny and useful for spatial stress testing, but their duration is below 15 seconds and they have no complete owner/normal interval labels. Ten positive reference frames per bundled clip are decoded, with original box coordinates and timestamp/FPS alignment checked. The short, out-of-domain collection cannot increase the count of verified unattended personal-item events.

10.6 Fixed overlapping tiles: a measurable component gain

The detector comparison uses the same pinned Grounding DINO weights, generic query, score thresholds and class NMS. Every input receives a full-image view plus four fixed corner tiles covering 60% of image width and height, at normalized offsets zero or .4. Tiles overlap by 20%. Tile boxes map back to normalized full-image coordinates and merge with full-image predictions using NMS at IoU .5. Reference boxes never select tile locations.

The 166 source-annotated frames comprise 136 CAVIAR training-source item frames and 30 highway-debris frames. Five views each yield 830 actual GPU image views, float32 with TF32 disabled, batch eight and no pacing.

Table 10.3: Full-image versus fixed full-plus-tiles item reference recall; partial references do not support precision.

Source	Input	IoU	Matches	Recall	Predicted items
CAVIAR train	full	0.3	69/136	50.7%	675
CAVIAR train	full	0.5	66/136	48.5%	675
CAVIAR train	full plus tiles	0.3	117/136	86.0%	1442
CAVIAR train	full plus tiles	0.5	116/136	85.3%	1442
Highway debris	full	0.3	0/30	0.0%	13
Highway debris	full	0.5	0/30	0.0%	13
Highway debris	full plus tiles	0.3	7/30	23.3%	40
Highway debris	full plus tiles	0.5	7/30	23.3%	40

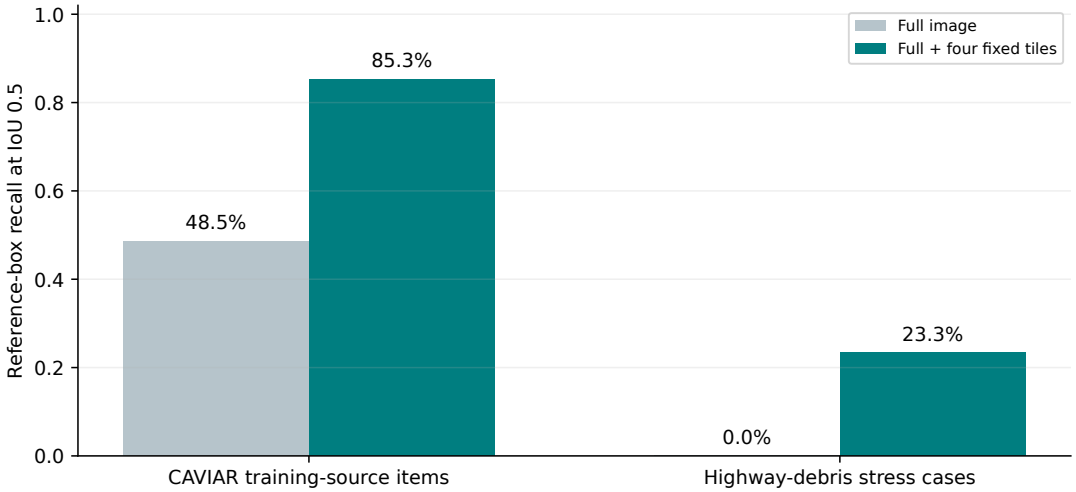


Figure 10.1: Full-image versus full-plus-tiles reference-box recall at IoU .5. CAVIAR is a training-source component diagnostic; highway debris is an out-of-domain stress test. Neither bar measures object precision or event detection.

CAVIAR recall at IoU .5 improves from 66/136 (48.5%) to 116/136 (85.3%); at .3 it improves 69/136 to 117/136. Highway-debris recall increases from 0/30 to 7/30 at both thresholds. Partial reference labels omit other objects, so unmatched detections cannot establish false positives. Adjacent source frames are correlated. The new tiled detector has not yet been validated as a full causal tracking/event replacement; the completed pilot's tracks and heads are not overwritten.

10.7 Aggregation correction and independent replay

The initial in-memory detector aggregation compared tuple tile coordinates with a list, accidentally omitting full-image baseline detections from the initial metric export. Raw GPU detections were intact. A cached-output replay with canonical tuple comparison restored the correct full-image baseline of 66/136. Invalid initial exports remain explicitly labelled, and the correction file explains the issue.

Verification now independently rebuilds full/tiled view selection and greedy NMS from serialized raw detections before recomputing Hungarian reference matches. It also recomputes all 38 development-policy scores/cutoffs/selections from the original dev predictions, replays 5,643 held-out smoking-policy predictions and 6,390 causal trace decisions, including 29 emitted memory alarms. All eight component metric cells match. The repository finishes this stage with 182 passing tests and clean lint/format/whitespace checks.

Evidence: `results/event_improvements_20261017/report.md`

Evidence: `results/event_improvements_20261017/verification.json`

Evidence: `results/event_improvements_20261017/component_correction.json`

Chapter 11

19 October: adapting SmolVLM to the five definitions

11.1 Purpose and the controlled comparison

The next requested experiment was to fine-tune SmolVLM or VadCLIP for the five existing objectives. This stage chose SmolVLM-500M-Instruct because its instruction-conditioned language model can share adaptation across all five definitions while retaining the source-specific annotations already assembled. It performs actual language-weight adaptation using low-rank adapters (LoRA). The historical supervised VadCLIP smoking experiment is retained without another training run; MiniCPM is also unchanged. Thus the project now includes learned CLIP visual representations, learned temporal/classifier heads, and a separately learned generative-model decision function. Those interventions have different trainable components and should not all be called visual fine-tuning.

The central controlled comparison is the original SmolVLM against its adapted versions under identical task queries, images, view order, image tokens and response scoring. The original checkpoint is `HuggingFaceTB/SmolVLM-500M-Instruct`, revision `a7da5b986cb59b408707209984f360a5f4ad7e47`; its base-weight SHA-256 is `d05b567eeaf534e83d375551f068ed57b5f52d37c657197f644af5ef9db091a2`. Both original and adapted models see the accepted target definitions. No target definition is changed to obtain a better score.

This experiment changes the interface from the earlier six-label JSON generation protocol to a task-conditioned binary query. An example asks about one known objective and its visible evidence, rather than requiring all categories in a free-generated response. That makes a genuine continuous first-answer score available. It also means that comparing the old JSON label accuracy directly with this study's AP would confound prompting, image processing, temporal representation and adaptation. The matched original model in this chapter is the proper weight-adaptation control.

Finding. Three LoRA candidates were trained successfully. Primary task routing improves four of the five historical target rankings, but loitering regresses, a single joint adapter regresses on the tiny historical sleep cohort, and the unattended-item development threshold transfers very poorly. These are research results, not a validated replacement for the existing event pipeline.

Evidence: `results/smolvlm_finetuning_20261019/protocol.json`

Evidence: `results/smolvlm_finetuning_20261019/run.json`

Evidence: `reports/smolvlm_finetuning_findings_20261019.md`

11.2 Manifest, units and source separation

11.2.1 Main study and additional transfer

The main manifest reuses the 16 October strategy training data and historical evaluation inputs. It contains 3,973 training examples, 1,812 development examples, 2,025 test examples and 806 separately reported external smoking photographs: 8,616 examples with 13,596 distinct decoded images. An example is a target-labelled image or ordered group of views, not necessarily a new independent event. Table 11.1 reports **normal/positive** counts. For instance, smoking training has 1,739 normal and 1,073 positive examples, totaling 2,812; it does not contain 2,812 normal examples.

Table 11.1: SmolVLM main partitions: normal/positive counts, not total/positive.

Target	Train N/+	Dev N/+	Test N/+	External N/+
Smoking	1739/1073	812/502	365/505	666/140
Dumping / littering	130/144	29/32	50/50	–
Loitering	282/41	85/13	64/11	–
Public sleep	49/25	14/7	12/6	–
Unattended	416/74	137/181	720/242	–

The smoking test separates 224 newer Kaggle photographs (112 positive, 112 normal) from 646 CCTV frames (393 positive, 253 normal). The historical Stanford40 evaluation adds 806 photographs, including 140 smoking positives. IWDD contributes 100 littering/dumping test clips with 50 positives; Street Scene has 75 loitering windows with 11 positives; the historical public-sleep cohort has 18 photographs with six positives; unattended items has 962 overlapping endpoints with 242 positives. These prevalences are essential when interpreting AP.

Three additional transfer collections remain outside training and development: 906 HMDB action clips (109 smoking, 797 other primary actions), 28 new-author public-sleep photographs (17 positive, 11 normal; 25 author groups), and 27 MEVA action windows (15 abandonment actions, 12 pre-event controls). Together these are 961 examples and 7,375 distinct decoded images. Main and additional held-out evaluation therefore covers 3,792 unique examples. Identity verification checks 20,971 distinct decoded image contents across the main and additional collections. Multiple methods and seeds reuse these same cases.

No new dataset is downloaded in this stage. The new contribution is model adaptation and controlled re-evaluation of previously collected sources. The existing source manifests remain authoritative about original acquisition, licenses, exclusions, aliases and source-group membership.

11.2.2 Leakage controls and the limits of these controls

The preparation retains recording, camera, author and source-family grouping rather than randomly distributing adjacent views. It rechecks decoded EXIF-corrected RGB pixel hashes, rejects cross-split image overlap, and checks full ordered input identities and conflicting known labels. Additional transfer inputs have no recorded group or exact-pixel overlap with the main learning/selection inputs. The same held-out identities and labels are checked against the two existing CLIP strategy pipelines before the contextual comparison later in this chapter.

These guards establish the recorded separation; they do not create independent surveillance environments. Loitering still comes from one Street Scene camera, with 20/6/6 recording groups in train/development/test. Historical unattended data has two training source families, one development family and two test families. Sparse views cannot resolve every ownership or identity ambiguity. Exact-pixel checks also do not prove the absence of every near-duplicate or shared actor/scene identity.

All these historical holdouts had already been inspected in earlier studies. Training, selection and threshold fitting use only the recorded learning/development partitions, but the experiment as a whole remains retrospective. Neither a new seed nor a fresh training run turns an already inspected source into a prospective test.

11.2.3 Label semantics and masked unknown objectives

Each example supervises exactly one annotated target. A smoking-negative image is not labelled negative for littering, loitering, sleep and unattended items. Unknown objectives have no loss term. This prevents the heterogeneous collections from generating large numbers of fabricated cross-task negatives.

IWDD remains an offline dumping/action task with full-clip inputs, not an exhaustively localized pedestrian-littering event. Historical sleep annotations describe provisional apparent public sleep under the existing visible-evidence definition; housing status is never inferred. HMDB's other primary actions are non-exhaustive smoking controls in a movie/web domain. MEVA abandonment actions do not establish a persistent unattended state. Historical unattended and duration labels retain the qualifications of their original preparation.

Evidence: `results/smolvlm_finetuning_20261019/input_label_audit.json`

Evidence: `results/smolvlm_finetuning_20261019/additional_transfer/protocol.json`

11.3 Image processing, view order and frozen visual features

Every image is EXIF-corrected, converted to RGB and resized with preserved aspect ratio so that its longest edge is at most 512 pixels. A centered square letterbox with gray value 127 precedes the publisher's 512-pixel processor. Image splitting is disabled. Each image produces 64 visual tokens through the frozen vision tower and multimodal connector. Their float32 outputs are cached; original and adapted language models receive identical visual tokens.

Static smoking and public-sleep examples use one image. Loitering and unattended-item examples retain all four existing ordered past views over approximately 20 seconds. Littering retains eight ordered views sampled across the source clip. The transfer HMDB smoking clips also use eight views; that is a transfer in both source and temporal input relative to the one-image smoking training examples. No future frames are added to causal temporal windows. IWDD full-clip classification remains offline and must not be presented as a causal online litter detector.

Caching is possible because the image encoder and connector never change. It removes repeated frozen computation without simulating a slower device or pacing the accuracy run. Native-image forward checks compare the cache/batched path with the actual publisher model for all five targets and all adapter states. Cache validity is an identity and numerical issue, not evidence that 64 image tokens capture tiny cigarettes or complete person/item histories.

The frozen base tensors are stored in BF16; converting their stored values to float32 does not recover extra pretrained mantissa precision or change their values. Training uses BF16 computation for the trainable language adapters. Frozen visual extraction, checkpoint selection, threshold fitting and final scoring use float32 with TF32 disabled. That numerical discipline follows the earlier CLIP batch-shape audit and avoids choosing a checkpoint under one scoring precision and deploying a cutoff under another.

11.4 What is learned and how the binary score is defined

11.4.1 Low-rank updates in language projections

For an adapted projection with frozen weight W, LoRA adds

$$W' = W + \frac{\alpha}{r}BA, \qquad r = 8, \quad \alpha = 16. \tag{11.1}$$

Dropout is 0.05 and no bias parameters are trained. The target projections are attention `q/k/v/o` and feed-forward `gate/up/down` in all 32 language layers: seven projections per layer, or 224 adapted modules. The resulting 4,341,760 trainable parameters are saved in approximately 17 MB adapter files. Vision, connector, token embeddings and the vocabulary output head remain frozen. Finite tensor values and nonzero learned LoRA B matrices are verified for all three saved adapters.

This is an actual change to the model's language computation, beyond fitting a classifier on fixed scene embeddings. It does not fine-tune the visual encoder, create tracked identities, localize a cigarette or add an ownership representation by itself. The distinction matters when comparing it with the earlier CLIP visual-stage updates and later temporal heads.

11.4.2 Answer-only training with a full-vocabulary loss

The response verbalizers are the first-token encodings of `Yes` (token 9230) and `No` (token 2838). For an input x asking about target t, let $z_v(x,t)$ be the next-token logit for vocabulary token v. The loss is

$$\mathcal{L}(x,t,y) = -z_{v_y}(x,t) + \log \sum_{v \in \mathcal{V}} \exp z_v(x,t), \tag{11.2}$$

where v_1 is Yes and v_0 is No. This is full-vocabulary cross entropy, not merely a two-logit loss. Every prompt/image position is excluded from the target loss. No explanation, end-of-sequence behavior, bounding box or unannotated objective is trained.

The continuous detector score is

$$s(x,t) = z_{\text{Yes}}(x,t) - z_{\text{No}}(x,t). \tag{11.3}$$

Raw classification chooses Yes at $s \geq 0$. Applying a sigmoid would give the relative Yes probability conditional on these two tokens, but would not calibrate event probability. To detect a model that prefers a third token, the study also records the total full-vocabulary probability mass on Yes and No. On development data the original mass has minimum 0.98398 and median 0.99930; the primary joint adapter has minimum approximately 0.987 and median approximately 1.0. Both already strongly follow the binary format. The study therefore does not support explaining its gains solely as repaired output syntax.

There is no learned abstention class. The inference wrapper returns unknown when required timed history is missing, but that input guard is different from a trained uncertainty or ownership decision. A confident binary margin cannot establish evidence that is absent from the input.

Table 11.2: Actual SmolVLM training and scoring settings.

C/ep	Recorded value
Adaptation	Language projections only; 4,341,760 trainable parameters
LoRA	Rank 8; alpha 16; dropout 0.05; no bias
Projections	q/k/v/o and gate/up/down in 32 language layers
Frozen	Vision tower, multimodal connector, embeddings, vocabulary head

Continued on next page

Table 11.2 continued

C/ep	Recorded value
Inputs	512-pixel gray letterbox; 64 visual tokens/image; no image splitting
Views	1 static; 4 ordered past views for temporal windows; 8 littering clip views
Loss	First-answer-token full-vocabulary cross entropy; all input positions masked
Optimizer	AdamW; weight decay 0.01; gradient norm clip 1.0
Schedule	5% warmup; linear decay; at most 6 epochs; patience 2
Primary search	Seed 42; learning rates 0.00003 and 0.0001
Replication	Seed 43; only the primary winning joint learning rate 0.0001
Batch	8 examples/microbatch; accumulation 4; effective batch 32
Coverage/epoch	3,973 unique examples once + 3,973 balanced resamples = 7,946 visits
Numerics	BF16 training; frozen visual tokens and all selection/final scores float32; TF32 off
Score	First-answer logit(Yes) minus logit(No); not a calibrated probability

11.5 Optimization, balancing and selection timing

11.5.1 Coverage and source-balanced resampling

Every epoch first includes all 3,973 training examples once. An equally sized resampled pass balances task/class, then dataset source and recording/author group, and both passes are shuffled together. This gives 7,946 example visits per epoch. It prevents rare targets from being skipped while ensuring full original coverage. The resampled half is balanced; the combined epoch retains the original corpus imbalance and should not be described as exactly equal training mass for the five tasks.

AdamW uses weight decay 0.01, 5% warmup, linear learning-rate decay and gradient-norm clipping at 1.0. Microbatch eight with four-step gradient accumulation gives effective batch 32. Primary seed 42 compares learning rates 3×10^{-5} and 10^{-4}. The planned limit is six epochs with patience two. Seed 43 repeats only the primary winning joint learning rate, 10^{-4}; it is a limited optimization replication, not a complete repeat of the primary grid.

All accuracy work runs on the available RTX 5000 Ada GPU with no pacing, CPU emulation or new latency benchmark. Training durations in the saved state files are execution provenance; they are not controlled inference-speed measurements. The nine completed epochs across the three runs comprise 71,514 example visits, repeatedly visiting the same 3,973 unique training examples.

11.5.2 Development metric and primary routes

For each task, AP is computed separately for its dataset-source development cohorts, then averaged within the task. The checkpoint metric is the equal average over all five tasks:

$$M_{\text{dev}} = \frac{1}{5} \sum_{t=1}^{5} \frac{1}{|S_t|} \sum_{s \in S_t} \text{AP}_{t,s} \, . \tag{11.4}$$

This avoids a pooled five-task metric being dominated by smoking example count. It does not guarantee good performance on the worst individual source.

The matched original development macro AP is 0.5657. The primary low-learning-rate checkpoint reaches 0.6584; the high-learning-rate checkpoint reaches 0.7026 and is the selected **joint** adapter. A separate per-task route chooses the best development source AP among the original and the two primary checkpoints, preferring the original on ties. That route uses the high learning rate for

smoking, littering, loitering and unattended items, and the low learning rate for public sleep. Task routing is a configuration containing two adapters, not a claim that one joint checkpoint achieves every routed score.

Checkpoint and route selection, as well as the raw, development-maximum-F1 and development-pooled-FPR-at-most-5% cutoffs, precede their held-out inference. Seed 43 never replaces the primary simply because one test result is better. The optional later safeguard has a different timing qualification described below.

Table 11.3: All three completed candidates; selection is based on development data.

Seed	Learning rate	Best epoch	Epochs run	Dev macro AP
42	3e-05	1	3	0.6584
42	0.0001	1	3	0.7026
43	0.0001	1	3	0.6548

Table 11.4: Primary per-task routes and exact development thresholds; raw margin is zero for every route.

Target	Route	Dev AP	Dev-F1 cutoff	Dev-5% cutoff
Smoking	High LR	0.7709	-2.629057	-0.663351
Dumping / littering	High LR	0.7764	-3.439997	1.996176
Loitering	High LR	0.3384	-2.286600	-0.170313
Public sleep	Low LR	0.9341	1.470860	1.470860
Unattended	High LR	0.7193	-7.749201	-5.379930

11.5.3 Early overfitting is visible in all candidates

All three candidates complete three epochs and select epoch one. For the primary high learning rate, training loss falls from 0.4789 to 0.2895 to 0.2235 while development macro AP falls from 0.7026 to 0.6485 to 0.6276. Additional optimization therefore fits the learning corpus more closely without improving the selected source-generalization diagnostic. Early stopping preserves the first checkpoint; reporting the final lowest-loss weights would misrepresent the selection rule.

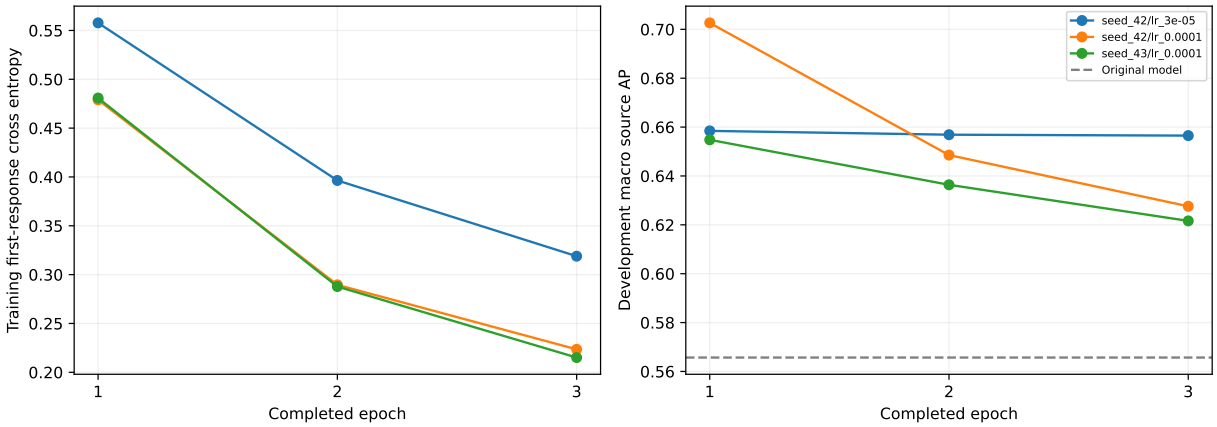


Figure 11.1: All three LoRA training histories. The original development baseline and selected first epochs distinguish optimization progress from source-transfer progress. These curves reuse the same development cohorts.

Evidence: `results/smolvlm_finetuning_20261019/seed_42/selection.json`

Evidence: `results/smolvlm_finetuning_20261019/seed_43/selection.json`

11.6 Held-out discrimination: gains and preserved regressions

Table 11.5: Matched primary held-out AP: identical inputs, definitions and scoring; historical source holdouts.

Source / task	Total / +	Constant AP	Original	Joint	Routed
Smoking photos (Kaggle)	224/112	0.5000	0.9670	0.9844	0.9844
Smoking CCTV c1	646/393	0.6084	0.5999	0.6225	0.6225
Littering / IWDD	100/50	0.5000	0.5922	0.7660	0.7660
Loitering / Street Scene	75/11	0.1467	0.4250	0.1822	0.1822
Public sleep / historical	18/6	0.3333	0.7437	0.6327	0.7885
Unattended / historical	962/242	0.2516	0.4762	0.6933	0.6933
Smoking / Stanford40	806/140	0.1737	0.5831	0.8900	0.8900
Smoking / HMDB	906/109	0.1203	0.7638	0.7837	0.7837
Public sleep / new authors	28/17	0.6071	0.8062	0.8441	0.8904
MEVA abandonment action	27/15	0.5556	0.6076	0.6457	0.6457

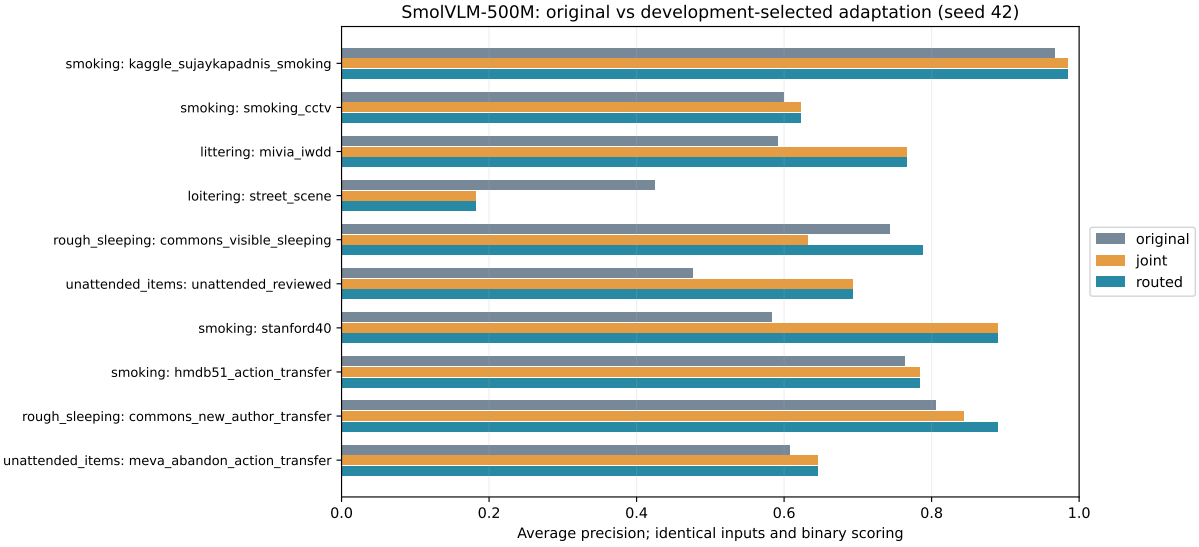


Figure 11.2: Original, joint and per-task routed SmolVLM AP on ten source/task cohorts. All use the same inputs and binary score within this study. Joint and routed results differ for public sleep.

The clearest gains are Stanford40 smoking photographs (AP 0.5831 to 0.8900), IWDD littering/-dumping clips (0.5922 to 0.7660), and historical unattended endpoints (0.4762 to 0.6933). Kaggle smoking photographs begin with high AP 0.9670 and rise to 0.9844. HMDB smoking transfer improves more modestly, from 0.7638 to 0.7837. New-author public-sleep AP rises from 0.8062 to 0.8904 under the selected low-learning-rate route; MEVA abandonment-action AP rises from 0.6076 to 0.6457.

The CCTV smoking gain is weak: 0.5999 to 0.6225 on a cohort whose constant-score AP is its positive fraction, $393/646 = 0.6084$. The adapted score only modestly exceeds that reference. High photo performance has again failed to establish useful surveillance-camera discrimination.

Loitering is a clear regression, from 0.4250 to 0.1822 despite a positive primary development gain. Source grouping did not prevent learning an unstable source-specific association. Four sparse images without persistent identity and activity context are also limited evidence for duration. The held-out result is retained under the primary route rather than replaced with the original after looking at the test.

Historical public sleep illustrates why the joint/routed distinction is necessary. The single high-learning-rate joint adapter falls from original AP 0.7437 to 0.6327. The development-selected low-learning-rate route reaches 0.7885. Thus the statement that four of five target rankings improve applies to the primary **routed configuration**; it is false for the joint checkpoint on this cohort. Six positive historical photographs remain too few for a strong generalization claim.

11.7 Fixed decisions and threshold-transfer failure

11.7.1 The development FPR budget does not transfer automatically

Each target has a threshold fitted only on its pooled development normals, allowing empirical FPR at most 5% and choosing the best F1 among permitted cutoffs. This is weaker than constraining every development source separately. Table 11.6 compares the original and routed methods at their own development-budget cutoffs. The values are endpoint/clip FPR, not alerts per normal hour.

Table 11.6: At the target-specific pooled development FPR budget of 5%; no held-out FPR guarantee.

Source / task	Orig rec.	Orig FPR	Route rec.	Route FPR	Route TP/FP
Smoking photos (Kaggle)	93.8%	12.5%	93.8%	1.8%	105/2
Smoking CCTV c1	0.0%	0.0%	1.8%	0.0%	7/0
Littering / IWDD	2.0%	4.0%	30.0%	8.0%	15/4
Loitering / Street Scene	0.0%	1.6%	9.1%	6.2%	1/4
Public sleep / historical	0.0%	0.0%	33.3%	8.3%	2/1
Unattended / historical	20.2%	1.9%	94.6%	70.8%	229/510
Smoking / Stanford40	77.1%	25.4%	78.6%	4.4%	110/29
Smoking / HMDB	63.3%	2.9%	69.7%	2.8%	76/22
Public sleep / new authors	0.0%	0.0%	41.2%	0.0%	7/0
MEVA abandonment action	0.0%	0.0%	66.7%	58.3%	10/7

Smoking photographs benefit: Kaggle recall remains 93.8% while FPR falls from 12.5% to 1.8%. Stanford40 recall is 78.6% with FPR 4.4%, compared with original 77.1% and 25.4%. HMDB reaches 69.7% recall with 2.8% FPR. CCTV, however, finds only seven of 393 positives: 1.8% recall with zero observed false positives. This is still unsuitable evidence for a useful CCTV smoking detector.

The important unattended failure is explicit: the routed cutoff finds 229 of 242 positives but falsely flags 510 of 720 normal endpoints. That is 94.6% recall at 70.8% FPR. The original method has much lower recall, 20.2%, but also much lower FPR, 1.9%. The AP gain has not produced a generally superior fixed decision. MEVA is similarly problematic at 66.7% recall and 58.3% FPR, with only 12 normal controls. Neither result supports an unattended alarm deployment claim.

Littering reaches 30.0% recall and 8.0% FPR; loitering reaches 9.1% and 6.2%; historical sleep reaches 33.3% and 8.3%. Small negative denominators make the latter percentages coarse. The new-author sleep set has seven true positives and zero false positives at the selected cutoff, but zero among 11 negatives does not establish a low general false-alarm rate.

11.7.2 Raw margin-zero decisions answer a different question

Table 11.7: Predeclared raw margin-zero decisions; primary routed versus matched original.

Source / task	Orig rec.	Orig FPR	Route rec.	Route FPR	Route F1
Smoking photos (Kaggle)	100.0%	98.2%	93.8%	1.8%	0.9589
Smoking CCTV c1	8.9%	9.9%	0.0%	0.0%	0.0000
Littering / IWDD	84.0%	70.0%	60.0%	26.0%	0.6452
Loitering / Street Scene	0.0%	1.6%	9.1%	4.7%	0.1333
Public sleep / historical	100.0%	100.0%	100.0%	25.0%	0.8000
Unattended / historical	26.4%	5.3%	64.9%	25.8%	0.5368
Smoking / Stanford40	100.0%	98.5%	78.6%	3.2%	0.8118
Smoking / HMDB	94.5%	61.2%	67.0%	1.6%	0.7487
Public sleep / new authors	100.0%	100.0%	70.6%	18.2%	0.7742
MEVA abandonment action	0.0%	0.0%	6.7%	8.3%	0.1176

The raw threshold does not fit a cutoff: Yes must have a logit at least as high as No. It therefore provides a useful check on decision-bias changes. Routed CCTV is entirely negative at this point, with zero recall and F1. Routed littering has 60.0% recall and 26.0% FPR; unattended items has 64.9% recall and 25.8% FPR. A different cutoff moves the tradeoff but does not repair missing ownership evidence or a shifted camera score distribution.

The appendix preserves all 60 source/method/seed summaries, including maximum-F1 decisions, and the source package includes the complete CSV. The report does not select the most flattering operating point separately for each test cohort. A future alert study must fit its policy on development streams and evaluate real alert starts, repeats, delay and reviewed normal hours; this LoRA stage contains no continuous event-policy evaluation.

11.8 Seed stability and the optional retrospective safeguard

Table 11.8: Joint-adapter AP under the fixed winning learning rate: optimization replication on the same sources.

Source / task	Seed 42	Seed 43
Smoking photos (Kaggle)	0.9844	0.9848
Smoking CCTV c1	0.6225	0.6295
Littering / IWDD	0.7660	0.7656
Loitering / Street Scene	0.1822	0.1314
Public sleep / historical	0.6327	0.9167
Unattended / historical	0.6933	0.7052
Smoking / Stanford40	0.8900	0.8934
Smoking / HMDB	0.7837	0.7862
Public sleep / new authors	0.8441	0.8498
MEVA abandonment action	0.6457	0.6620

Seed 43 repeats the selected high learning rate, choosing epoch one with development macro AP 0.6548. Main joint-adapter APs broadly repeat for photographs, CCTV, dumping and unattended endpoints. Loitering remains poor and falls further to 0.1314. Historical sleep changes from seed-42 joint AP 0.6327 to seed-43 joint AP 0.9167. That variation on six positives is a warning about sensitivity, not permission to promote seed 43 after inspecting held-out results. Both seeds share the same cameras and labels, so agreement cannot establish independent-domain replication.

After primary held-out results had been inspected and the first seed-43 development epoch was available, an optional repeated-gain rule was specified. It keeps a primary per-task adapter only

when both its primary development AP gain and the seed-43 joint development AP gain over the original are strictly positive. Otherwise it restores the original model and its original development cutoffs. The function reads development scores only, but the decision to introduce this function is post hoc. Its result is an exploratory retrospective safeguard, not a new prospective model-selection success.

Table 11.9: Exploratory retrospective routing safeguard; not a prospectively validated selection rule.

Target	Primary dev gain	Seed-43 dev gain	Optional route
Smoking	0.0872	0.0828	High LR
Dumping / littering	0.1589	0.1017	High LR
Loitering	0.0308	-0.1620	Original
Public sleep	0.3084	0.3426	Low LR
Unattended	0.1253	0.0805	High LR

The rule restores the original for loitering because the primary gain is +0.0308 but the second-seed development gain is −0.1620. The other four primary task routes remain. The public-sleep primary is a low-learning-rate adapter whereas seed 43 repeats the high-learning-rate joint adapter, so the check supports a broad adaptation-family gain rather than exact replication of every routed configuration. It does not solve the unattended FPR failure, create temporal labels or add a camera holdout. The primary tables above remain unchanged.

Evidence: `results/smolvlm_finetuning_20261019/repeated_gain_protocol.json`

Evidence: `results/smolvlm_finetuning_20261019/repeated_gain_routes.json`

11.9 Context against the existing CLIP strategies

Table 11.10: Same 3,792 held-out identities and labels: different trained pipelines, not an equal-input architecture ablation.

Source / task	MobileCLIP	SigLIP	Smol routed
Smoking photos (Kaggle)	0.9975	0.9998	0.9844
Smoking CCTV c1	0.6902	0.7225	0.6225
Littering / IWDD	0.8064	0.8934	0.7660
Loitering / Street Scene	0.4894	0.5505	0.1822
Public sleep / historical	0.9306	0.8833	0.7885
Unattended / historical	0.9803	0.9250	0.6933
Smoking / Stanford40	0.8803	0.9373	0.8900
Smoking / HMDB	0.5764	0.6639	0.7837
Public sleep / new authors	0.9521	0.9330	0.8904
MEVA abandonment action	0.5452	0.6205	0.6457

The comparison uses the 16 October primary CLIP strategy routes, the same 3,973/1,812 learning/development partitions, and exactly 3,792 matched held-out identities and labels. The recorded comparison provenance checks that identity match. Automatic crops, resolution, image representation, optimization and trainable components differ, so these are complete trained-pipeline comparisons rather than isolated architecture effects.

SigLIP and MobileCLIP remain stronger on most historical cohorts. For example, CCTV AP is 0.6902/0.7225 versus SmolVLM 0.6225; IWDD is 0.8064/0.8934 versus 0.7660; loitering is 0.4894/0.5505 versus 0.1822; and historical unattended endpoints are 0.9803/0.9250 versus 0.6933. High historical CLIP unattended AP must still be read with its own threshold failures documented earlier.

SmolVLM is stronger on HMDB transfer in this matched protocol: AP 0.7837 versus 0.5764/0.6639. It is also modestly higher on MEVA action transfer, 0.6457 versus 0.5452/0.6205, on only 27 examples with action-proxy labels. On Stanford40 it lies between MobileCLIP and SigLIP. These source-specific findings justify retaining the adapted VLM as a research alternative, not declaring a five-objective winner.

The later CLIP smoking video pilot uses 530 HMDB training clips, 205 held-out clips and a different 16-view temporal/crop protocol. Its 0.9645 ensemble result cannot be placed against the 906-case SmolVLM transfer score as an equal-split comparison. VadCLIP uses another supervised offline smoking split of 1,056 frames, so it also stays outside this matched-cohort table.

Evidence: `results/smolvlm_finetuning_20261019/clip_comparison_provenance.json`

11.10 Verification, exported adapters and inference limits

The completed verification reconstructs all selections and frozen cutoffs from development predictions and replays 7,248 development plus 15,168 held-out scores. It checks full example coverage in every completed epoch, 13,596 main plus 7,375 transfer decoded image identities, finite parameters and changed LoRA B matrices for all three adapters, and 20 native-image GPU cases: original plus three adapters, each across the five targets. The largest native-versus-saved score error is 7.25×10^{-5}, below the recorded 0.002 tolerance. These checks establish that the implementation and saved metrics agree within their numerical scope; they do not validate the annotation semantics.

The software snapshot has 187 passing unit tests, Ruff lint/format checks and a clean whitespace diff. Six inference-interface checks cover four real native scoring cases and two missing-timestamp temporal guards. These are functional score/interface checks, not accuracy successes on six independently sampled examples. Some selected positive examples are classified absent at the strict development-budget cutoff; that fact is consistent with the measured recall limitations.

Portable exports replace a machine-local base-model cache path in adapter configuration with the public model identity and pinned revision; the adapter weight bytes do not change. The model bundle records selected adapter paths, weight/config hashes, primary routes and all cutoffs. Its explicit status is `research_inference_configuration`. The optional repeated-gain routes retain their retrospective qualification. Each candidate also saves its processor and complete training history.

The real-input wrapper supports the primary routing by default, an original-model control, a joint-adapter control, and optional repeated-gain routing. Duration-dependent targets require timed context and return unknown when it is absent. Providing four timestamps passes an input requirement; it does not prove stable identity, continuous dwell or an owner leaving. No visual boxes, calibrated event probabilities, explanation quality, learned abstention, long normal-stream rates or computation latency are measured by this stage.

Evidence: `results/smolvlm_finetuning_20261019/verification.json`

Evidence: `results/smolvlm_finetuning_20261019/software_checks.json`

Evidence: `results/smolvlm_finetuning_20261019/cli_checks.json`

Evidence: `results/smolvlm_finetuning_20261019/model_bundle.json`

Evidence: `results/smolvlm_finetuning_20261019/portable_adapter_export.json`

Evidence: `anomaly_eval/training/smolvlm.py`

Evidence: `scripts/run_smolvlm_finetuning.py`

Evidence: `scripts/verify_smolvlm_finetuning.py`

Evidence: `scripts/predict_smolvlm.py`

Evidence: `docs/smolvlm_finetuning.md`

Chapter 12

What the results establish and what remains unresolved

12.1 Smoking

Useful progress comes from task-labelled photos, person-region supervision, better crop coverage and a strong adapted-action ensemble. However, the main surveillance issue remains. Tiny smokers lose action detail under global resizing, confusable mouth/hand actions remain common, and camera-specific score offsets make conservative cutoffs miss most CCTV positives. Static historical frames cannot test causal action history. The .9645 HMDB result is a retrospective movie/web action result on 205 clips with 530 training clips; it is not a deployment smoking score.

The next evidence should contain independently labelled surveillance videos from cameras not already inspected, person regions and matched drinking/eating/coughing/phone controls. Evaluate detector/crop coverage and event decisions separately. A photo collection with thousands of images from similar viewpoints can improve representation learning without providing that source-transfer evidence.

SmolVLM LoRA adds strong photo-domain gains and a modest HMDB transfer gain, but CCTV AP is only 0.6225 against constant-score AP 0.6084. Its conservative cutoff finds seven of 393 CCTV positives. Language adaptation therefore repeats the photo-to-camera gap rather than resolving it.

12.2 Littering

The strongest established result is improved ranking on the official 100-clip IWDD holdout: first supervised AP .858/.885 and later selected strategy .806/.893 for MobileCLIP/SigLIP. This is meaningful action supervision with real hard negatives, but includes larger dumping and uniform full-clip views. It does not establish small pedestrian litter disposal, localization of the responsible person/object, or online detection delay.

More useful data would distinguish disposal from benign placing, picking up, carrying and object transfer, with complete temporal boundaries. Generic garbage presence and positive-only throwing boxes should remain component candidates. A full event pipeline must discover the action without annotation-end anchoring or oracle frame selection.

The matched SmolVLM experiment improves IWDD AP from 0.5922 to 0.7660, below the earlier CLIP strategy pipelines on identical cases. At its development-budget cutoff it finds 30% of positives with 8% FPR. This is further evidence that the source labels can support action learning,

with the same limits on pedestrian litter and online localization.

12.3 Loitering

Global CLIP fine-tuning fails on the first 75-window Street Scene test. Sparse geometry later improves retrospective AP, suggesting that movement/presence history matters. The separate thermal experiment demonstrates descriptor value beyond duration alone but also a large development-to-date-test drop. A provided complete track from one thermal camera excludes real detection and identity errors and does not guarantee RGB transfer.

The next model needs automatic person identities, stable histories, source-compatible duration/activity labels and new camera/day controls. Merely increasing person-box frames does not label loitering. A person who waits for a legitimate activity or moves through the area is a necessary control, with unknown retained when the required history is unavailable.

SmolVLM LoRA regresses from 0.4250 to 0.1822 AP on the same 75 historical windows; the second seed is also weak. An optional retrospective repeated-gain rule restores the original for this task, but cannot be described as a prospectively validated generalization policy. More language-model optimization does not supply tracked duration evidence.

12.4 Rough sleeping / apparent public sleep

The synthetic oracle posture test shows that some models can distinguish lying from standing under supplied crops. The public-photo study provides more relevant visible context and author-held-out controls, but remains small and provisionally reviewed. High AP on six initial positive test photos or 17 new-author positives cannot validate persistent sleep, repeated-camera behavior or varied rest contexts.

The appropriate next evidence is independently reviewed public-scene video with persistence and matched awake-rest/lying/bench controls under the accepted visible-evidence definition. Housing status remains outside the target. A claim that a person is lying is neither a confirmed sleep event nor a reason to infer personal circumstances.

SmolVLM reinforces the small-sample warning: historical sleep AP is 0.6327 for the primary joint adapter, 0.7885 for the development-selected low-learning-rate route and 0.9167 for the second-seed joint adapter. Those are different configurations, not interchangeable estimates of one model. None provides persistent-video validation.

12.5 Unattended items

The data now includes reviewed historical intervals, published AVSS temporal annotations and qualified MEVA actions. The detector component improves strongly with open-vocabulary proposals and fixed tiling. Explicit same-item persistence/owner-away state outperforms global classifier alarm behavior on two clean AVSS events. Nevertheless, ownership remains a proximity hypothesis, normal exposure is only a few minutes, alarm delay is long, and crowded/occluded/pickup scenarios are not adequately covered.

The next study should evaluate item recall, track identity, owner association, stationarity and alert accounting as separate components before claiming a better end-to-end model. New independently reviewed cameras need long normal streams and diverse positive events. Fixed tiling must run blindly on those complete streams rather than only on reference-positive frames before any event benefit can be asserted.

SmolVLM improves endpoint AP from 0.4762 to 0.6933 while its development-budget threshold yields 510 false positives among 720 normal historical endpoints. MEVA action-transfer FPR is 58.3%. This is a concrete failed threshold transfer, and the adapter is not promoted as an unattended alarm replacement. Object persistence and owner history remain necessary evidence.

12.6 Generalization lessons across all stages

The experiments support five practical lessons. First, correct label semantics matter more than forcing every folder into an event taxonomy. Second, source-aware splits and duplicate guards are necessary but do not manufacture camera diversity. Third, learning a ranking and transferring its threshold are distinct problems; AP gains can coexist with nearly universal false alarms. Fourth, temporal tasks need identified histories rather than simply more independent-looking frames. Fifth, additional architecture complexity should remain optional when controlled ablations do not improve the event result.

No result justifies a universal model choice across the five targets. Different candidates use different supervision, sensors, units, historical exposure and definitions. The correct comparison is within each documented protocol. The next decisive result should come from a new frozen source benchmark, rather than another retrospective headline selected from a familiar holdout.

12.7 Proposed next validation gates, not work already completed

Future work should independently review target event boundaries, ownership and matched controls; freeze new source-camera blocks before predictions are inspected; validate automatic component coverage/identity; train on source-balanced confirmed examples with all alternatives retained; and report event recall, onset delay, normal false alerts, repeats, unknown-time alerts and exposure-based uncertainty. These are recommendations at the reporting cutoff. They are not presented as experiments already run.

Three zero-alert hours is only the conditional Poisson exposure calculation for an upper rate below one/hour; it is not a sufficient deployment certification. Source diversity, seasonal conditions, crowded scenes and changes in prevalence must also be represented. Unknown/abstain outputs remain appropriate when the operational evidence is missing.

Chapter 13

Software, artifacts, reproducibility and reporting

13.1 Repository organization and portable execution

The common package `anomaly_eval/` contains data/provenance, datasets, model adapters, resource controls, metrics and training code. The primary entry point is `anomaly-eval` or `python -m anomaly_eval`; the notebook is a front end rather than an independent evaluation implementation. Data, results, cache, bundle and configuration roots can be supplied through command-line options, a workspace JSON configuration or documented environment variables. Machine-specific paths are diagnostic metadata rather than experiment identities.

Existing pretrained caches and pinned revisions are retained. The GPU launcher selects matching project-local driver libraries and `.venv-gpu`. CPU preparation can add isolated OpenCV dependencies to `PYTHONPATH`. Model accuracy is unpaced; explicit concurrency/batch choices use the GPU's actual capacity. Download source rate limits are respected independently of compute utilization.

13.2 Reproduce a new study without overwriting completed output

Completed training/evaluation directories are intentionally immutable. Use new output paths for changed protocols. The following are representative commands; the linked study guides contain acquisition dependencies and each dated launcher's fixed-path behavior.

```
# Original shared image quality run, available GPU, all confirmed inputs.
scripts/evaluate_gpu.sh run --device cuda --batch-size 32

# New actual visual fine-tune: use a new output directory.
scripts/evaluate_gpu.sh train-clip \
  --manifest data/clip_training_20261015/samples.jsonl \
  --model siglip2-b16-224 --output results/new_training_run \
  --batch-size 32 --workers 6 --threads 8 --epochs 10 \
  --learning-rates 1e-5 3e-5 --seed 42

# New policy replay output; does not retrain CLIP or rerun the VLMs.
PYTHONPATH="$PWD" .venv/bin/python scripts/run_event_improvements.py \
  --output results/new_event_policy_replay
```

The primary inference interface can classify a static image, timed views or an offline dumping

clip using the trained routing. Duration-dependent objectives return unknown without sufficient history. Strategy and video-pilot experimental prediction scripts replay saved checkpoints/features under their documented scope. In particular, the pilot cached-feature CLI is not a complete arbitrary-new-video decoder/tracker service.

13.3 What is saved for every major stage

The frozen image studies retain cohorts, definitions, ordered identities, processor/model revisions, raw responses, summaries and calibrated thresholds. Fine-tuning adds actual visual/head checkpoints, training histories, selected learning rates/epochs, three-seed selections, parameter-delta evidence, the original AMP results and final float32 scoring. The strategy screen adds all candidate-development summaries, fitted heads, feature/weight identities, source transfer policies and the separate thermal estimator. The video pilot adds causal recordings/windows, detector outputs, automatic tracks, region arrays, per-head selections, event policies, alarm-time metrics and review queues.

The 17 October round adds all 38 smoking-policy summaries, held-out score replays, item-state trace decisions, raw per-tile detections, independently rebuilt merged detections, annotation-compatibility audits, the acquisition receipt and the explicitly invalid initial aggregate. It preserves historical completed runs unchanged.

The 19 October SmolVLM study adds pinned base/manifest identities, original development scores, frozen visual caches, three actual language-adapter checkpoints, complete epoch histories, task routes, exact cutoffs, ten-cohort source results, response-token mass checks, a second optimization seed, portable adapter metadata, native-image comparisons and an explicitly retrospective optional routing rule. The inference bundle retains its research status.

13.4 SmolVLM reproduction and controls

The isolated additional dependencies are PEFT 0.17.1 and Accelerate 1.10.1 under `data/tools/smol_training_deps`; the existing CUDA PyTorch and Transformers environment is retained. The fixed default study output is already complete. For another experiment, use the training script's output option and a new directory; do not overwrite these checkpoints. The launcher help and guide document the exact interface.

```
# Inspect/verify existing results, without changing their selection.
bash scripts/smolvlm_gpu.sh verify

# Real input inference with the default primary task routing.
bash scripts/smolvlm_gpu.sh predict --task smoking --images /path/frame.jpg

# Timed history required for unattended items.
bash scripts/smolvlm_gpu.sh predict --task unattended_items \
  --images /path/t0.jpg /path/t7.jpg /path/t13.jpg /path/t20.jpg \
  --timestamps 0 7 13 20

# Distinct controls; do not combine routing options.
bash scripts/smolvlm_gpu.sh predict --original \
  --task smoking --images /path/frame.jpg
bash scripts/smolvlm_gpu.sh predict --joint \
  --task smoking --images /path/frame.jpg
bash scripts/smolvlm_gpu.sh predict --repeated-gains \
  --task loitering \
  --images /path/t0.jpg /path/t7.jpg /path/t13.jpg /path/t20.jpg \
  --timestamps 0 7 13 20
```

The repeated-gain option is exploratory and retrospective. Missing required temporal evidence returns unknown; valid timestamps alone do not verify ownership or persistent identity. Raw scores, selected cutoffs and route names must be retained when interpreting a decision. The guide also records training, transfer evaluation, export and checkpoint-replay commands; these are separate from the reporting build.

13.5 Verification progression

Table 13.1: Historical completion audits; counts are replayed predictions, not independent events.

Stage	Test snapshot	Prediction/identity evidence
Corrected GPU baseline	144 tests	13,400 planned predictions checked; 20 summaries recomputed.
Expanded accuracy	Existing tests retained	40/40 successful CUDA summaries; 55,156 predictions replayed.
Actual fine-tuning	153 tests	46,926 held-out + 29,394 dev predictions; six changed visual checkpoints.
Strategy screen	159 tests	117,612 predictions; 330 selections; 20,980 input image identities; ten live cases.
Causal video pilot	171 tests	99,528 predictions; 50 selections/heads; 25,452 input images and live encoder checks.
Latest improvements	182 tests	5,643 smoking scores; 38 dev policies; 6,390 trace decisions; eight component cells.
SmolVLM LoRA	187 tests; six CLI checks	7,248 dev + 15,168 held-out scores; 20,971 decoded images; three changed adapters; 20 native GPU cases.

Counts of passing tests are historical snapshots at the stated stages, not independent experimental replications. The strongest verification is layered: unit tests check semantics such as masking, causal padding, source leakage and threshold ties; source/pixel manifests check identity; metric replays check summaries against predictions; and live encoder/checkpoint cases check saved predictions against actual model execution. A successful script exit alone would be weaker evidence.

This documentation task checks the saved artifact status and source hashes, regenerates report tables from structured outputs, validates derived count/metric relationships and compiles the LaTeX. It does not rerun the historical model evaluations or claim new accuracy measurements. Separate report-build validation records page count, references, compiler warnings and the source-package contents.

13.6 Presentations and this LaTeX report

The first presentation summarized the 13 October comparison; the 14 October expanded deck contained 16 slides. The 17 October progress review contains 32 editable slides and a matching vector PDF, plus speaker notes, source hashes, a summary slide and rendered previews. It covers the training and iterative studies without repeating latency testing. The updated 19 October presentation adds the SmolVLM methods, complete source-level comparisons, threshold failures, replication and current assessment, with an editable deck and matching PDF. Historical decks are preserved.

This report expands that content into a technical account with equations, data semantics, acquisition exclusions, hyperparameters, selection timing, failed interventions and appendices. Numeric tables are generated from saved JSON/CSV sources, with full source SHA-256 identities in `source_manifest.json`. Pinned model identities and acquisition URLs are reproduced from

recorded metadata, not represented as a new 19 October availability survey.

No TeX compiler was initially installed. A portable official Tectonic 0.17.0 binary is installed under the project's ignored data/tools tree after its archive digest is checked against the release identity. Required TeX packages are fetched to a workspace-local cache; no system driver, system Python or system LaTeX installation is changed. The source ZIP includes the LaTeX chapters, generated tables/figures, source registry and build instructions, so it can be compiled separately with a normal LaTeX installation or the recorded compiler bootstrap.

Evidence: `docs/evaluation.md`

Evidence: `docs/training.md`

Evidence: `docs/strategy_study.md`

Evidence: `docs/video_pilot.md`

Evidence: `docs/event_improvements.md`

Evidence: `docs/smolvlm_finetuning.md`

Evidence: `scripts/smolvlm_gpu.sh`

Appendix A

Historical audit and superseded adapter experiments

The archived audit is part of the work completed, but its historical 80-image results and validation-only temporal ablations are not the corrected test protocol. The fifteen findings explain why later source labels, metrics, cached identities, checkpoints and split rules were strengthened. Their titles are preserved as audit observations; the accompanying source contains the reproductions and detailed limitations.

Table A.1: Fifteen archived audit findings; original reproductions are in the registered audit source.

ID	Severity	Historical issue
F01	High	fractional positives are silently converted to negatives
F02	High	reported optimal thresholds do not match their metrics
F03	High	the CPU best-weight snapshot changes during further training
F04	High	validation is reused for model/epoch/branch/threshold selection
F05	High	dataset labels do not fully match the specified events
F06	High	source and negative distribution confound the image comparison
F07	High	the requested five-anomaly coverage is incomplete
F08	Medium	abstentions are excluded from most headline metrics
F09	Medium	macro-F1 has a changing class denominator
F10	Medium	bootstrap ignores the group structure and does not establish a winner
F11	Medium	padded frames influence real predictions
F12	Medium	session labels/order and feature-cache reuse need stronger provenance
F13	Medium	notebook reuse can silently show stale results
F14	Medium	the old VadCLIP wrapper is not a working upstream integration
F15	Medium	hardware/runtime tests do not select the best deployment pipeline

The following older six-run VadCLIP suite varied graph use, attention window and representative length. It used a 12-session train/four-session validation split with no separate final test. The archived audit identified fractional-label truncation, threshold inconsistency and CPU snapshot/-padding issues. Representative target counts differ with visual length. Therefore these archived AP/F1 entries document what was run, not valid evidence that one architecture generalizes better. They remain separate from the corrected 1,056-frame held-out test in Chapter 4.

Table A.2: Superseded validation-only adapter records; target/compression and implementation defects preclude a clean architecture claim.

Run	Length	Window	Graph	Reps	+ reps	Old AP	Old opt F1
full	256	8	on	383	115	0.5253	0.4891
no graph	256	8	off	383	115	0.5105	0.4934

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Table A.2 continued

Run	Length	Window	Graph	Reps	+ reps	Old AP	Old opt F1
win 1	256	1	on	383	115	0.5419	0.4635
win 16	256	16	on	383	115	0.5361	0.4870
vl 128	128	8	on	255	60	0.6374	0.5778
vl 512	512	8	on	639	276	0.5814	0.6033

Appendix B

Fine-tuning replications and operating points

The following tables reproduce the primary three operating points and all seed-specific routed results. They preserve the development routing rather than replacing it with the best test candidate. "M" denotes MobileCLIP2-S2 and "S" denotes SigLIP2-B/16-224 in compact tables. AP, F1, recall and FPR use the exact cohort/split described in Chapter 6; public sleep and historical unattended labels retain their provisional qualifications.

Table B.1: All primary routed operating points; historical/provisional cohort qualifications apply.

Model	Target	Cutoff rule	Cutoff	PPV	Recall	F1	FPR
MobileCLIP	Smoking	raw	0.000	0.0%	0.0%	0.000	1.2%
MobileCLIP	Smoking	dev max f1	-4.416	60.7%	93.1%	0.735	93.7%
MobileCLIP	Smoking	dev fpr budget	-0.760	50.0%	3.1%	0.058	4.7%
MobileCLIP	Dumping / littering	raw	0.000	78.8%	82.0%	0.804	22.0%
MobileCLIP	Dumping / littering	dev max f1	-1.372	78.9%	90.0%	0.841	24.0%
MobileCLIP	Dumping / littering	dev fpr budget	4.225	90.5%	38.0%	0.535	4.0%
MobileCLIP	Loitering	raw	0.000	0.0%	0.0%	0.000	9.4%
MobileCLIP	Loitering	dev max f1	-0.472	0.0%	0.0%	0.000	12.5%
MobileCLIP	Loitering	dev fpr budget	-0.472	0.0%	0.0%	0.000	12.5%
MobileCLIP	Public sleep	raw	0.000	83.3%	83.3%	0.833	8.3%
MobileCLIP	Public sleep	dev max f1	0.779	100.0%	83.3%	0.909	0.0%
MobileCLIP	Public sleep	dev fpr budget	0.779	100.0%	83.3%	0.909	0.0%
MobileCLIP	Unattended	raw	0.000	100.0%	25.6%	0.408	0.0%
MobileCLIP	Unattended	dev max f1	-5.249	25.6%	100.0%	0.408	97.6%
MobileCLIP	Unattended	dev fpr budget	-3.311	47.0%	96.3%	0.631	36.5%
SigLIP	Smoking	raw	0.000	68.3%	10.9%	0.189	7.9%
SigLIP	Smoking	dev max f1	-1.445	60.3%	91.3%	0.727	93.3%
SigLIP	Smoking	dev fpr budget	0.778	100.0%	0.3%	0.005	0.0%
SigLIP	Dumping / littering	raw	0.000	75.5%	80.0%	0.777	26.0%
SigLIP	Dumping / littering	dev max f1	-0.245	76.8%	86.0%	0.811	26.0%
SigLIP	Dumping / littering	dev fpr budget	1.790	85.7%	60.0%	0.706	10.0%
SigLIP	Loitering	raw	0.000	0.0%	0.0%	0.000	0.0%
SigLIP	Loitering	dev max f1	-2.409	2.2%	9.1%	0.035	70.3%
SigLIP	Loitering	dev fpr budget	-1.556	0.0%	0.0%	0.000	12.5%

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Table B.1 continued

Model	Target	Cutoff rule	Cutoff	PPV	Recall	F1	FPR
SigLIP	Public sleep	raw	0.000	85.7%	100.0%	0.923	8.3%
SigLIP	Public sleep	dev max f1	2.163	100.0%	33.3%	0.500	0.0%
SigLIP	Public sleep	dev fpr budget	2.163	100.0%	33.3%	0.500	0.0%
SigLIP	Unattended	raw	0.000	96.7%	61.2%	0.749	0.7%
SigLIP	Unattended	dev max f1	-5.255	25.2%	100.0%	0.403	99.6%
SigLIP	Unattended	dev fpr budget	-4.791	25.3%	100.0%	0.403	99.4%

Table B.2: All seed-specific routed results at maximum-dev-F1 cutoffs; seed 42 remains primary.

Model	Seed	Target	Dev route	AP	F1	Recall	FPR
MobileCLIP	42	Smoking	finetuned	0.564	0.735	93.1%	93.7%
MobileCLIP	42	Dumping / littering	finetuned	0.858	0.841	90.0%	24.0%
MobileCLIP	42	Loitering	finetuned	0.093	0.000	0.0%	12.5%
MobileCLIP	42	Public sleep	finetuned	0.976	0.909	83.3%	0.0%
MobileCLIP	42	Unattended	finetuned	0.652	0.408	100.0%	97.6%
MobileCLIP	43	Smoking	linear probe	0.594	0.752	98.0%	97.2%
MobileCLIP	43	Dumping / littering	linear probe	0.861	0.847	94.0%	28.0%
MobileCLIP	43	Loitering	finetuned	0.099	0.097	27.3%	75.0%
MobileCLIP	43	Public sleep	finetuned	1.000	0.909	83.3%	0.0%
MobileCLIP	43	Unattended	finetuned	0.697	0.405	100.0%	98.6%
MobileCLIP	44	Smoking	finetuned	0.590	0.756	100.0%	100.0%
MobileCLIP	44	Dumping / littering	finetuned	0.799	0.760	76.0%	24.0%
MobileCLIP	44	Loitering	finetuned	0.111	0.000	0.0%	9.4%
MobileCLIP	44	Public sleep	finetuned	0.976	0.909	83.3%	0.0%
MobileCLIP	44	Unattended	finetuned	0.513	0.406	100.0%	98.5%
SigLIP	42	Smoking	finetuned	0.650	0.727	91.3%	93.3%
SigLIP	42	Dumping / littering	linear probe	0.885	0.811	86.0%	26.0%
SigLIP	42	Loitering	zero shot	0.097	0.035	9.1%	70.3%
SigLIP	42	Public sleep	linear probe	0.915	0.500	33.3%	0.0%
SigLIP	42	Unattended	finetuned	0.896	0.403	100.0%	99.6%
SigLIP	43	Smoking	finetuned	0.665	0.756	100.0%	100.0%
SigLIP	43	Dumping / littering	linear probe	0.885	0.811	86.0%	26.0%
SigLIP	43	Loitering	zero shot	0.097	0.035	9.1%	70.3%
SigLIP	43	Public sleep	linear probe	0.915	0.500	33.3%	0.0%
SigLIP	43	Unattended	finetuned	0.902	0.403	100.0%	99.4%
SigLIP	44	Smoking	finetuned	0.692	0.755	99.5%	99.6%
SigLIP	44	Dumping / littering	linear probe	0.885	0.811	86.0%	26.0%
SigLIP	44	Loitering	finetuned	0.095	0.000	0.0%	7.8%
SigLIP	44	Public sleep	linear probe	0.915	0.500	33.3%	0.0%
SigLIP	44	Unattended	linear probe	0.855	0.402	100.0%	100.0%

Appendix C

All 330 strategy-development configurations

This is a complete development table: 33 candidates for each of five targets and two backbones. It is not a test-selected leaderboard. Strategy names preserve interpolation coefficients and representation recipes; the "setting" column gives linear C or an MLP selected epoch. The primary per-task choice maximizes mean development-source AP. Complete held-out family operating-point data remain in the original CSV and are included in the source package's numeric exports.

Table C.1: All 330 strategy candidates: development scores only; M=MobileCLIP, S=SigLIP.

M/S	Target	Representation	Head	C/ep	Mean AP	Worst AP
M	Smoking	endpoint a0	zero_shot	–	0.4450	0.1929
M	Smoking	full pool a0	linear	0.1	0.7467	0.3703
M	Smoking	full pool a0	linear	1	0.7526	0.3786
M	Smoking	full pool a0	linear	10	0.7520	0.3922
M	Smoking	full pool a0.25	linear	0.1	0.7663	0.3981
M	Smoking	full pool a0.25	linear	1	0.7679	0.4047
M	Smoking	full pool a0.25	linear	10	0.7582	0.3964
M	Smoking	full pool a0.5	linear	0.1	0.7947	0.4642
M	Smoking	full pool a0.5	linear	1	0.7940	0.4705
M	Smoking	full pool a0.5	linear	10	0.7744	0.4477
M	Smoking	full pool a1	linear	0.1	0.7604	0.4125
M	Smoking	full pool a1	linear	1	0.7597	0.4312
M	Smoking	full pool a1	linear	10	0.7612	0.4638
M	Smoking	endpoint a0	linear	0.1	0.7467	0.3703
M	Smoking	endpoint a0	linear	1	0.7527	0.3788
M	Smoking	endpoint a0	linear	10	0.7520	0.3922
M	Smoking	full ordered a0	linear	0.1	0.7467	0.3703
M	Smoking	full ordered a0	linear	1	0.7526	0.3786
M	Smoking	full ordered a0	linear	10	0.7490	0.3909
M	Smoking	full ordered a1	linear	0.1	0.7604	0.4125
M	Smoking	full ordered a1	linear	1	0.7597	0.4312
M	Smoking	full ordered a1	linear	10	0.7612	0.4638
M	Smoking	full crop pool a0	linear	0.1	0.7650	0.4320
M	Smoking	full crop pool a0	linear	1	0.7837	0.4725
M	Smoking	full crop pool a0	linear	10	0.7999	0.5095
M	Smoking	full crop ordered a0	linear	0.1	0.7650	0.4320
M	Smoking	full crop ordered a0	linear	1	0.7837	0.4725
M	Smoking	full crop ordered a0	linear	10	0.7977	0.5052

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Table C.1 continued

M/S	Target	Representation	Head	C/ep	Mean AP	Worst AP
M	Smoking	full geometry a0	linear	0.1	0.7501	0.3790
M	Smoking	full geometry a0	linear	1	0.7499	0.3765
M	Smoking	full geometry a0	linear	10	0.7499	0.3874
M	Smoking	full pool a0	mlp	19	0.7455	0.3726
M	Smoking	full crop pool a0	mlp	28	0.7933	0.4986
M	Dumping / littering	endpoint a0	zero_shot	–	0.4671	0.4671
M	Dumping / littering	full pool a0	linear	0.1	0.8491	0.8491
M	Dumping / littering	full pool a0	linear	1	0.8462	0.8462
M	Dumping / littering	full pool a0	linear	10	0.8880	0.8880
M	Dumping / littering	full pool a0.25	linear	0.1	0.8864	0.8864
M	Dumping / littering	full pool a0.25	linear	1	0.8986	0.8986
M	Dumping / littering	full pool a0.25	linear	10	0.9103	0.9103
M	Dumping / littering	full pool a0.5	linear	0.1	0.9231	0.9231
M	Dumping / littering	full pool a0.5	linear	1	0.9185	0.9185
M	Dumping / littering	full pool a0.5	linear	10	0.9184	0.9184
M	Dumping / littering	full pool a1	linear	0.1	0.9123	0.9123
M	Dumping / littering	full pool a1	linear	1	0.9096	0.9096
M	Dumping / littering	full pool a1	linear	10	0.9112	0.9112
M	Dumping / littering	endpoint a0	linear	0.1	0.8486	0.8486
M	Dumping / littering	endpoint a0	linear	1	0.8659	0.8659
M	Dumping / littering	endpoint a0	linear	10	0.8974	0.8974
M	Dumping / littering	full ordered a0	linear	0.1	0.8337	0.8337
M	Dumping / littering	full ordered a0	linear	1	0.8380	0.8380
M	Dumping / littering	full ordered a0	linear	10	0.8839	0.8839
M	Dumping / littering	full ordered a1	linear	0.1	0.9065	0.9065
M	Dumping / littering	full ordered a1	linear	1	0.9134	0.9134
M	Dumping / littering	full ordered a1	linear	10	0.9175	0.9175
M	Dumping / littering	full crop pool a0	linear	0.1	0.8241	0.8241
M	Dumping / littering	full crop pool a0	linear	1	0.8609	0.8609
M	Dumping / littering	full crop pool a0	linear	10	0.9062	0.9062
M	Dumping / littering	full crop ordered a0	linear	0.1	0.8201	0.8201
M	Dumping / littering	full crop ordered a0	linear	1	0.8400	0.8400
M	Dumping / littering	full crop ordered a0	linear	10	0.8668	0.8668
M	Dumping / littering	full geometry a0	linear	0.1	0.7218	0.7218
M	Dumping / littering	full geometry a0	linear	1	0.8470	0.8470
M	Dumping / littering	full geometry a0	linear	10	0.9153	0.9153
M	Dumping / littering	full pool a0	mlp	1	0.8556	0.8556
M	Dumping / littering	full crop pool a0	mlp	12	0.8415	0.8415
M	Loitering	endpoint a0	zero_shot	–	0.1747	0.1747
M	Loitering	full pool a0	linear	0.1	0.1058	0.1058
M	Loitering	full pool a0	linear	1	0.1096	0.1096
M	Loitering	full pool a0	linear	10	0.1199	0.1199
M	Loitering	full pool a0.25	linear	0.1	0.1340	0.1340
M	Loitering	full pool a0.25	linear	1	0.1429	0.1429
M	Loitering	full pool a0.25	linear	10	0.1720	0.1720
M	Loitering	full pool a0.5	linear	0.1	0.1926	0.1926
M	Loitering	full pool a0.5	linear	1	0.2216	0.2216
M	Loitering	full pool a0.5	linear	10	0.2251	0.2251
M	Loitering	full pool a1	linear	0.1	0.2915	0.2915
M	Loitering	full pool a1	linear	1	0.2792	0.2792
M	Loitering	full pool a1	linear	10	0.2620	0.2620
M	Loitering	endpoint a0	linear	0.1	0.1183	0.1183

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Table C.1 continued

M/S	Target	Representation	Head	C/ep	Mean AP	Worst AP
M	Loitering	endpoint a0	linear	1	0.1254	0.1254
M	Loitering	endpoint a0	linear	10	0.1374	0.1374
M	Loitering	full ordered a0	linear	0.1	0.1156	0.1156
M	Loitering	full ordered a0	linear	1	0.1216	0.1216
M	Loitering	full ordered a0	linear	10	0.1377	0.1377
M	Loitering	full ordered a1	linear	0.1	0.2332	0.2332
M	Loitering	full ordered a1	linear	1	0.2346	0.2346
M	Loitering	full ordered a1	linear	10	0.2105	0.2105
M	Loitering	full crop pool a0	linear	0.1	0.2170	0.2170
M	Loitering	full crop pool a0	linear	1	0.1972	0.1972
M	Loitering	full crop pool a0	linear	10	0.1884	0.1884
M	Loitering	full crop ordered a0	linear	0.1	0.2410	0.2410
M	Loitering	full crop ordered a0	linear	1	0.2018	0.2018
M	Loitering	full crop ordered a0	linear	10	0.1777	0.1777
M	Loitering	full geometry a0	linear	0.1	0.4858	0.4858
M	Loitering	full geometry a0	linear	1	0.3374	0.3374
M	Loitering	full geometry a0	linear	10	0.2079	0.2079
M	Loitering	full pool a0	mlp	5	0.1052	0.1052
M	Loitering	full crop pool a0	mlp	5	0.1913	0.1913
M	Public sleep	endpoint a0	zero_shot	–	0.5420	0.5420
M	Public sleep	full pool a0	linear	0.1	0.9683	0.9683
M	Public sleep	full pool a0	linear	1	0.9683	0.9683
M	Public sleep	full pool a0	linear	10	0.9683	0.9683
M	Public sleep	full pool a0.25	linear	0.1	1.0000	1.0000
M	Public sleep	full pool a0.25	linear	1	1.0000	1.0000
M	Public sleep	full pool a0.25	linear	10	1.0000	1.0000
M	Public sleep	full pool a0.5	linear	0.1	1.0000	1.0000
M	Public sleep	full pool a0.5	linear	1	1.0000	1.0000
M	Public sleep	full pool a0.5	linear	10	1.0000	1.0000
M	Public sleep	full pool a1	linear	0.1	1.0000	1.0000
M	Public sleep	full pool a1	linear	1	1.0000	1.0000
M	Public sleep	full pool a1	linear	10	1.0000	1.0000
M	Public sleep	endpoint a0	linear	0.1	0.9683	0.9683
M	Public sleep	endpoint a0	linear	1	0.9683	0.9683
M	Public sleep	endpoint a0	linear	10	0.9683	0.9683
M	Public sleep	full ordered a0	linear	0.1	0.9683	0.9683
M	Public sleep	full ordered a0	linear	1	0.9683	0.9683
M	Public sleep	full ordered a0	linear	10	0.9683	0.9683
M	Public sleep	full ordered a1	linear	0.1	1.0000	1.0000
M	Public sleep	full ordered a1	linear	1	1.0000	1.0000
M	Public sleep	full ordered a1	linear	10	1.0000	1.0000
M	Public sleep	full crop pool a0	linear	0.1	0.9683	0.9683
M	Public sleep	full crop pool a0	linear	1	0.9821	0.9821
M	Public sleep	full crop pool a0	linear	10	0.9821	0.9821
M	Public sleep	full crop ordered a0	linear	0.1	0.9683	0.9683
M	Public sleep	full crop ordered a0	linear	1	0.9821	0.9821
M	Public sleep	full crop ordered a0	linear	10	0.9821	0.9821
M	Public sleep	full geometry a0	linear	0.1	0.8955	0.8955
M	Public sleep	full geometry a0	linear	1	0.9240	0.9240
M	Public sleep	full geometry a0	linear	10	0.9478	0.9478
M	Public sleep	full pool a0	mlp	1	1.0000	1.0000
M	Public sleep	full crop pool a0	mlp	1	1.0000	1.0000

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Table C.1 continued

M/S	Target	Representation	Head	C/ep	Mean AP	Worst AP
M	Unattended	endpoint a0	zero_shot	–	0.4967	0.4967
M	Unattended	full pool a0	linear	0.1	0.6441	0.6441
M	Unattended	full pool a0	linear	1	0.6221	0.6221
M	Unattended	full pool a0	linear	10	0.5933	0.5933
M	Unattended	full pool a0.25	linear	0.1	0.7052	0.7052
M	Unattended	full pool a0.25	linear	1	0.7207	0.7207
M	Unattended	full pool a0.25	linear	10	0.6968	0.6968
M	Unattended	full pool a0.5	linear	0.1	0.7797	0.7797
M	Unattended	full pool a0.5	linear	1	0.7714	0.7714
M	Unattended	full pool a0.5	linear	10	0.7633	0.7633
M	Unattended	full pool a1	linear	0.1	0.8310	0.8310
M	Unattended	full pool a1	linear	1	0.8477	0.8477
M	Unattended	full pool a1	linear	10	0.8641	0.8641
M	Unattended	endpoint a0	linear	0.1	0.6711	0.6711
M	Unattended	endpoint a0	linear	1	0.6275	0.6275
M	Unattended	endpoint a0	linear	10	0.7049	0.7049
M	Unattended	full ordered a0	linear	0.1	0.5108	0.5108
M	Unattended	full ordered a0	linear	1	0.5405	0.5405
M	Unattended	full ordered a0	linear	10	0.6042	0.6042
M	Unattended	full ordered a1	linear	0.1	0.9222	0.9222
M	Unattended	full ordered a1	linear	1	0.9442	0.9442
M	Unattended	full ordered a1	linear	10	0.9469	0.9469
M	Unattended	full crop pool a0	linear	0.1	0.7582	0.7582
M	Unattended	full crop pool a0	linear	1	0.7826	0.7826
M	Unattended	full crop pool a0	linear	10	0.5570	0.5570
M	Unattended	full crop ordered a0	linear	0.1	0.7149	0.7149
M	Unattended	full crop ordered a0	linear	1	0.8159	0.8159
M	Unattended	full crop ordered a0	linear	10	0.5884	0.5884
M	Unattended	full geometry a0	linear	0.1	0.6529	0.6529
M	Unattended	full geometry a0	linear	1	0.6987	0.6987
M	Unattended	full geometry a0	linear	10	0.6460	0.6460
M	Unattended	full pool a0	mlp	5	0.6525	0.6525
M	Unattended	full crop pool a0	mlp	14	0.7569	0.7569
S	Smoking	endpoint a0	zero_shot	–	0.5667	0.3957
S	Smoking	full pool a0	linear	0.1	0.7509	0.3737
S	Smoking	full pool a0	linear	1	0.7668	0.3705
S	Smoking	full pool a0	linear	10	0.7770	0.3857
S	Smoking	full pool a0.25	linear	0.1	0.7712	0.3841
S	Smoking	full pool a0.25	linear	1	0.7775	0.3793
S	Smoking	full pool a0.25	linear	10	0.7850	0.3943
S	Smoking	full pool a0.5	linear	0.1	0.7825	0.3955
S	Smoking	full pool a0.5	linear	1	0.7838	0.3891
S	Smoking	full pool a0.5	linear	10	0.7813	0.3848
S	Smoking	full pool a1	linear	0.1	0.7951	0.4105
S	Smoking	full pool a1	linear	1	0.7920	0.3984
S	Smoking	full pool a1	linear	10	0.7828	0.3828
S	Smoking	endpoint a0	linear	0.1	0.7509	0.3737
S	Smoking	endpoint a0	linear	1	0.7668	0.3705
S	Smoking	endpoint a0	linear	10	0.7783	0.3919
S	Smoking	full ordered a0	linear	0.1	0.7510	0.3727
S	Smoking	full ordered a0	linear	1	0.7668	0.3705
S	Smoking	full ordered a0	linear	10	0.7816	0.3902

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Table C.1 continued

M/S	Target	Representation	Head	C/ep	Mean AP	Worst AP
S	Smoking	full ordered a1	linear	0.1	0.7951	0.4105
S	Smoking	full ordered a1	linear	1	0.7920	0.3984
S	Smoking	full ordered a1	linear	10	0.7828	0.3828
S	Smoking	full crop pool a0	linear	0.1	0.7721	0.4271
S	Smoking	full crop pool a0	linear	1	0.7966	0.4517
S	Smoking	full crop pool a0	linear	10	0.8124	0.4863
S	Smoking	full crop ordered a0	linear	0.1	0.7721	0.4271
S	Smoking	full crop ordered a0	linear	1	0.7966	0.4517
S	Smoking	full crop ordered a0	linear	10	0.8028	0.4629
S	Smoking	full geometry a0	linear	0.1	0.7564	0.3906
S	Smoking	full geometry a0	linear	1	0.7652	0.3714
S	Smoking	full geometry a0	linear	10	0.7757	0.3789
S	Smoking	full pool a0	mlp	31	0.7777	0.3714
S	Smoking	full crop pool a0	mlp	39	0.8209	0.5053
S	Dumping / littering	endpoint a0	zero_shot	–	0.6288	0.6288
S	Dumping / littering	full pool a0	linear	0.1	0.8704	0.8704
S	Dumping / littering	full pool a0	linear	1	0.8846	0.8846
S	Dumping / littering	full pool a0	linear	10	0.9185	0.9185
S	Dumping / littering	full pool a0.25	linear	0.1	0.8789	0.8789
S	Dumping / littering	full pool a0.25	linear	1	0.8858	0.8858
S	Dumping / littering	full pool a0.25	linear	10	0.9207	0.9207
S	Dumping / littering	full pool a0.5	linear	0.1	0.8861	0.8861
S	Dumping / littering	full pool a0.5	linear	1	0.8954	0.8954
S	Dumping / littering	full pool a0.5	linear	10	0.9146	0.9146
S	Dumping / littering	full pool a1	linear	0.1	0.8823	0.8823
S	Dumping / littering	full pool a1	linear	1	0.8966	0.8966
S	Dumping / littering	full pool a1	linear	10	0.9010	0.9010
S	Dumping / littering	endpoint a0	linear	0.1	0.8608	0.8608
S	Dumping / littering	endpoint a0	linear	1	0.8763	0.8763
S	Dumping / littering	endpoint a0	linear	10	0.9145	0.9145
S	Dumping / littering	full ordered a0	linear	0.1	0.8552	0.8552
S	Dumping / littering	full ordered a0	linear	1	0.8692	0.8692
S	Dumping / littering	full ordered a0	linear	10	0.9059	0.9059
S	Dumping / littering	full ordered a1	linear	0.1	0.8848	0.8848
S	Dumping / littering	full ordered a1	linear	1	0.8880	0.8880
S	Dumping / littering	full ordered a1	linear	10	0.9060	0.9060
S	Dumping / littering	full crop pool a0	linear	0.1	0.8275	0.8275
S	Dumping / littering	full crop pool a0	linear	1	0.8581	0.8581
S	Dumping / littering	full crop pool a0	linear	10	0.9017	0.9017
S	Dumping / littering	full crop ordered a0	linear	0.1	0.8399	0.8399
S	Dumping / littering	full crop ordered a0	linear	1	0.8608	0.8608
S	Dumping / littering	full crop ordered a0	linear	10	0.8996	0.8996
S	Dumping / littering	full geometry a0	linear	0.1	0.6962	0.6962
S	Dumping / littering	full geometry a0	linear	1	0.8542	0.8542
S	Dumping / littering	full geometry a0	linear	10	0.9263	0.9263
S	Dumping / littering	full pool a0	mlp	13	0.8651	0.8651
S	Dumping / littering	full crop pool a0	mlp	12	0.8122	0.8122
S	Loitering	endpoint a0	zero_shot	–	0.2290	0.2290
S	Loitering	full pool a0	linear	0.1	0.0845	0.0845
S	Loitering	full pool a0	linear	1	0.0860	0.0860
S	Loitering	full pool a0	linear	10	0.0948	0.0948
S	Loitering	full pool a0.25	linear	0.1	0.0889	0.0889

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Table C.1 continued

M/S	Target	Representation	Head	C/ep	Mean AP	Worst AP
S	Loitering	full pool a0.25	linear	1	0.0935	0.0935
S	Loitering	full pool a0.25	linear	10	0.1044	0.1044
S	Loitering	full pool a0.5	linear	0.1	0.0968	0.0968
S	Loitering	full pool a0.5	linear	1	0.1014	0.1014
S	Loitering	full pool a0.5	linear	10	0.1228	0.1228
S	Loitering	full pool a1	linear	0.1	0.1144	0.1144
S	Loitering	full pool a1	linear	1	0.1293	0.1293
S	Loitering	full pool a1	linear	10	0.1514	0.1514
S	Loitering	endpoint a0	linear	0.1	0.0926	0.0926
S	Loitering	endpoint a0	linear	1	0.0934	0.0934
S	Loitering	endpoint a0	linear	10	0.0986	0.0986
S	Loitering	full ordered a0	linear	0.1	0.0842	0.0842
S	Loitering	full ordered a0	linear	1	0.0848	0.0848
S	Loitering	full ordered a0	linear	10	0.0896	0.0896
S	Loitering	full ordered a1	linear	0.1	0.1045	0.1045
S	Loitering	full ordered a1	linear	1	0.1167	0.1167
S	Loitering	full ordered a1	linear	10	0.1399	0.1399
S	Loitering	full crop pool a0	linear	0.1	0.2598	0.2598
S	Loitering	full crop pool a0	linear	1	0.2129	0.2129
S	Loitering	full crop pool a0	linear	10	0.1942	0.1942
S	Loitering	full crop ordered a0	linear	0.1	0.2785	0.2785
S	Loitering	full crop ordered a0	linear	1	0.2206	0.2206
S	Loitering	full crop ordered a0	linear	10	0.1919	0.1919
S	Loitering	full geometry a0	linear	0.1	0.5223	0.5223
S	Loitering	full geometry a0	linear	1	0.5063	0.5063
S	Loitering	full geometry a0	linear	10	0.3094	0.3094
S	Loitering	full pool a0	mlp	8	0.0871	0.0871
S	Loitering	full crop pool a0	mlp	15	0.2120	0.2120
S	Public sleep	endpoint a0	zero_shot	–	0.8524	0.8524
S	Public sleep	full pool a0	linear	0.1	1.0000	1.0000
S	Public sleep	full pool a0	linear	1	1.0000	1.0000
S	Public sleep	full pool a0	linear	10	1.0000	1.0000
S	Public sleep	full pool a0.25	linear	0.1	1.0000	1.0000
S	Public sleep	full pool a0.25	linear	1	1.0000	1.0000
S	Public sleep	full pool a0.25	linear	10	1.0000	1.0000
S	Public sleep	full pool a0.5	linear	0.1	1.0000	1.0000
S	Public sleep	full pool a0.5	linear	1	1.0000	1.0000
S	Public sleep	full pool a0.5	linear	10	1.0000	1.0000
S	Public sleep	full pool a1	linear	0.1	1.0000	1.0000
S	Public sleep	full pool a1	linear	1	1.0000	1.0000
S	Public sleep	full pool a1	linear	10	1.0000	1.0000
S	Public sleep	endpoint a0	linear	0.1	1.0000	1.0000
S	Public sleep	endpoint a0	linear	1	1.0000	1.0000
S	Public sleep	endpoint a0	linear	10	1.0000	1.0000
S	Public sleep	full ordered a0	linear	0.1	1.0000	1.0000
S	Public sleep	full ordered a0	linear	1	1.0000	1.0000
S	Public sleep	full ordered a0	linear	10	1.0000	1.0000
S	Public sleep	full ordered a1	linear	0.1	1.0000	1.0000
S	Public sleep	full ordered a1	linear	1	1.0000	1.0000
S	Public sleep	full ordered a1	linear	10	1.0000	1.0000
S	Public sleep	full crop pool a0	linear	0.1	1.0000	1.0000
S	Public sleep	full crop pool a0	linear	1	1.0000	1.0000

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Table C.1 continued

M/S	Target	Representation	Head	C/ep	Mean AP	Worst AP
S	Public sleep	full crop pool a0	linear	10	1.0000	1.0000
S	Public sleep	full crop ordered a0	linear	0.1	1.0000	1.0000
S	Public sleep	full crop ordered a0	linear	1	1.0000	1.0000
S	Public sleep	full crop ordered a0	linear	10	1.0000	1.0000
S	Public sleep	full geometry a0	linear	0.1	0.9094	0.9094
S	Public sleep	full geometry a0	linear	1	0.9617	0.9617
S	Public sleep	full geometry a0	linear	10	0.9617	0.9617
S	Public sleep	full pool a0	mlp	2	1.0000	1.0000
S	Public sleep	full crop pool a0	mlp	1	1.0000	1.0000
S	Unattended	endpoint a0	zero_shot	–	0.4672	0.4672
S	Unattended	full pool a0	linear	0.1	0.6561	0.6561
S	Unattended	full pool a0	linear	1	0.7002	0.7002
S	Unattended	full pool a0	linear	10	0.7779	0.7779
S	Unattended	full pool a0.25	linear	0.1	0.6751	0.6751
S	Unattended	full pool a0.25	linear	1	0.7250	0.7250
S	Unattended	full pool a0.25	linear	10	0.8025	0.8025
S	Unattended	full pool a0.5	linear	0.1	0.6905	0.6905
S	Unattended	full pool a0.5	linear	1	0.7498	0.7498
S	Unattended	full pool a0.5	linear	10	0.8216	0.8216
S	Unattended	full pool a1	linear	0.1	0.7272	0.7272
S	Unattended	full pool a1	linear	1	0.8065	0.8065
S	Unattended	full pool a1	linear	10	0.8376	0.8376
S	Unattended	endpoint a0	linear	0.1	0.6085	0.6085
S	Unattended	endpoint a0	linear	1	0.7093	0.7093
S	Unattended	endpoint a0	linear	10	0.8502	0.8502
S	Unattended	full ordered a0	linear	0.1	0.5221	0.5221
S	Unattended	full ordered a0	linear	1	0.6074	0.6074
S	Unattended	full ordered a0	linear	10	0.8534	0.8534
S	Unattended	full ordered a1	linear	0.1	0.7060	0.7060
S	Unattended	full ordered a1	linear	1	0.8496	0.8496
S	Unattended	full ordered a1	linear	10	0.9681	0.9681
S	Unattended	full crop pool a0	linear	0.1	0.5440	0.5440
S	Unattended	full crop pool a0	linear	1	0.6116	0.6116
S	Unattended	full crop pool a0	linear	10	0.6748	0.6748
S	Unattended	full crop ordered a0	linear	0.1	0.4900	0.4900
S	Unattended	full crop ordered a0	linear	1	0.5747	0.5747
S	Unattended	full crop ordered a0	linear	10	0.7216	0.7216
S	Unattended	full geometry a0	linear	0.1	0.6319	0.6319
S	Unattended	full geometry a0	linear	1	0.7142	0.7142
S	Unattended	full geometry a0	linear	10	0.7430	0.7430
S	Unattended	full pool a0	mlp	3	0.6707	0.6707
S	Unattended	full crop pool a0	mlp	51	0.6430	0.6430

Appendix D

All 50 final video-pilot heads

Final source-guarded heads and their epoch/seed, worst-source AP, mean-source AP and selected-route flag are reproduced below. Earlier prefix-split fits are diagnostic only. A hard-negative checkpoint selected before epoch four is not evidence of mining benefit. These are frozen-pretrained-encoder heads, distinct from the earlier six actual visual fine-tunes.

Table D.1: All final source-guarded pilot heads; asterisk marks the primary development route.

M/S	Target	Head	Seed	Ep.	Worst AP	Mean AP	*
M	Smoking	scene linear	–	–	0.3733	0.6659	
M	Smoking	region linear	–	–	0.4544	0.7368	
M	Smoking	region temporal seed 42	42	15	0.5427	0.7926	
M	Smoking	region temporal seed 43	43	22	0.5162	0.7723	
M	Smoking	region temporal seed 44	44	15	0.5173	0.7709	
M	Smoking	region track temporal seed 42	42	15	0.5492	0.7862	
M	Smoking	region track temporal seed 43	43	14	0.5274	0.7755	
M	Smoking	region track temporal seed 44	44	21	0.5175	0.7922	
M	Smoking	person supervised seed 42	42	60	0.5704	0.7967	
M	Smoking	person supervised seed 43	43	60	0.5723	0.7978	
M	Smoking	person supervised seed 44	44	60	0.5723	0.7994	
M	Smoking	person supervised hardneg seed 42	42	60	0.5736	0.7975	*
M	Smoking	person supervised hardneg seed 43	43	60	0.5670	0.8004	
M	Smoking	person supervised hardneg seed 44	44	60	0.5719	0.8035	
M	Unattended	scene linear	–	–	0.8130	0.8130	
M	Unattended	region linear	–	–	0.8782	0.8782	*
M	Unattended	region temporal seed 42	42	1	0.8146	0.8146	
M	Unattended	region temporal seed 43	43	4	0.7552	0.7552	
M	Unattended	region temporal seed 44	44	1	0.8316	0.8316	
M	Unattended	region track temporal seed 42	42	1	0.8259	0.8259	
M	Unattended	region track temporal seed 43	43	15	0.8297	0.8297	
M	Unattended	region track temporal seed 44	44	1	0.8318	0.8318	
M	Unattended	grounded track temporal seed 42	42	1	0.8067	0.8067	
M	Unattended	grounded track temporal seed 43	43	15	0.8377	0.8377	
M	Unattended	grounded track temporal seed 44	44	8	0.7828	0.7828	
S	Smoking	scene linear	–	–	0.3723	0.6987	
S	Smoking	region linear	–	–	0.4293	0.7264	
S	Smoking	region temporal seed 42	42	59	0.5810	0.7890	*
S	Smoking	region temporal seed 43	43	23	0.5393	0.7818	
S	Smoking	region temporal seed 44	44	24	0.5435	0.7995	
S	Smoking	region track temporal seed 42	42	43	0.5710	0.7765	

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Table D.1 continued

M/S	Target	Head	Seed	Ep.	Worst AP	Mean AP	*
S	Smoking	region track temporal seed 43	43	24	0.5530	0.7980	
S	Smoking	region track temporal seed 44	44	47	0.5774	0.7776	
S	Smoking	person supervised seed 42	42	9	0.4921	0.8164	
S	Smoking	person supervised seed 43	43	59	0.5531	0.8369	
S	Smoking	person supervised seed 44	44	60	0.5542	0.8372	
S	Smoking	person supervised hardneg seed 42	42	2	0.4890	0.8146	
S	Smoking	person supervised hardneg seed 43	43	60	0.5471	0.8348	
S	Smoking	person supervised hardneg seed 44	44	60	0.5506	0.8339	
S	Unattended	scene linear	–	–	0.7520	0.7520	
S	Unattended	region linear	–	–	0.7576	0.7576	
S	Unattended	region temporal seed 42	42	21	0.9700	0.9700	
S	Unattended	region temporal seed 43	43	39	0.9643	0.9643	
S	Unattended	region temporal seed 44	44	12	0.9477	0.9477	
S	Unattended	region track temporal seed 42	42	21	0.9552	0.9552	
S	Unattended	region track temporal seed 43	43	35	0.9710	0.9710	
S	Unattended	region track temporal seed 44	44	19	0.9120	0.9120	
S	Unattended	grounded track temporal seed 42	42	20	0.9730	0.9730	*
S	Unattended	grounded track temporal seed 43	43	35	0.9701	0.9701	
S	Unattended	grounded track temporal seed 44	44	2	0.8805	0.8805	

Appendix E

All 38 latest smoking policies

These policies reuse the existing saved heads and fixed equal sigmoid averages. Each cutoff satisfies the empirical development-source constraint; values are not calibrated probabilities. The selected flag is determined from worst/mean development recall with AP ties. A separate AP-selected alternative remains in the source JSON. All relevant held-out comparisons are retrospective.

Table E.1: All 38 smoking-policy development summaries; equal sigmoid averages are not calibrated probabilities.

M/S	Policy	n	Cutoff	Worst recall	Mean recall	Worst AP	*
M	scene linear	1	0.60429	0.0%	50.6%	0.3733	
M	region linear	1	0.58476	0.0%	54.3%	0.4544	
M	region temporal seed 42	1	0.59187	5.3%	64.6%	0.5427	
M	region temporal seed 43	1	0.88623	12.7%	62.0%	0.5162	
M	region temporal seed 44	1	0.56992	1.5%	61.2%	0.5173	
M	region track temporal seed 42	1	0.40928	12.2%	61.2%	0.5492	
M	region track temporal seed 43	1	0.59090	12.7%	60.1%	0.5274	
M	region track temporal seed 44	1	0.64229	8.1%	64.1%	0.5175	
M	person supervised seed 42	1	0.68699	15.8%	56.6%	0.5704	
M	person supervised seed 43	1	0.71093	10.2%	56.5%	0.5723	
M	person supervised seed 44	1	0.69508	11.7%	57.2%	0.5723	
M	person supervised hardneg seed 42	1	0.67886	16.5%	58.7%	0.5736	*
M	person supervised hardneg seed 43	1	0.68530	14.5%	56.3%	0.5670	
M	person supervised hardneg seed 44	1	0.67189	15.3%	56.2%	0.5719	
M	region temporal ensemble	3	0.55311	11.5%	64.9%	0.5307	
M	region track temporal ensemble	3	0.52465	12.5%	61.3%	0.5344	
M	person supervised ensemble	3	0.69209	13.0%	57.8%	0.5711	
M	person supervised hardneg ensemble	3	0.67862	15.5%	56.5%	0.5712	
M	all head ensemble	14	0.52609	15.3%	65.8%	0.5717	
S	scene linear	1	0.58306	0.0%	57.1%	0.3723	
S	region linear	1	0.53811	0.0%	59.6%	0.4293	
S	region temporal seed 42	1	0.87444	15.0%	64.7%	0.5810	
S	region temporal seed 43	1	0.54643	13.5%	65.6%	0.5393	
S	region temporal seed 44	1	0.63610	13.7%	66.3%	0.5435	
S	region track temporal seed 42	1	0.75900	14.0%	65.7%	0.5710	
S	region track temporal seed 43	1	0.79908	13.7%	64.4%	0.5530	
S	region track temporal seed 44	1	0.85386	14.5%	62.1%	0.5774	
S	person supervised seed 42	1	0.57632	0.0%	69.2%	0.4921	
S	person supervised seed 43	1	0.63885	4.3%	70.3%	0.5531	
S	person supervised seed 44	1	0.62491	5.9%	70.7%	0.5542	

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Table E.1 continued

M/S	Policy	n	Cutoff	Worst recall	Mean recall	Worst AP	*
S	person supervised hardneg seed 42	1	0.53053	0.0%	68.9%	0.4890	
S	person supervised hardneg seed 43	1	0.63121	5.6%	70.6%	0.5471	
S	person supervised hardneg seed 44	1	0.61131	6.1%	70.8%	0.5506	
S	region temporal ensemble	3	0.65598	14.2%	64.5%	0.5584	
S	region track temporal ensemble	3	0.78296	15.8%	63.7%	0.5715	
S	person supervised ensemble	3	0.61597	3.1%	70.0%	0.5497	
S	person supervised hardneg ensemble	3	0.59184	4.8%	70.4%	0.5479	
S	all head ensemble	14	0.54582	16.8%	66.4%	0.5789	*

Appendix F

Detailed event results and source accounting

This table retains global, historical, known-criteria and censored event outcomes from the pilot, including nonprimary candidates where the event policy was defined. Camera/scene counts and normal exposure are more informative than the number of overlapping decision windows. Zero normal exposure would make a normal hourly rate unavailable rather than zero. Unknown alerts are distinct from normal false alarms.

Table F.1: Pilot event outcomes for all scored family routes; delay is median onset-to-emission time.

Model	Protocol/cohort	Head	Hit / N	FP / dup / unk	Normal s	Delay s
Mobile CLIP	historical	scene linear	1/5	2/0/3	230.0	0.0
Mobile CLIP	fresh avss	scene linear	3/3	6/6/0	264.1	23.2
Mobile CLIP	fresh avss clean primary	scene linear	2/2	4/5/0	165.7	20.6
Mobile CLIP	fresh avss right censored diagnostic	scene linear	1/1	2/1/0	98.3	23.2
Mobile CLIP	fresh avss clean known 30 second rule	scene linear	1/2	0/0/0	165.7	31.2
Mobile CLIP	fresh avss clean state rule only	scene linear	2/2	0/0/0	165.7	34.8
Mobile CLIP	historical	region linear	1/5	0/0/4	230.0	12.5
Mobile CLIP	fresh avss	region linear	3/3	4/4/0	264.1	18.2
Mobile CLIP	fresh avss clean primary	region linear	2/2	1/3/0	165.7	25.8
Mobile CLIP	fresh avss right censored diagnostic	region linear	1/1	3/1/0	98.3	18.2
Mobile CLIP	fresh avss clean known 30 second rule	region linear	2/2	0/0/0	165.7	36.3
Mobile CLIP	fresh avss clean state rule only	region linear	2/2	0/0/0	165.7	34.8
Mobile CLIP	historical	region temporal seed 44	1/5	0/0/1	230.0	3.5
Mobile CLIP	fresh avss	region temporal seed 44	3/3	1/1/0	264.1	55.2
Mobile CLIP	fresh avss clean primary	region temporal seed 44	2/2	1/1/0	165.7	37.6

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Table F.1 continued

Model	Protocol/cohort	Head	Hit / N	FP / dup / unk	Normal s	Delay s
Mobile CLIP	fresh avss right censored diagnostic	region temporal seed 44	1/1	0/0/0	98.3	88.2
Mobile CLIP	fresh avss clean known 30 second rule	region temporal seed 44	1/2	0/0/0	165.7	44.9
Mobile CLIP	fresh avss clean state rule only	region temporal seed 44	2/2	0/0/0	165.7	34.8
Mobile CLIP	historical	region track temporal seed 44	1/5	0/0/1	230.0	7.0
Mobile CLIP	fresh avss	region track temporal seed 44	0/3	0/0/0	264.1	–
Mobile CLIP	fresh avss clean primary	region track temporal seed 44	0/2	0/0/0	165.7	–
Mobile CLIP	fresh avss right censored diagnostic	region track temporal seed 44	0/1	0/0/0	98.3	–
Mobile CLIP	fresh avss clean known 30 second rule	region track temporal seed 44	0/2	0/0/0	165.7	–
Mobile CLIP	fresh avss clean state rule only	region track temporal seed 44	2/2	0/0/0	165.7	34.8
Mobile CLIP	historical	grounded track temporal seed 43	1/5	5/0/5	230.0	7.5
Mobile CLIP	fresh avss	grounded track temporal seed 43	2/3	0/0/0	264.1	37.1
Mobile CLIP	fresh avss clean primary	grounded track temporal seed 43	2/2	0/0/0	165.7	37.1
Mobile CLIP	fresh avss right censored diagnostic	grounded track temporal seed 43	0/1	0/0/0	98.3	–
Mobile CLIP	fresh avss clean known 30 second rule	grounded track temporal seed 43	0/2	0/0/0	165.7	–
Mobile CLIP	fresh avss clean state rule only	grounded track temporal seed 43	2/2	0/0/0	165.7	34.8
SigLIP	historical	scene linear	0/5	0/0/0	230.0	–
SigLIP	fresh avss	scene linear	2/3	0/0/0	264.1	34.3
SigLIP	fresh avss clean primary	scene linear	1/2	0/0/0	165.7	18.4
SigLIP	fresh avss right censored diagnostic	scene linear	1/1	0/0/0	98.3	50.2
SigLIP	fresh avss clean known 30 second rule	scene linear	0/2	0/0/0	165.7	–
SigLIP	fresh avss clean state rule only	scene linear	2/2	0/0/0	165.7	34.8
SigLIP	historical	region linear	1/5	1/0/5	230.0	1.0
SigLIP	fresh avss	region linear	2/3	4/1/0	264.1	62.8
SigLIP	fresh avss clean primary	region linear	1/2	3/1/0	165.7	47.9
SigLIP	fresh avss right censored diagnostic	region linear	1/1	1/0/0	98.3	77.7
SigLIP	fresh avss clean known 30 second rule	region linear	2/2	0/0/0	165.7	36.3
SigLIP	fresh avss clean state rule only	region linear	2/2	0/0/0	165.7	34.8
SigLIP	historical	region temporal seed 42	0/5	5/0/9	230.0	–
SigLIP	fresh avss	region temporal seed 42	3/3	1/1/0	264.1	49.2
SigLIP	fresh avss clean primary	region temporal seed 42	2/2	1/1/0	165.7	35.3

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Table F.1 continued

Model	Protocol/cohort	Head	Hit / N	FP / dup / unk	Normal s	Delay s
SigLIP	fresh avss right censored diagnostic	region temporal seed 42	1/1	0/0/0	98.3	49.2
SigLIP	fresh avss clean known 30 second rule	region temporal seed 42	0/2	0/0/0	165.7	–
SigLIP	fresh avss clean state rule only	region temporal seed 42	2/2	0/0/0	165.7	34.8
SigLIP	historical	region track temporal seed 43	0/5	7/0/5	230.0	–
SigLIP	fresh avss	region track temporal seed 43	3/3	11/2/0	264.1	9.7
SigLIP	fresh avss clean primary	region track temporal seed 43	2/2	7/1/0	165.7	5.6
SigLIP	fresh avss right censored diagnostic	region track temporal seed 43	1/1	4/1/0	98.3	21.7
SigLIP	fresh avss clean known 30 second rule	region track temporal seed 43	2/2	0/0/0	165.7	34.8
SigLIP	fresh avss clean state rule only	region track temporal seed 43	2/2	0/0/0	165.7	34.8
SigLIP	historical	grounded track temporal seed 42	0/5	6/0/3	230.0	–
SigLIP	fresh avss	grounded track temporal seed 42	3/3	9/5/0	264.1	17.9
SigLIP	fresh avss clean primary	grounded track temporal seed 42	2/2	6/3/0	165.7	12.1
SigLIP	fresh avss right censored diagnostic	grounded track temporal seed 42	1/1	3/2/0	98.3	21.7
SigLIP	fresh avss clean known 30 second rule	grounded track temporal seed 42	2/2	0/1/0	165.7	34.8
SigLIP	fresh avss clean state rule only	grounded track temporal seed 42	2/2	0/0/0	165.7	34.8

Table F.2: Latest state ablations by source/split; historical intervals have different source criteria.

Source	Policy	Hit / N	FP / dup / unk	Normal s	Delay s	95% upper/hr
historical train	original tracks item memory	0/4	1/0/2	78.00	–	218.9
historical train	robust tracks item memory	0/4	1/0/2	78.00	–	218.9
historical train	original tracks global debounce	0/4	1/0/1	78.00	–	218.9
historical train	robust tracks global debounce	0/4	1/0/1	78.00	–	218.9
historical test	original tracks item memory	4/5	0/1/1	230.00	19.24	46.9
historical test	robust tracks item memory	4/5	0/0/1	230.00	18.99	46.9
historical test	original tracks global debounce	4/5	0/0/1	230.00	19.24	46.9
historical test	robust tracks global debounce	4/5	0/0/1	230.00	18.99	46.9
historical dev	original tracks item memory	0/2	0/0/1	28.00	–	385.2
historical dev	robust tracks item memory	0/2	0/0/1	28.00	–	385.2
historical dev	original tracks global debounce	0/2	0/0/1	28.00	–	385.2
historical dev	robust tracks global debounce	0/2	0/0/1	28.00	–	385.2
avss primary retrospective	original tracks item memory	2/2	0/0/0	165.74	34.82	65.1
avss primary retrospective	robust tracks item memory	2/2	0/0/0	165.74	35.07	65.1

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Table F.2 continued

Source	Policy	Hit / N	FP / dup / unk	Normal s	Delay s	95% upper/hr
avss primary retrospective	original tracks global debounce	2/2	0/0/0	165.74	34.82	65.1
avss primary retrospective	robust tracks global debounce	2/2	0/0/0	165.74	35.07	65.1
avss censored	original tracks item memory	1/1	0/2/0	98.32	10.68	109.7
avss censored	robust tracks item memory	1/1	0/2/0	98.32	10.68	109.7
avss censored	original tracks global debounce	1/1	0/1/0	98.32	10.68	109.7
avss censored	robust tracks global debounce	1/1	0/1/0	98.32	10.68	109.7

Appendix G

Complete SmolVLM histories and source results

These tables retain all nine completed epoch records and all 60 source/method/seed summaries. O, J and R distinguish original, one joint adapter and per-task routing. Original rows repeat across seeds to support paired comparisons; they are the same frozen checkpoint. Rows and cohort counts are not independent events or extra cameras. Threshold definitions and retrospective qualifications are given in Chapter 11. The bundled numeric CSV also retains every raw, development-F1 and development-budget field and TP/FP counts.

Table G.1: Complete epoch histories. Falling training loss does not imply better source transfer.

Seed	Learning rate	Ep.	Train loss	Dev macro AP	Visits
42	3e-05	1	0.5578	0.6584	7946
42	3e-05	2	0.3964	0.6569	7946
42	3e-05	3	0.3189	0.6565	7946
42	0.0001	1	0.4789	0.7026	7946
42	0.0001	2	0.2895	0.6485	7946
42	0.0001	3	0.2235	0.6276	7946
43	0.0001	1	0.4808	0.6548	7946
43	0.0001	2	0.2878	0.6364	7946
43	0.0001	3	0.2151	0.6216	7946

Table G.2: All 60 cohort/method/seed summaries. O=original, J=joint, R=per-task routed; AUC denotes AUROC. Rec./FPR use the dev-5% point.

Source	Seed	M	AP	AUC	Raw F1	Dev-F1 F1	Rec.	FPR
Smoking photos (Kaggle)	42	O	0.9670	0.9619	0.6707	0.6667	93.8%	12.5%
Smoking photos (Kaggle)	42	J	0.9844	0.9754	0.9589	0.9636	93.8%	1.8%
Smoking photos (Kaggle)	42	R	0.9844	0.9754	0.9589	0.9636	93.8%	1.8%
Smoking CCTV c1	42	O	0.5999	0.4835	0.1545	0.7214	0.0%	0.0%
Smoking CCTV c1	42	J	0.6225	0.5016	0.0000	0.7537	1.8%	0.0%
Smoking CCTV c1	42	R	0.6225	0.5016	0.0000	0.7537	1.8%	0.0%
Littering / IWDD	42	O	0.5922	0.5888	0.6614	0.6466	2.0%	4.0%
Littering / IWDD	42	J	0.7660	0.8032	0.6452	0.7895	30.0%	8.0%
Littering / IWDD	42	R	0.7660	0.8032	0.6452	0.7895	30.0%	8.0%
Loitering / Street Scene	42	O	0.4250	0.8452	0.0000	0.5161	0.0%	1.6%
Loitering / Street Scene	42	J	0.1822	0.4148	0.1333	0.1818	9.1%	6.2%
Loitering / Street Scene	42	R	0.1822	0.4148	0.1333	0.1818	9.1%	6.2%

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Table G.2 continued

Source	Seed	M	AP	AUC	Raw F1	Dev-F1 F1	Rec.	FPR
Public sleep / historical	42	O	0.7437	0.8194	0.5000	0.6154	0.0%	0.0%
Public sleep / historical	42	J	0.6327	0.7917	0.6667	0.4000	33.3%	16.7%
Public sleep / historical	42	R	0.7885	0.8889	0.8000	0.4444	33.3%	8.3%
Unattended / historical	42	O	0.4762	0.6711	0.3721	0.4604	20.2%	1.9%
Unattended / historical	42	J	0.6933	0.8132	0.5368	0.4588	94.6%	70.8%
Unattended / historical	42	R	0.6933	0.8132	0.5368	0.4588	94.6%	70.8%
Smoking / Stanford40	42	O	0.5831	0.8374	0.2991	0.2960	77.1%	25.4%
Smoking / Stanford40	42	J	0.8900	0.9594	0.8118	0.7311	78.6%	4.4%
Smoking / Stanford40	42	R	0.8900	0.9594	0.8118	0.7311	78.6%	4.4%
Smoking / HMDB	42	O	0.7638	0.9014	0.2943	0.2295	63.3%	2.9%
Smoking / HMDB	42	J	0.7837	0.9053	0.7487	0.6332	69.7%	2.8%
Smoking / HMDB	42	R	0.7837	0.9053	0.7487	0.6332	69.7%	2.8%
Public sleep / new authors	42	O	0.8062	0.7005	0.7556	0.7143	0.0%	0.0%
Public sleep / new authors	42	J	0.8441	0.7273	0.6429	0.5600	41.2%	0.0%
Public sleep / new authors	42	R	0.8904	0.7701	0.7742	0.5833	41.2%	0.0%
MEVA abandonment action	42	O	0.6076	0.5278	0.0000	0.4800	0.0%	0.0%
MEVA abandonment action	42	J	0.6457	0.5611	0.1176	0.7368	66.7%	58.3%
MEVA abandonment action	42	R	0.6457	0.5611	0.1176	0.7368	66.7%	58.3%
Smoking photos (Kaggle)	43	O	0.9670	0.9619	0.6707	0.6667	93.8%	12.5%
Smoking photos (Kaggle)	43	J	0.9848	0.9762	0.9636	0.9511	94.6%	1.8%
Smoking photos (Kaggle)	43	R	0.9848	0.9762	0.9636	0.9511	94.6%	1.8%
Smoking CCTV c1	43	O	0.5999	0.4835	0.1545	0.7214	0.0%	0.0%
Smoking CCTV c1	43	J	0.6295	0.5137	0.0000	0.7590	0.8%	0.0%
Smoking CCTV c1	43	R	0.6295	0.5137	0.0000	0.7590	0.8%	0.0%
Littering / IWDD	43	O	0.5922	0.5888	0.6614	0.6466	2.0%	4.0%
Littering / IWDD	43	J	0.7656	0.8028	0.7748	0.7308	8.0%	0.0%
Littering / IWDD	43	R	0.7656	0.8028	0.7748	0.7308	8.0%	0.0%
Loitering / Street Scene	43	O	0.4250	0.8452	0.0000	0.5161	0.0%	1.6%
Loitering / Street Scene	43	J	0.1314	0.2983	0.0000	0.1017	0.0%	1.6%
Loitering / Street Scene	43	R	0.4250	0.8452	0.0000	0.5161	0.0%	1.6%
Public sleep / historical	43	O	0.7437	0.8194	0.5000	0.6154	0.0%	0.0%
Public sleep / historical	43	J	0.9167	0.9444	0.7692	0.5000	33.3%	0.0%
Public sleep / historical	43	R	0.9167	0.9444	0.7692	0.5000	33.3%	0.0%
Unattended / historical	43	O	0.4762	0.6711	0.3721	0.4604	20.2%	1.9%
Unattended / historical	43	J	0.7052	0.8048	0.6045	0.4112	94.2%	61.7%
Unattended / historical	43	R	0.7052	0.8048	0.6045	0.4112	94.2%	61.7%
Smoking / Stanford40	43	O	0.5831	0.8374	0.2991	0.2960	77.1%	25.4%
Smoking / Stanford40	43	J	0.8934	0.9619	0.7889	0.6564	81.4%	6.2%
Smoking / Stanford40	43	R	0.8934	0.9619	0.7889	0.6564	81.4%	6.2%
Smoking / HMDB	43	O	0.7638	0.9014	0.2943	0.2295	63.3%	2.9%
Smoking / HMDB	43	J	0.7862	0.9028	0.7379	0.6585	69.7%	2.9%
Smoking / HMDB	43	R	0.7862	0.9028	0.7379	0.6585	69.7%	2.9%
Public sleep / new authors	43	O	0.8062	0.7005	0.7556	0.7143	0.0%	0.0%
Public sleep / new authors	43	J	0.8498	0.7433	0.6154	0.3810	23.5%	0.0%
Public sleep / new authors	43	R	0.8498	0.7433	0.6154	0.3810	23.5%	0.0%
MEVA abandonment action	43	O	0.6076	0.5278	0.0000	0.4800	0.0%	0.0%
MEVA abandonment action	43	J	0.6620	0.6056	0.1250	0.7317	53.3%	50.0%

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Table G.2 continued

Source	Seed	M	AP	AUC	Raw F1	Dev-F1 F1	Rec.	FPR
MEVA abandonment action	43	R	0.6620	0.6056	0.1250	0.7317	53.3%	50.0%

Appendix H

Pinned identities, artifact registry and acquisition references

Full SHA-256 values are stored in the accompanying JSON manifest; the table shows compact prefixes for reading. Paths are repository-relative. The registry identifies what this documentation build read, including code inspected for implementation details. It is not a claim that every current code file is byte-identical to every original execution snapshot; those execution-specific identities remain in their original artifacts.

Table H.1: Pinned image-model identities from the adapter registry.

Model	Repository	Revision
MobileCLIP	timm/MobileCLIP2-S2-OpenCLIP	`ac6b37c8fc40b62b623d09ed228a6f0c9bc29fe6`
SigLIP	google/siglip2-base-patch16-224	`75de2d55ec2d0b4efc50b3e9ad70dba96a7b2fa2`
SmolVLM	HuggingFaceTB/SmolVLM-500M-Instruct	`a7da5b986cb59b408707209984f360a5f4ad7e47`
MiniCPM	openbmb/MiniCPM-V-4	`6f69a8235885be89608f3a71ffd40695b379e1e7`

H.1 Preserved run-directory inventory

These statuses are read from each root metadata file where present. "Not recorded" is not inferred to mean success. Prepared, interrupted, error and incomplete-data runs are preserved as diagnostics; the narrative identifies which completed studies provide the authoritative stage results. Separate source jobs assembled into a combined report are not extra independent datasets.

Table H.2: Preserved root run metadata status; not a combined performance leaderboard.

Result directory	Metadata	Recorded status
`anomaly_headtohead`	run.json	not recorded
`clip_all_confirmed_20261013`	evaluation.json	interrupted
`clip_all_confirmed_20261013_v2`	evaluation.json	complete for available tasks
`clip_finetuning_20261015`	run.json	complete
`cpu_latency_20261013`	run.json	not recorded
`evaluation_20261011T104039_496754Z`	evaluation.json	interrupted
`evaluation_20261011T120031_096651Z`	evaluation.json	prepared
`evaluation_20261011T120249_353844Z`	evaluation.json	interrupted
`evaluation_20261011T120315_062384Z`	evaluation.json	complete for available tasks

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Table H.2 continued

Result directory	Metadata	Recorded status
evaluation_20261012T034523_356317Z	evaluation.json	prepared
evaluation_20261012T035016_680032Z	evaluation.json	interrupted
evaluation_20261012T040547_431301Z	evaluation.json	interrupted
evaluation_20261012T040605_929193Z	evaluation.json	error
evaluation_20261012T165345_130231Z	evaluation.json	error
evaluation_20261013T061827_722009Z	evaluation.json	error
event_improvements_20261017	run.json	complete
expanded_accuracy_20261013	state.json	complete
five_task_readiness_20261013	evaluation.json	incomplete data
gpu_all_models_20261013	evaluation.json	complete for available tasks
gpu_full_accuracy_20261013	evaluation.json	complete for available tasks
gpu_minicpm_full_accuracy_20261013	evaluation.json	complete for available tasks
model_comparison_20261013	evaluation.json	interrupted
model_comparison_20261013_v2	evaluation.json	complete for available tasks
smolvlm_finetuning_20261019	run.json	complete
strategy_study_20261016	run.json	complete
video_pilot_20261016	run.json	complete
workflow_gpu_ready_pilot	evaluation.json	complete for available tasks

H.2 Source registry

Table H.3: Documentation source registry; complete SHA-256 identities are in source_manifest.json.

ID	Repository-relative source	SHA-256 prefix
S001	anomaly_eval/config/anomalies.json	589882c0a43d846b
S002	anomaly_eval/models.py	799f40199789e3ba
S003	anomaly_eval/training/clip.py	c400ce680f762926
S004	anomaly_eval/training/data.py	5424b130aa53df77
S005	anomaly_eval/training/event_memory.py	6b5150df1b6096cb
S006	anomaly_eval/training/smolvlm.py	7902175ed111964e
S007	anomaly_eval/training/strategies.py	a92a363a55a4e5be
S008	anomaly_eval/training/temporal.py	e1d62cc77bb64681
S009	anomaly_eval/training/video.py	ebefa15bb1206191
S010	data/clip_training_20261015/commons_source_audit.json	aef85b7a0c7b2620
S011	data/clip_training_20261015/dataset.json	2abb33d2d852eeec
S012	data/clip_training_20261015/iwdd_source_audit.json	e5b8a85233b1e3d1
S013	data/clip_training_20261015/stanford_source_audit.json	a32d853aeb257eab
S014	data/event_improvements_20261017/component_protocol.json	0547d4d78376b4bb
S015	data/event_improvements_20261017/receipt.json	3ea715d29628785f
S016	data/strategy_sources_20261016/meva/index.json	8825968c7e43a9d3
S017	data/strategy_sources_20261016/small_sources_receipts.json	693ad195b4b01efa
S018	data/strategy_study_20261016/dataset.json	738025df2f939460
S019	data/video_pilot_20261016/data_verification.json	706494b7b35c53e6
S020	data/video_pilot_20261016/detector.json	6901655eb0cdbebd
S021	data/video_pilot_20261016/fresh_cohort.json	c759448c14a2444b
S022	data/video_pilot_20261016/fresh_recordings.jsonl	58b640e8e906b2d5
S023	data/video_pilot_20261016/grounding_all.json	5d16644a3da50282
S024	data/video_pilot_20261016/source_guard.json	371c7477a3c9ec0d
S025	docs/evaluation.md	00db02e3c7cf204e
S026	docs/event_improvements.md	9f38600c38313640

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Table H.3 continued

ID	Repository-relative source	SHA-256 prefix
S027	docs/smolvlm_finetuning.md	ceb323f2f2cba81f
S028	docs/strategy_study.md	567bf7f347dfb6d0
S029	docs/training.md	c4ecc6fa6e3f55d4
S030	docs/video_pilot.md	19b9277e57b3ae39
S031	reports/anomaly_benchmark_report.md	55e7710474e7354f
S032	reports/archive/2026-10-11/evaluation_audit.md	f077b00644d3eca8
S033	reports/archive/2026-10-11/evaluation_validation_summary.json	4c7a8967184ffdc2
S034	reports/clip_finetuning_findings_20261016.md	734201d3930c873a
S035	reports/clip_training_protocol_20261015.md	673c4c5b69f019f3
S036	reports/data_readiness_20261013.json	1d2a3cf7ac3edf5c
S037	reports/event_improvements_findings_20261017.md	f57c2e781e054a78
S038	reports/hardware_20261013.json	f2ffc6760ed3748b
S039	reports/slides/progress_validation_20261017.json	0edfb0dbb6ab41e1
S040	reports/smolvlm_finetuning_findings_20261019.md	ee68ae7119060393
S041	reports/strategy_findings_20261016.md	940ab943c63d4435
S042	reports/strategy_protocol_20261016.md	08261f1b12abda00
S043	reports/vadclip_source_validation.json	6e1e8d99bdb8502b
S044	reports/video_pilot_findings_20261016.md	08d109020bcd93c8
S045	reports/video_pilot_protocol_20261016.md	029ec05c5583a4a8
S046	reports/workflow_review_20261013.md	afe2cfb6ad54abf1
S047	results/anomaly_headtohead/run.json	3a1942b128f568a1
S048	results/clip_all_confirmed_20261013/evaluation.json	c38190048b077759
S049	results/clip_all_confirmed_20261013_v2/evaluation.json	2be4dfc8650c046c
S050	results/clip_finetuning_20261015/comparison.csv	27af4c8fb13878e2
S051	results/clip_finetuning_20261015/run.json	ac4f15585af8dd99
S052	results/clip_finetuning_20261015/seed_42/mobileclip2-s2/float32_evaluation/results.json	db0c73c60eb5c31a
S053	results/clip_finetuning_20261015/seed_42/mobileclip2-s2/selection.json	3e15fc97805d3488
S054	results/clip_finetuning_20261015/seed_42/siglip2-b16-224/float32_evaluation/results.json	460b098feb1d3c02
S055	results/clip_finetuning_20261015/seed_42/siglip2-b16-224/selection.json	62820ea45cee7d26
S056	results/clip_finetuning_20261015/seed_43/mobileclip2-s2/float32_evaluation/results.json	3f60d6c625b27794
S057	results/clip_finetuning_20261015/seed_43/mobileclip2-s2/selection.json	b2594aceb8747804
S058	results/clip_finetuning_20261015/seed_43/siglip2-b16-224/float32_evaluation/results.json	9888428d4c30e4a6
S059	results/clip_finetuning_20261015/seed_43/siglip2-b16-224/selection.json	a8507c66816b5fed
S060	results/clip_finetuning_20261015/seed_44/mobileclip2-s2/float32_evaluation/results.json	afada61f1de46d1d
S061	results/clip_finetuning_20261015/seed_44/mobileclip2-s2/selection.json	b07d6d538c932634
S062	results/clip_finetuning_20261015/seed_44/siglip2-b16-224/float32_evaluation/results.json	59761c9f8cbc2b1d
S063	results/clip_finetuning_20261015/seed_44/siglip2-b16-224/selection.json	04a7c8f977522543
S064	results/clip_finetuning_20261015/verification.json	510410415c81480e
S065	results/cpu_latency_20261013/minicpm-v4/summary.json	b44cc7a7c41a9bea
S066	results/cpu_latency_20261013/mobileclip2-s2/summary.json	cbe6858112ab9321
S067	results/cpu_latency_20261013/run.json	b9497cc0e967c09f

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Table H.3 continued

ID	Repository-relative source	SHA-256 prefix
S068	results/cpu_latency_20261013/siglip2-b16-224/summary.json	8c20f68fccdccad3
S069	results/cpu_latency_20261013/smolvlm-500m/summary.json	0419b64d3acf3026
S070	results/evaluation_20261011T104039_496754Z/evaluation.json	8ea7b7f59e4de28b
S071	results/evaluation_20261011T120031_096651Z/evaluation.json	21385e876c00fd08
S072	results/evaluation_20261011T120249_353844Z/evaluation.json	c58cfd292ca83f0c
S073	results/evaluation_20261011T120315_062384Z/evaluation.json	878d8af398f241d5
S074	results/evaluation_20261012T034523_356317Z/evaluation.json	08a7bdeb13385f2d
S075	results/evaluation_20261012T035016_680032Z/evaluation.json	a0421cb1f1e34ab2
S076	results/evaluation_20261012T040547_431301Z/evaluation.json	a0421cb1f1e34ab2
S077	results/evaluation_20261012T040605_929193Z/evaluation.json	960841c767ff8966
S078	results/evaluation_20261012T165345_130231Z/evaluation.json	cc46d5af7ce79ae1
S079	results/evaluation_20261012T165345_130231Z/report.md	4524008fae4559a5
S080	results/evaluation_20261013T061827_722009Z/evaluation.json	fcd5f73c8ae9714d
S081	results/event_improvements_20261017/component_correction.json	7214307a3ce826f2
S082	results/event_improvements_20261017/component_results.json	c17ed0fc199353d5
S083	results/event_improvements_20261017/event_results.json	553119083ed95417
S084	results/event_improvements_20261017/protocol.json	ac309f57cacd6a4e
S085	results/event_improvements_20261017/report.md	f57c2e781e054a78
S086	results/event_improvements_20261017/run.json	2159e8ce72cd3855
S087	results/event_improvements_20261017/smoking_results.json	d363c41b3d186cc9
S088	results/event_improvements_20261017/smoking_selection.json	9beab75476dab925
S089	results/event_improvements_20261017/validation.json	7e1492e63b8da925
S090	results/event_improvements_20261017/verification.json	8d47672cb508354c
S091	results/expanded_accuracy_20261013/quality_summary.json	bd5adf87592a779b
S092	results/expanded_accuracy_20261013/state.json	efaaa8bf6833b32a
S093	results/expanded_accuracy_20261013/verification.json	ad313d179a83660f
S094	results/five_task_readiness_20261013/evaluation.json	7d77fa220dd048b0
S095	results/gpu_all_models_20261013/evaluation.json	50b0b93b7ea0ed8c
S096	results/gpu_all_models_20261013/report.md	677e5c93d29144db
S097	results/gpu_all_models_20261013/smoking_frame/dev/minicpm-v4/summary.json	9aa833a2a3dc3dce
S098	results/gpu_all_models_20261013/smoking_frame/dev/mobileclip2-s2/summary.json	81ef7998a3c6ca16
S099	results/gpu_all_models_20261013/smoking_frame/dev/siglip2-b16-224/summary.json	675e56ba96403504
S100	results/gpu_all_models_20261013/smoking_frame/dev/smolvlm-500m/summary.json	f0c9e86f86829442
S101	results/gpu_all_models_20261013/unattended_items_frame/test/minicpm-v4/summary.json	be005bd3154ea150
S102	results/gpu_all_models_20261013/unattended_items_frame/test/mobileclip2-s2/summary.json	29f8f32074fbf431
S103	results/gpu_all_models_20261013/unattended_items_frame/test/siglip2-b16-224/summary.json	24959f408523ed34
S104	results/gpu_all_models_20261013/unattended_items_frame/test/smolvlm-500m/summary.json	1bd5994a794d33f0
S105	results/gpu_all_models_20261013/unattended_items_window/test/minicpm-v4/summary.json	a99de748065a6bca
S106	results/gpu_all_models_20261013/unattended_items_window/test/mobileclip2-s2/summary.json	f1925a4e95608add
S107	results/gpu_all_models_20261013/unattended_items_window/test/siglip2-b16-224/summary.json	7f4793928d1041cc
S108	results/gpu_all_models_20261013/unattended_items_window/test/smolvlm-500m/summary.json	bd0b8af8d51362a7

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Table H.3 continued

ID	Repository-relative source	SHA-256 prefix
S109	results/gpu_all_models_20261013/vadclip/adapter/eval.json	53480bc4b00d1d19
S110	results/gpu_full_accuracy_20261013/evaluation.json	266956f97179412c
S111	results/gpu_minicpm_full_accuracy_20261013/evaluation.json	c916084379862df9
S112	results/model_comparison_20261013/evaluation.json	13c4e481f1354c75
S113	results/model_comparison_20261013_v2/evaluation.json	8de485383b1b60b3
S114	results/smolvlm_finetuning_20261019/additional_transfer/protocol.json	4a667a0a939383f6
S115	results/smolvlm_finetuning_20261019/cli_checks.json	b8833204438ad077
S116	results/smolvlm_finetuning_20261019/clip_comparison_provenance.json	491cfd51323d7605
S117	results/smolvlm_finetuning_20261019/environment.json	6f15970af209bfb0
S118	results/smolvlm_finetuning_20261019/figures/learning_curves.pdf	837fdde5bb093791
S119	results/smolvlm_finetuning_20261019/figures/matched_ap.pdf	8a50b3364f508a75
S120	results/smolvlm_finetuning_20261019/implementation_manifest.json	482b8eb82dfaa060
S121	results/smolvlm_finetuning_20261019/input_label_audit.json	21730eb54e96fe44
S122	results/smolvlm_finetuning_20261019/metrics.csv	e937cd9c7ec41ca1
S123	results/smolvlm_finetuning_20261019/model_bundle.json	c4f10d45feecda27
S124	results/smolvlm_finetuning_20261019/portable_adapter_export.json	d084a366d4c084ae
S125	results/smolvlm_finetuning_20261019/protocol.json	e2b9ddc811daeceb
S126	results/smolvlm_finetuning_20261019/repeated_gain_protocol.json	7dfa0c31c058dc36
S127	results/smolvlm_finetuning_20261019/repeated_gain_routes.json	c8833fd6b23bd288
S128	results/smolvlm_finetuning_20261019/report_data.json	91d32c25a1d71ee3
S129	results/smolvlm_finetuning_20261019/response_mass.json	add9c85dc8d0dcf3
S130	results/smolvlm_finetuning_20261019/run.json	b66a427586d11455
S131	results/smolvlm_finetuning_20261019/seed_42/lr_0.0001/adapter/adapter_config.json	7c5e157b0b455b43
S132	results/smolvlm_finetuning_20261019/seed_42/lr_0.0001/history.json	15b429993b498eeb
S133	results/smolvlm_finetuning_20261019/seed_42/lr_0.0001/state.json	e90ab11f1a040465
S134	results/smolvlm_finetuning_20261019/seed_42/lr_3e-05/adapter/adapter_config.json	7c5e157b0b455b43
S135	results/smolvlm_finetuning_20261019/seed_42/lr_3e-05/history.json	c8d46af1aad1ddd2
S136	results/smolvlm_finetuning_20261019/seed_42/lr_3e-05/state.json	bf1a56f43765bf43
S137	results/smolvlm_finetuning_20261019/seed_42/selection.json	a6025c69113aee4f
S138	results/smolvlm_finetuning_20261019/seed_43/lr_0.0001/adapter/adapter_config.json	a66ce50439265a87
S139	results/smolvlm_finetuning_20261019/seed_43/lr_0.0001/history.json	f5e08567d711bc3e
S140	results/smolvlm_finetuning_20261019/seed_43/lr_0.0001/state.json	9b82c02690e0fdc3
S141	results/smolvlm_finetuning_20261019/seed_43/selection.json	0259512a4409921e
S142	results/smolvlm_finetuning_20261019/software_checks.json	7b3e0c1688599464
S143	results/smolvlm_finetuning_20261019/verification.json	17de97649f6789d6
S144	results/strategy_study_20261016/comparison.csv	1451de20fec4a23d
S145	results/strategy_study_20261016/mobileclip2-s2/development_results.json	2ec38b5e140bcc88

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Table H.3 continued

ID	Repository-relative source	SHA-256 prefix
S146	results/strategy_study_20261016/mobileclip2-s2/selection.json	0cf75a9a614d3c6e
S147	results/strategy_study_20261016/mobileclip2-s2/transfer_evaluation/per_source_results.json	5462f52a7ecba313
S148	results/strategy_study_20261016/report_data.json	f2ad1868f968373a
S149	results/strategy_study_20261016/run.json	31f0f1424a033048
S150	results/strategy_study_20261016/siglip2-b16-224/development_results.json	1aaa42385ea28801
S151	results/strategy_study_20261016/siglip2-b16-224/selection.json	1aca5b02c95b3b41
S152	results/strategy_study_20261016/siglip2-b16-224/transfer_evaluation/per_source_results.json	fbfc44585ab6f8f3
S153	results/strategy_study_20261016/thermal_trajectories/results.json	fa3ce229a07b41fe
S154	results/strategy_study_20261016/verification.json	ecd2984c5861159b
S155	results/vadclip/runs/full/eval.json	b80eb22a5dfb82c9
S156	results/vadclip/runs/no_graph/eval.json	6ce3846625a6d0b1
S157	results/vadclip/runs/vl_128/eval.json	d0e9386f5505e66a
S158	results/vadclip/runs/vl_512/eval.json	39dc97cc97f36ec1
S159	results/vadclip/runs/win_1/eval.json	beb67019aaabef2c
S160	results/vadclip/runs/win_16/eval.json	46feb1360d4d6ba9
S161	results/video_pilot_20261016/comparison.csv	c474b6adb4955ebf
S162	results/video_pilot_20261016/error_audit/pilot_region_coverage.json	57a619664b80ab33
S163	results/video_pilot_20261016/events.csv	cee168b1e111e7f3
S164	results/video_pilot_20261016/mobileclip2-s2/event_policies.json	5f38904bccd14c9f
S165	results/video_pilot_20261016/mobileclip2-s2/joint_selection.json	3348807a3182eae0
S166	results/video_pilot_20261016/run.json	9207ebb42c2803e0
S167	results/video_pilot_20261016/siglip2-b16-224/joint_selection.json	fa85f8a1cb9c59e6
S168	results/video_pilot_20261016/verification.json	85a94ff384de4e75
S169	results/workflow_gpu_ready_pilot/evaluation.json	6af247bed20d2a48
S170	scripts/benchmark_item_tiles.py	a5dd00d522e5c6b7
S171	scripts/benchmark_loitering_trajectories.py	3295870fb8b00cf6
S172	scripts/bootstrap_report_latex.py	f7e23ec670d5325a
S173	scripts/build_technical_runthrough.py	22cf8e2de2ec9745
S174	scripts/evaluate_video_pilot.py	377727fcd2ce65c0
S175	scripts/fetch_item_component.py	3e45a01a59606a06
S176	scripts/predict_smolvlm.py	52c88551955303ef
S177	scripts/prepare_item_component.py	8b06b0678db5b7d2
S178	scripts/report_event_improvements.py	445563e442f08fd6
S179	scripts/run_event_improvements.py	e47654a9f4c27216
S180	scripts/run_smolvlm_finetuning.py	75afb3ac09195f37
S181	scripts/run_strategy_screen.py	3e64f3363f1ecda9
S182	scripts/smolvlm_gpu.sh	bf97aa57e9945dff
S183	scripts/train_grounded_pilot.py	03b88ff04f6d2e60
S184	scripts/train_person_pilot.py	f0fedd3b7731c799
S185	scripts/train_video_pilot.py	52db372eed7ea451
S186	scripts/verify_smolvlm_finetuning.py	83564369a4109fc1

H.3 Recorded public source locations

These URLs identify source releases, author pages or model repositories recorded by the project. They are not a new survey of current access conditions, and original dataset label/license metadata remains authoritative.

1. Stanford40 action images: http://vision.stanford.edu/Datasets/40actions.html.
2. MIVIA-IWDD author release: https://github.com/MiviaLab/IllegalWasteDumping.
3. Street Scene source: https://www.merl.com/research/?research=license-request&sw=StreetScene.
4. Added smoking photographs: https://www.kaggle.com/datasets/sujaykapadnis/smoking.
5. Recorded HMDB archive source: https://huggingface.co/datasets/Serrelab/hmdb51.
6. Thermal loitering annotations: https://github.com/johnnynunez/RS-WACV24_Loitering.
7. MEVA public data: https://mevadata.org/.
8. Commons media and per-image metadata: https://commons.wikimedia.org/.
9. CAVIAR source: https://homepages.inf.ed.ac.uk/rbf/CAVIAR/.
10. ABODA source: https://github.com/kevinlin311tw/ABODA.
11. AVSS/AOD author package: http://www-vpu.eps.uam.es/publications/AODsurvey/.
12. Additional author debris release: https://github.com/hangsuuuu/video-of-abandoned-object-detection.
13. Grounding DINO tiny weights: https://huggingface.co/IDEA-Research/grounding-dino-tiny.
14. Official Tectonic installation/release documentation: https://tectonic-typesetting.github.io/en-US/install.html.

H.4 Final scope statement

The work produced standardized source-qualified evaluations, multiple trained research bundles, stronger representations/components and a more defensible account of generalization failures. It did not complete independent human verification of all target labels, an untouched multi-camera benchmark covering all five events, or a production streaming application. Those distinctions define what the existing results can support.